Inadequacy of Ordinary Differentiation:

Consider a contravariant tensor

$$T = T^i(x^i)$$

defined on a curve

$$C: x^i = x^i(t)$$

Consider the transformation law:

$$\bar{T}^i = T^r \frac{\partial \bar{x}^i}{\partial x^r}$$

Differentiating w.r.t 't' gives:

$$\frac{d\bar{T}^i}{dt} = \frac{d}{dt}\left(T^r \frac{\partial \bar{x}^i}{\partial x^r}\right)$$

$$= \frac{dT^r}{dt}\frac{\partial \bar{x}^i}{\partial x^r} + T^r \frac{d}{dt}\left(\frac{\partial \bar{x}^i}{\partial x^r}\right)$$

$$\frac{d\bar{T}^i}{dt} = \frac{dT^r}{dt}\frac{\partial \bar{x}^i}{\partial x^r} + \left[T^r \frac{\partial^2 \bar{x}^i}{\partial x^s \partial x^r}\cdot\frac{dx^s}{dt}\right]$$

$$\frac{d}{dt} = \frac{\partial}{\partial x^s}\cdot\frac{dx^s}{dt}$$

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which shows that the ordinary derivative of tensor T along the curve is a contravariant tensor when and only when the $\bar{x}^i$ are the linear functions of the x^r.

Theorem:

The derivative of a tensor is a tensor if and only if coordinate changes are restricted to linear transformations.

Example: Let C is a curve defined by $C: x^i = x^i(t)$ then

$\vec{T} = \frac{d\vec{x}}{dt}$ is tangent field along curve C,

and $k = \left\|\frac{d\vec{T}}{dt}\right\|$ is the curvature.

but 'k' is not an invariant in curvilinear coordinates since $\frac{d\vec{T}}{dt}$ is not a general tensor. Clearly, we require a general concept of tensor differentiation.

Covariant Derivative :

The covariant derivative of an arbitrary tensor is a tensor of which the covariant order exceeds that of the original tensor by exactly one.

Rank of a tensor derivative:

The derivative of a zero order tensor becomes a first order tensor. Generally, the derivative of an m- order tensor forms an $m+1$ order tensor.

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However, if the derivative index is a dummy index, then the rank of the derivative will be lower than that of the original tensor. For example

$A_{ij,j}$ → rank is one.

b/c there is only one free index i.

The Covariant Derivative

- One of the (many) reasons why tensor calculus is so useful is that it allows a proper description of physically meaningful derivatives of vectors.
- We have not mentioned this before, but taking a derivative of a vector (or of any tensor) is a non-trivial thing.
- As usual, in a rectangular orthonormal coordinate system this is usually not an issue. But when the coordinates are curved (like in polar coordinates, for example) then this becomes a big issue.

$\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$

$\frac{d\vec{A}}{dt} = \frac{dA_1}{dt}\hat{i} + \frac{dA_2}{dt}\hat{j} + \frac{dA_3}{dt}\hat{k}$

but in curvilinear coordinates $\hat{e}_i(\vec{x}_i)$ → so their derivatives exist

- It is even more of an issue when the manifold is curved (like in general relativity), the techniques we will discuss here will be equally well applicable for curved manifolds.
- We want to construct a derivative of a vector in such a way that it makes physical sense even in the case of curved coordinates.

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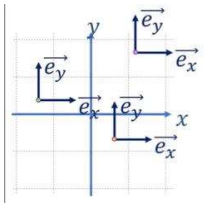
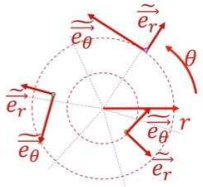
The Covariant Derivative:

Consider the problem of taking the derivative of a vector. In ordinary Cartesian coordinates, this is not complicated because the basis vectors are constant. So we can get the derivative of a vector by differentiating its components. However, in general things are not so straightforward. In curved spaces the basis vectors themselves may vary from point to point. This means that when we take derivative of a vector, we will have to

differentiate the basis vectors as well. For this purpose consider a 2-dimensional rectangular coordinate system and polar coordinate system as shown below.





where $\vec{e}_1 = \frac{\partial \vec{r}}{\partial x^1}$, $\vec{e}_2 = \frac{\partial \vec{r}}{\partial x^2}$

$\vec{\tilde{e}}_1 = \frac{\partial \vec{r}}{\partial \tilde{x}^1}$ and $\vec{\tilde{e}}_2 = \frac{\partial \vec{r}}{\partial \tilde{x}^2}$ $\begin{cases} x^1, x^2 \to x, y \\ \tilde{x}^1, \tilde{x}^2 \to r, \theta \end{cases}$

Now consider the rate of change of some vector field $\vec{V}$ in both coordinate system.

where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\hat{e}_1 = \partial \vec{r}/\partial u^1 \,/\, |\partial \vec{r}/\partial u^1|$

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In rectangular coordinate system

$$\vec{V} = V^1 \vec{e}_1 + V^2 \vec{e}_2 = \sum_i V^i \vec{e}_i$$

$$\Rightarrow \vec{V} = V^i \vec{e}_i = V^i \frac{\partial \vec{r}}{\partial x^i}$$

$$\Rightarrow \frac{\partial}{\partial x^1}(\vec{V}) = \frac{\partial}{\partial x^1}(V^i \vec{e}_i)$$

$$= \frac{\partial V^i}{\partial x^1}\vec{e}_i + V^i \frac{\partial \vec{e}_i}{\partial x^1}$$

But here $\vec{e}_1 = \hat{i}$ and $\vec{e}_2 = \hat{j}$ means constant vectors, Implies

$$\frac{\partial \vec{V}}{\partial x^1} = \frac{\partial V^i}{\partial x^1}\vec{e}_i$$

and similarly:

$$\frac{\partial \vec{V}}{\partial x^2} = \frac{\partial V^i}{\partial x^2}\vec{e}_i$$

Now consider the vector $\vec{V}$ in polar coordinates as:

$$\vec{V} = \tilde{V}^1 \vec{\tilde{e}}_1 + \tilde{V}^2 \vec{\tilde{e}}_2$$

$$\Rightarrow \vec{V} = \tilde{V}^i \vec{\tilde{e}}_i$$

$$\Rightarrow \frac{\partial \vec{V}}{\partial \tilde{x}^1} = \frac{\partial}{\partial \tilde{x}^1}\left(\tilde{V}^i \vec{\tilde{e}}_i\right)$$

$$= \frac{\partial \tilde{V}^i}{\partial \tilde{x}^1} \vec{\tilde{e}}_i + \tilde{V}^i \frac{\partial}{\partial \tilde{x}^1} \vec{\tilde{e}}_i$$

and similarly:

$$\frac{\partial \vec{V}}{\partial \tilde{x}^2} = \frac{\partial \tilde{V}^i}{\partial \tilde{x}^2} \vec{\tilde{e}}_i + \tilde{V}^i \frac{\partial \vec{\tilde{e}}_i}{\partial \tilde{x}^2}$$

Now recall that

$$x = r\cos\theta \quad , \quad y = r\sin\theta$$

$$\vec{\tilde{e}}_r = \cos\theta \vec{e}_1 + \sin\theta \vec{e}_2 = \vec{\tilde{e}}_1$$

$$\vec{\tilde{e}}_\theta = -r\sin\theta \vec{e}_1 + r\cos\theta \vec{e}_2 = \vec{\tilde{e}}_2$$

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$$\Rightarrow \frac{\partial \vec{\tilde{e}}_1}{\partial \tilde{x}^1} = 0 \quad ,$$

$$\frac{\partial \vec{\tilde{e}}_2}{\partial \tilde{x}^1} = \frac{\partial \vec{\tilde{e}}_1}{\partial \tilde{x}^2} = \frac{1}{r} \vec{\tilde{e}}_2$$

and $$\frac{\partial \vec{\tilde{e}}_2}{\partial \tilde{x}^2} = -r \vec{\tilde{e}}_1$$

$$\Rightarrow \frac{\partial \vec{V}}{\partial \tilde{x}^1} = \frac{\partial \tilde{V}^1}{\partial \tilde{x}^1} \vec{\tilde{e}}_1 + \left[\frac{\partial \tilde{V}^2}{\partial \tilde{x}^1} + \frac{\tilde{V}^2}{\tilde{x}^1}\right] \vec{\tilde{e}}_2$$

and $$\frac{\partial \vec{V}}{\partial r} = \frac{\partial \tilde{V}^r}{\partial r} \vec{\tilde{e}}_r + \left[\frac{\partial \tilde{V}^\theta}{\partial r} + \frac{\tilde{V}^\theta}{r}\right] \vec{\tilde{e}}_\theta$$

$$\frac{\partial \vec{V}}{\partial \theta} = \left(\frac{\partial \tilde{V}^1}{\partial \tilde{x}^2} - \tilde{x}^1 \tilde{V}^2\right) \vec{\tilde{e}}_1 + \left(\frac{\partial \tilde{V}^2}{\partial \tilde{x}^2} + \frac{\tilde{V}^1}{\tilde{x}^1}\right) \vec{\tilde{e}}_2$$

$$= \left[\frac{\partial \tilde{V}^r}{\partial \theta} - r\tilde{V}^\theta\right] \vec{\tilde{e}}_r + \left[\frac{\partial \tilde{V}^\theta}{\partial \theta} + \frac{\tilde{V}^r}{r}\right] \vec{\tilde{e}}_\theta$$

So, in general

$$\frac{\partial \vec{V}}{\partial x^i} = \frac{\partial V^j}{\partial x^i}\vec{e}_j + V^j \frac{\partial \vec{e}_j}{\partial x^i}$$

where

$$\frac{\partial \vec{e}_j}{\partial x^i} = \Gamma^k_{ij}\,\vec{e}_k$$

$$= \Gamma^1_{ij}\vec{e}_1 + \Gamma^2_{ij}\vec{e}_2$$

$$\Rightarrow \frac{\partial \vec{V}}{\partial x^i} = \frac{\partial V^i}{\partial x^i}\vec{e}_i + V^i \Gamma^k_{ij}\vec{e}_k$$

These Γ^k_{ij} are the christoffel symbols and the derivative on R.H.S is called covariant derivative.

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Christoffel Symbols of the First Kind:

In 1869 Christoffel introduced certain functions involving derivatives of the components of the fundamental tensor 'g_{ij}'. These functions are called christoffel symbols or an affine connection.

Basically, these quantities represent correction terms, which a derivative operator needs to differentiate a tensor and give a result that is another tensor.

Definition:

The quantities or functions defined by

$$\Gamma_{ijk} = \frac{1}{2}\left[\frac{\partial}{\partial x^i}(g_{jk}) + \frac{\partial}{\partial x^j}(g_{ki}) - \frac{\partial}{\partial x^k}(g_{ij})\right]$$

are the Christoffel Symbols of the first kind.

Transformation Law:

The transformation law for the Γ_{ijk} can be derived from that of g_{ij}. Since g_{ij} is a second rank tensor:

$$\tilde{g}_{ij} = \frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j} g_{rs}$$

differentiating both sides:

$$\tilde{g}_{ij,k} = \frac{\partial}{\partial \tilde{x}^k}\left[\left(\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}\right) g_{rs}\right]$$

$\tilde{g}_{ij,k} \rightarrow \partial \tilde{g}_{ij}/\partial x^k$

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$$\Rightarrow \tilde{g}_{ij,k} = \frac{\partial g_{rs}}{\partial \tilde{x}^k}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j} + g_{rs}\frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j} + g_{rs}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial^2 x^s}{\partial \tilde{x}^k \partial \tilde{x}^j}$$

using chain rule for:

$$\frac{\partial g_{rs}}{\partial \tilde{x}^k} = \frac{\partial g_{rs}}{\partial x^t}\frac{\partial x^t}{\partial \tilde{x}^k} = g_{rs,t}\frac{\partial x^t}{\partial \tilde{x}^k}$$

$$\Rightarrow \tilde{g}_{ij,k} = g_{rs,t}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k} + g_{rs}\left[\overbrace{\frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i}}^{(1)}\frac{\partial x^s}{\partial \tilde{x}^j} + \overbrace{\frac{\partial^2 x^s}{\partial \tilde{x}^k \partial \tilde{x}^j}}^{(2)}\frac{\partial x^r}{\partial \tilde{x}^i}\right]$$

Implies: $(\tilde{g}_{ij,k},\ \tilde{g}_{jk,i},\ \tilde{g}_{ki,j})$

(By interchanging $i \to j$, $j \to k$, $k \to i$)

$$\tilde{g}_{jk,i} = g_{st,r}\frac{\partial x^s}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k}\frac{\partial x^r}{\partial \tilde{x}^j} + g_{sr}\left[\overbrace{\frac{\partial^2 x^s}{\partial \tilde{x}^i \partial \tilde{x}^j}}^{(1)}\frac{\partial x^r}{\partial \tilde{x}^k} + \frac{\partial^2 x^r}{\partial \tilde{x}^i \partial \tilde{x}^k}\frac{\partial x^s}{\partial \tilde{x}^j}\right]$$

Similarly:

$$\tilde{g}_{ki,j} = g_{tr,s}\frac{\partial x^t}{\partial \tilde{x}^k}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j} + g_{sr}\left[\overbrace{\frac{\partial^2 x^s}{\partial \tilde{x}^j \partial \tilde{x}^k}}^{(2)}\frac{\partial x^r}{\partial \tilde{x}^i} + \frac{\partial^2 x^r}{\partial \tilde{x}^j \partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^k}\right]$$

Consider:

$$\bar{\Gamma}^{i}_{jk} = \frac{1}{2}\left(-\bar{g}_{ij,k} + \bar{g}_{jk,i} + \bar{g}_{ki,j}\right)$$

now replacing all values of the metric tensor derivative, we have:

$$\bar{\Gamma}^{i}_{jk} = \frac{1}{2}\left[-g_{rs,t} + g_{st,r} + g_{tr,s}\right]\frac{\partial x^r}{\partial \bar{x}^i}\frac{\partial x^s}{\partial \bar{x}^j}\frac{\partial x^t}{\partial \bar{x}^k}$$

$$+\frac{1}{2} g_{rs}\left[\frac{\partial^2 x^s}{\partial \bar{x}^i \partial \bar{x}^j}\frac{\partial x^r}{\partial \bar{x}^k} + \frac{\partial^2 x^r}{\partial \bar{x}^j \partial \bar{x}^i}\frac{\partial x^s}{\partial \bar{x}^k}\right]$$

$$\Rightarrow \bar{\Gamma}^{i}_{jk} = \Gamma^{r}_{st}\frac{\partial x^r}{\partial \bar{x}^i}\frac{\partial x^s}{\partial \bar{x}^j}\frac{\partial x^t}{\partial \bar{x}^k} + g_{rs}\frac{\partial^2 x^r}{\partial \bar{x}^i \partial \bar{x}^j}\frac{\partial x^s}{\partial \bar{x}^k}$$

(by interchanging the dummy indices r and s in last term.)

$\Rightarrow$ Christoffel Symbol is not a general tensor.

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Question: Compute the christoffel symbols corresponding to the Euclidean metric for spherical coordinates.

$$[g_{\mu\nu}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & (x^1)^2 \sin^2 x^2 \end{bmatrix}$$

Solution: The formula for christoffel symbols is:

$$\Gamma^{i}_{jk} = \frac{1}{2}\left[-g_{ij,k} + g_{jk,i} + g_{ki,j}\right]$$

$$\Gamma_{221} = \frac{1}{2}\left[-g_{22,1} + g_{21,2} + g_{12,2}\right]$$

$$= \frac{1}{2}(-2x^1 + 0 + 0) = -x^1 \text{ or } \boxed{\Gamma_{221} = -x^1}$$

$$\Gamma_{212} = \frac{1}{2}\left(-\overrightarrow{g_{21,2}}^{0} + g_{22,1} + \overrightarrow{g_{12,2}}^{0}\right)$$

$$= \frac{1}{2}(2x^1) = x^1 \text{ or } \boxed{\Gamma_{212} = x^1}$$

$$\Gamma_{122} = \frac{1}{2}\left(-\overrightarrow{g_{12,2}}^{0} + g_{22,1} + \overrightarrow{g_{21,2}}^{0}\right)$$

$$= \frac{1}{2}(x^1) = x^1$$

$$\Gamma_{223} = \frac{1}{2}\left(-g_{22,3} + g_{23,2} + g_{32,2}\right) = 0$$

Similarly $\Gamma_{232} = \Gamma_{322} = 0$

$$\Gamma_{331} = \frac{1}{2}\left(-g_{33,1} + \overrightarrow{g_{31,3}}^{0} + \overrightarrow{g_{13,3}}^{0}\right)$$

$$= \frac{1}{2}\left(-2x^1 \sin^2(x^2)\right) = -x^1 \sin^2 x^2 \text{ or } = x \sin^2\theta$$

Similarly:

$$\Gamma_{133} = x^1 \sin^2 x^2 = \Gamma_{313}$$

$$\Gamma_{332} = \frac{1}{2}\left(-g_{33,2} + \overrightarrow{g_{32,3}}^{0} + \overrightarrow{g_{23,3}}^{0}\right)$$

$$= \frac{1}{2}\left(-(x^1)^2 2 \sin x^2 \cos x^2\right)$$

$$= -(x^1)^2 \sin x^2 \cos x^2 \text{ or } \Gamma_{332} = -x \sin\theta \cos\theta$$

Similarly:

$$\Gamma_{323} = (x^1)^2 \sin x^2 \cos x^2$$

and

$$\Gamma_{233} = (x^1)^2 \sin x^2 \cos x^2$$

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Properties of Christofell Symbol of first kind:

i) $\Gamma_{ijk} = \Gamma_{jik}$ (symmetric in first two indices)

ii) All Γ_{ijk} vanish if all g_{ij} are constant.

Question:
Verify that $\Gamma_{ijk} = \Gamma_{jik}$.

Solution:
By definition,

$$\Gamma_{ijk} = \frac{1}{2}\left(-g_{ij,k} + g_{jk,i} + g_{ki,j}\right)$$

Interchanging i and j:

$$\Gamma_{jik} = \frac{1}{2}\left(-g_{ji,k} + g_{ik,j} + g_{kj,i}\right)$$

Since metric tensor is symmetric.
This implies:

$$g_{ij,k} = g_{ji,k}\ ,\ g_{jk,i} = g_{kj,i}\ ,\ g_{ki,j} = g_{ik,j}$$

$$\Rightarrow \Gamma_{ijk} = \Gamma_{jik}.$$

Question: Show that if $[g_{ij}]$ is a diagonal matrix, then for all fixed subscripts α and $\beta \neq \alpha$ in the range $1,2,\ldots,n$,

1) $\Gamma_{\alpha\alpha\alpha} = \frac{1}{2} g_{\alpha\alpha,\alpha}$ (not summed on α)

2) $-\Gamma_{\alpha\alpha\beta} = \Gamma_{\alpha\beta\alpha} = \Gamma_{\beta\alpha\alpha} = \frac{1}{2} g_{\alpha\alpha\beta}$ (not summed on α)

3) All remaining Christoffel Symbols Γ_{ijk} are zero.

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Solution:

(1) $\Gamma_{\alpha\alpha\alpha} = \frac{1}{2}\left[-g_{\alpha\alpha,\alpha} + g_{\alpha\alpha,\alpha} + g_{\alpha\alpha,\alpha}\right]$

$$\Gamma_{\alpha\alpha\alpha} = \frac{1}{2} g_{\alpha\alpha,\alpha}$$

(2) $-\Gamma_{\alpha\alpha\beta} = -\frac{1}{2}\left(-g_{\alpha\alpha,\beta} + g_{\alpha\beta,\alpha}^{\nearrow 0} + g_{\beta\alpha,\alpha}^{\nearrow 0}\right)$

$$= \frac{1}{2} g_{\alpha\alpha,\beta}$$

$$\Gamma_{\alpha\beta\alpha} = \Gamma_{\beta\alpha\alpha} = \frac{1}{2}\left(-g_{\alpha\beta,\alpha}^{\nearrow 0} + g_{\beta\alpha,\alpha}^{\nearrow 0} + g_{\alpha\alpha,\beta}\right)$$

$$= \frac{1}{2} g_{\alpha\alpha,\beta}$$

(3) Let i,j,k be distinct subscripts.

Then $g_{ij} = 0$ and $g_{ij,k} = 0$

implying that

$$\Gamma_{ijk} = 0$$

$$\Gamma_{ijk} = \frac{1}{2}\left[-g_{ij,k}^{\nearrow 0} + g_{jk,i}^{\nearrow 0} + g_{ki,j}^{\nearrow 0}\right]$$

$$i \neq j \neq k$$

Question: If all Γ_{ijk} vanish in some coordinate system, then do the metric tensor has constant components in every coordinate system?

Solution: As $\Gamma_{ijk} = \frac{1}{2}(-g_{ij,k} + g_{jk,i} + g_{ki,j})$

$$\Gamma_{jki} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ij,k})$$

Adding both:

$$\Gamma_{ijk} + \Gamma_{jki} = \frac{1}{2}(2g_{ik,j}) = g_{ik,j}$$

$$\Rightarrow g_{ik,j} = \Gamma_{ijk} + \Gamma_{jki}$$

Now, the conclusion would be valid if the Γ_{ijk} vanished in every coordinate system. But Γ_{ijk} is not a tensor and the conclusion is false. For instance, all $\Gamma_{ijk} = 0$ for the Euclidean metric in rectangular coordinates, but $g_{22} = (x^1)^2, g_{33} = (x^1)^2 \sin^2(x^2)$ in spherical coordinates.

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Christoffel Symbols of the Second Kind:

The functions

$$\Gamma^i_{jk} = g^{ir}\Gamma_{jkr}$$

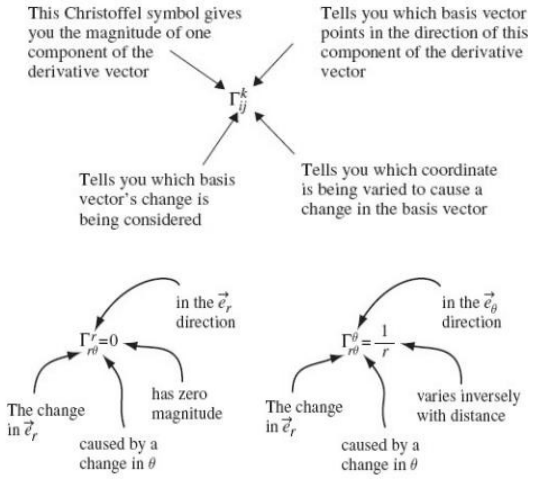
are the Christoffel symbols of the second kind. It is simply the result of raising the third subscript of the Christoffel Symbol of the first kind.

Properties:

The basic properties of Γ_{ijk} carry over to Γ^i_{jk}.

1) $\Gamma^i_{jk} = \Gamma^i_{kj}$ (Symmetric in the lower indices.)

2) All Γ^i_{jk} vanish if all g_{ij} are constant.



Transformation law:

Starting with:

$$\tilde{\Gamma}^{i}_{jk} = \tilde{g}^{ir}\,\tilde{\Gamma}_{jkr} = \left(g^{st}\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial \tilde{x}^r}{\partial x^t}\right)\Gamma_{jkr}$$

Substitute for $\tilde{\Gamma}_{jkr}$ from the transformation law of Christoffel symbol of first kind, we have

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$$\tilde{\Gamma}^{i}_{jk} = \left(g^{st}\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial \tilde{x}^r}{\partial x^t}\right)\left(\Gamma_{uvw}\frac{\partial x^u}{\partial \tilde{x}^j}\frac{\partial x^v}{\partial \tilde{x}^k}\frac{\partial x^w}{\partial \tilde{x}^r}\right) + \left(g^{st}\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial \tilde{x}^r}{\partial x^t}\right)\left(g_{uv}\frac{\partial^2 x^u}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial x^v}{\partial \tilde{x}^r}\right)$$

$$= g^{st}\Gamma_{uvw}\,\delta^w_t\,\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial x^u}{\partial \tilde{x}^j}\frac{\partial x^v}{\partial \tilde{x}^k} + g^{st}g_{uv}\,\delta^v_t\,\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial^2 x^u}{\partial \tilde{x}^j \partial \tilde{x}^k}$$

$$\Rightarrow \tilde{\Gamma}^{i}_{jk} = g^{st}\Gamma_{uvt}\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial x^u}{\partial \tilde{x}^j}\frac{\partial x^v}{\partial \tilde{x}^k} + g^{st}g_{ut}\frac{\partial \tilde{x}^i}{\partial x^s}\frac{\partial^2 x^u}{\partial \tilde{x}^j \partial \tilde{x}^k}$$

Since $g^{st}\Gamma_{uvt} = \Gamma^s_{uv}$ and $g^{st}g_{ut} = \delta^s_u$,

Implies:

$$\Rightarrow \tilde{\Gamma}^{i}_{jk} = \Gamma^{r}_{st}\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k} + \frac{\partial^2 x^r}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial \tilde{x}^i}{\partial x^r}$$

$\Rightarrow \Gamma^i_{jk}$ is not a general tensor and is merely an affine tensor.

Question: Calculate the Christoffel Symbols of the second kind for the Euclidean metric in polar coordinates.

$$[g_{\mu\nu}] = \begin{bmatrix} 1 & 0 \\ 0 & (x^1)^2 \end{bmatrix} \quad \begin{pmatrix} x^1 \to r \\ x^2 \to \theta \end{pmatrix}$$

Solution:

Using the formula:

$$\Gamma^i_{jk} = \frac{g^{ir}}{2}\left[-g_{jk,r} + g_{kr,j} + g_{rj,k}\right]$$

$$\Gamma^1_{11} = \frac{g^{1r}}{2}\left[-g_{11,1} + g_{1r,1} + g_{r1,1}\right]$$

$$= \frac{g^{11}}{2}\left[-g_{11,1} + g_{11,1} + g_{11,1}\right]$$

$$= \frac{1}{2}\left[0 + 0 + 0\right] = 0$$

$$\Gamma^1_{21} = \frac{g^{1r}}{2}\left[-g_{21,r} + g_{1r,2} + g_{r2,1}\right]$$

$$= \frac{g^{11}}{2}\left[-g_{21,1}^{\to 0} + g_{11,2}^{\nearrow 0} + g_{12,1}^{\to 0}\right] = 0$$

Similarly $\Gamma^1_{12} = 0$

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$$\Gamma^1_{22} = \frac{g^{1r}}{2}\left[-g_{22,1} + g_{2r,2} + g_{r2,2}\right]$$

$$= \frac{g^{11}}{2}\left[-g_{22,1} + g_{22,2} + g_{22,2}\right]$$

$$= \frac{1}{2}\left[-2(x^1) + 0 + \quad\right] = -x^1$$

$$\Gamma^2_{21} = \frac{g^{2r}}{2}\left[-g_{21,r} + g_{1r,2} + g_{r2,1}\right]$$

$$= \frac{g^{22}}{2}\left[-g_{21,2}^{\to 0} + g_{12,2}^{\to 0} + g_{22,1}\right]$$

$$= \frac{g^{22}}{2}(2x^1) = (x^1)^{-2}(x^1) = \frac{1}{x^1} \text{ or } \frac{1}{r}$$

and $\Gamma^2_{12} = \frac{1}{x^1}$

$$\Gamma^2_{11} = \frac{g^{2r}}{2}\left[-g_{11,r} + g_{1r,2} + g_{r1,1}\right]$$

$$= \frac{g^{22}}{2}\left[-g_{11,2}^{\to 0} + g_{12,2}^{\to 0} + g_{21,1}^{\to 0}\right] = 0$$

$$\Gamma^2_{22} = \frac{g^{2k}}{2}\left[-g_{22,k} + g_{2k,2} + g_{k2,2}\right]$$

$$= \frac{g^{22}}{2}\left[-\cancel{g_{22,2}}^{\,0} + \cancel{g_{22,2}}^{\,0} + \cancel{g_{22,2}}^{\,0}\right]$$

$$= 0$$

Question: If $[g_{ij}]$ is a diagonal matrix, Show that for all fixed indices in the range $1, 2, \ldots, n$.

a) $\Gamma^{\alpha}_{\alpha\beta} = \Gamma^{\alpha}_{\beta\alpha} = \frac{\partial}{\partial x^{\beta}}\left(\frac{1}{2}\ln|g_{\alpha\alpha}|\right)$

consider

$$\Gamma^{\alpha}_{\alpha\beta} = \frac{g^{\alpha k}}{2}\left[-\cancel{g_{\alpha\beta,k}}^{\,0} + g_{\alpha k,\beta} + g_{\beta k,\alpha}\right]$$

$$= \frac{g^{\alpha\alpha}}{2}\left[g_{\alpha\alpha,\beta} + \cancel{g_{\beta\alpha,\alpha}}^{\,0}\right] = \frac{1}{2}g^{\alpha\alpha}\, g_{\alpha\alpha,\beta}$$

$$= \frac{1}{2}\frac{g_{\alpha\alpha,\beta}}{g_{\alpha\alpha}} = \frac{\partial}{\partial x^{\beta}}\left(\frac{1}{2}\ln g_{\alpha\alpha}\right)$$

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b) $\Gamma^{\alpha}_{\beta\beta} = \frac{-1}{2g_{\alpha\alpha}}\, g_{\beta\beta,\alpha}$

$$\Gamma^{\alpha}_{\beta\beta} = \frac{g^{\alpha k}}{2}\left[-g_{\beta\beta,k} + g_{\beta k,\beta} + g_{\beta k,\beta}\right]$$

$$= \frac{g^{\alpha\alpha}}{2}\left[-g_{\beta\beta,\alpha} + \cancel{g_{\beta\alpha,\beta}}^{\,0} + \cancel{g_{\beta\alpha,\beta}}^{\,0}\right]$$

$$= -\frac{g^{\alpha\alpha}}{2}\, g_{\beta\beta,\alpha}$$

$$= -\frac{1}{2g_{\alpha\alpha}}\, g_{\beta\beta,\alpha}$$

Question: Christoffel Symbols of the second kind for the Euclidean metric in spherical coordinates.

here, $[g_{\alpha\beta}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & (x^1)^2\sin^2 x^2 \end{bmatrix}$

Then the non zero Christoffel symbols of second kind are:

$$\Gamma^2_{21} = \Gamma^2_{12} = \frac{\partial}{\partial x^1}\left(\frac{1}{2}\ln(x^1)^2\right) = \frac{1}{x^1}$$

$$\Gamma^3_{31} = \Gamma^3_{13} = \frac{\partial}{\partial x^1}\left(\frac{1}{2}\ln\left((x^1)^2\sin^2 x^2\right)\right) = \frac{1}{x^1}$$

$$\Gamma^3_{32} = \Gamma^3_{23} = \frac{\partial}{\partial x^2}\left(\frac{1}{2}\ln\left((x^1)^2\sin^2 x^2\right)\right) = \cot x^2$$

Similarly:

$$\Gamma^1_{22} = -\frac{1}{2(1)}\frac{\partial}{\partial x^1}(x^1)^2 = -x^1$$

$$\Gamma^1_{33} = -\frac{1}{2(1)}\frac{\partial}{\partial x^1}\left((x^1)^2\sin^2 x^2\right) = -x^1\sin^2 x^2$$

$$\Gamma^2_{33} = -\frac{1}{2(x^1)^2}\frac{\partial}{\partial x^2}\left((x^1)^2\sin^2 x^2\right) = -\sin x^2\cos x^2$$

Question: Use the equation

$$\frac{\partial^2 x^k}{\partial \bar{x}^i \partial \bar{x}^j} = \bar{\Gamma}^s_{ij}\frac{\partial x^k}{\partial \bar{x}^s} - \Gamma^k_{st}\frac{\partial x^s}{\partial \bar{x}^i}\frac{\partial x^t}{\partial \bar{x}^j} \rightarrow (\star)$$

to find the most general 3-dimensional transformation of coordinates such that x^i correspond to rectangular and $\bar{x}^i$ correspond to any other coordinate system for which the Christoffel symbols are:

$$\bar{\Gamma}^1_{11} = 1, \quad \bar{\Gamma}^2_{22} = 2 \text{ and } \bar{\Gamma}^3_{33} = 3$$

all other are zero.

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Solution: $x^i \rightarrow$ rectangular Coordinates

$$\Rightarrow [g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \Gamma^k_{st} = 0$$

so Eq.$\star$ implies:

$$\frac{\partial^2 x^k}{\partial \bar{x}^i \partial \bar{x}^j} = \bar{\Gamma}^s_{ij}\frac{\partial x^k}{\partial \bar{x}^s}$$

let $\frac{\partial x^k}{\partial \bar{x}^s} = \bar{u}^k_s$

$$\Rightarrow \frac{\partial \bar{u}^k_j}{\partial \bar{x}^i} = \bar{\Gamma}^s_{ij}\bar{u}^k_s$$

Moreover, replace $\bar{u}^k_j = \bar{u}_j$

$$\Rightarrow \frac{\partial \bar{u}_j}{\partial \bar{x}^i} = \bar{\Gamma}^s_{ij}\bar{u}_s$$

$$\Rightarrow \frac{\partial \bar{u}_1}{\partial \bar{x}^1} = \bar{\Gamma}^1_{11}\bar{u}_1 \Rightarrow \frac{\partial \bar{u}_1}{\partial \bar{x}^1} = \bar{u}_1$$

$$\Rightarrow \frac{\partial \bar{u}_1}{u_1} = \partial \bar{x}^1 \Rightarrow \ln \bar{u}_1 = \bar{x}^1 + c_1$$

$$\Rightarrow \bar{u}_1 = b_1 e^{\bar{x}^1}$$

and Similarly:

$$\bar{u}_2 = b_2 e^{2\bar{x}^2} \quad \text{and} \quad \bar{u}_3 = b_3 e^{3\bar{x}^3}$$

now, $\frac{\partial x^{\bar{1}}}{\partial \bar{x}^1} = b_1 e^{\bar{x}^1}$

$\Rightarrow \partial x^{\bar{1}} = b_1 e^{\bar{x}^1} \partial \bar{x}^1$

$\Rightarrow x^{\bar{1}} = b_1 e^{\bar{x}^1} + \phi(\bar{x}^2, \bar{x}^3)$

and similarly:

$\frac{\partial x^{\bar{2}}}{\partial \bar{x}^2} = b_2 e^{2\bar{x}^2}$ and $\frac{\partial x^{\bar{3}}}{\partial \bar{x}^3} = b_3 e^{3\bar{x}^3}$

$\Rightarrow x^{\bar{2}} = b_2 e^{2\bar{x}^2} + \phi(\bar{x}^1, \bar{x}^3)$

and $x^{\bar{3}} = b_3 e^{3\bar{x}^3} + \phi(\bar{x}^1, \bar{x}^2)$

$\Rightarrow x^{\bar{1}} = b_1^{\bar{1}} e^{\bar{x}^1} + b_2^{\bar{1}} e^{\bar{x}^2} + b_3^{\bar{1}} e^{\bar{x}^3} + b_4^{\bar{1}}$

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Question: If

$\bar{x}^1 = -e^{d_1 x^1} + e^{d_2 x^2} + e^{d_3 x^3}$

$\bar{x}^2 = 2e^{d_1 x^1} - e^{d_2 x^2} + e^{d_3 x^3}$

$\bar{x}^3 = e^{d_1 x^1} - 2e^{d_2 x^2} + 3e^{d_3 x^3}$

and if all $\bar{\Gamma}^{\bar{i}}_{jk} = 0$, find the Γ^i_{jk}.

Use the formula:

$\Gamma^i_{jk} = \frac{\partial^2 \bar{x}^{\bar{s}}}{\partial x^j \partial x^k} \frac{\partial x^i}{\partial \bar{x}^{\bar{s}}}$

Solution: Given

$\Gamma^i_{jk} = \frac{\partial^2 \bar{x}^{\bar{s}}}{\partial x^j \partial x^k} \frac{\partial x^i}{\partial \bar{x}^{\bar{s}}}$

$\frac{\partial \bar{x}^1}{\partial x^1} = -e^{d_1 x^1} \cdot \frac{\partial}{\partial x^1}(d_1 x^1) = \phi_1(x^1)$

$\Rightarrow \frac{\partial \bar{x}^1}{\partial x^2 \partial x^1} = 0$ and $\frac{\partial \bar{x}^1}{\partial x^3 \partial x^1} = 0$

Similarly

$\frac{\partial \bar{x}^1}{\partial x^1 \partial x^2} = 0$ and $\frac{\partial \bar{x}^1}{\partial x^1 \partial x^3} = 0$

$$\Rightarrow \frac{\partial^2 \tilde{x}^k}{\partial x^j \partial x^k} \neq 0 \quad \text{for} \quad j = k$$

$$= 0 \quad \text{for} \quad j \neq K$$

$$\Rightarrow \Gamma^i_{jj} = \frac{\partial}{\partial x^j}\left(\frac{\partial \tilde{x}^k}{\partial x^j}\right) \cdot \frac{\partial x^i}{\partial \tilde{x}^k}$$

and $\Gamma^i_{jk} = 0$ when $j \neq K$.

Question: let the coordinate system x^i be connected to a system of rectangular coordinates $\tilde{x}^i$ by:

$$\tilde{x}^1 = e^{(x^1 + x^2)}$$

and $\tilde{x}^2 = e^{(x^1 - x^2)}$

Then using $\Gamma^i_{jk} = \frac{\partial^2 \tilde{x}^k}{\partial x^j \partial x^k} \frac{\partial x^i}{\partial \tilde{x}^k}$

Find the non-zero Christoffel Symbols.

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Solution: Given

$$\Gamma^i_{jk} = \frac{\partial^2 \tilde{x}^k}{\partial x^j \partial x^k} \frac{\partial x^i}{\partial \tilde{x}^k}$$

$$\frac{\partial \tilde{x}^1}{\partial x^1} = e^{x^1 + x^2}(1) = e^{x^1 + x^2}$$

$$\frac{\partial \tilde{x}^1}{\partial x^2} = e^{x^1 + x^2}(1) = e^{x^1 + x^2}$$

$$\frac{\partial^2 \tilde{x}^1}{(\partial x^1)^2} = e^{x^1 + x^2}, \quad \frac{\partial^2 \tilde{x}^1}{\partial x^1 \partial x^2} = e^{x^1 + x^2}$$

and $\frac{\partial^2 \tilde{x}^1}{(\partial x^2)^2} = e^{x^1 + x^2}$

$$\frac{\partial \tilde{x}^2}{\partial x^1} = e^{(x^1 - x^2)}(1) = e^{x^1 - x^2}$$

$$\frac{\partial \tilde{x}^2}{\partial x^2} = -e^{x^1 - x^2}$$

$$\frac{\partial^2 \tilde{x}^2}{(\partial x^1)^2} = e^{x^1 - x^2}, \quad \frac{\partial^2 \tilde{x}^2}{(\partial x^2)^2} = e^{x^1 - x^2}$$

$$\frac{\partial^2 \tilde{x}^2}{\partial x^1 \partial x^2} = -e^{x^1 - x^2} = \frac{\partial^2 \tilde{x}^2}{\partial x^2 \partial x^1}$$

We also require inverse transformations:

$\tilde{x}^1 = e^{x^1+x^2}$ and $\tilde{x}^2 = e^{x^1-x^2}$

$\Rightarrow x^1+x^2 = \ln\tilde{x}^1$ and $x^1-x^2 = \ln\tilde{x}^2$

$\Rightarrow 2x^1 = \ln\tilde{x}^1+\ln\tilde{x}^2$

$\Rightarrow x^1 = \frac{1}{2}\left[\ln\tilde{x}^1 + \ln\tilde{x}^2\right]$

and $2x^2 = \ln\tilde{x}^1 - \ln\tilde{x}^2$

$\Rightarrow x^2 = \frac{1}{2}\left[\ln\tilde{x}^1 - \ln\tilde{x}^2\right]$

$\Rightarrow \frac{\partial x^1}{\partial\tilde{x}^1} = \frac{1}{2\tilde{x}^1} = \frac{1}{2e^{x^1+x^2}}$

$\frac{\partial x^1}{\partial\tilde{x}^2} = \frac{1}{2\tilde{x}^2} = \frac{1}{2e^{x^1-x^2}}$

and $\frac{\partial x^2}{\partial\tilde{x}^1} = \frac{1}{2\tilde{x}^1} = \frac{1}{2e^{x^1+x^2}}$

$\frac{\partial x^2}{\partial\tilde{x}^2} = -\frac{1}{2\tilde{x}^2} = -\frac{1}{2e^{x^1-x^2}}$

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$\Rightarrow \Gamma^1_{11} = \frac{\partial^2\tilde{x}^k}{\partial x^1\partial x^1}\cdot\frac{\partial x^1}{\partial\tilde{x}^k}$

$= e^{x^1+x^2}\left(\frac{1}{2e^{x^1+x^2}}\right) + \left(e^{x^1-x^2}\right)\left(\frac{1}{2e^{x^1-x^2}}\right)$

$\Gamma^1_{11} = \frac{1}{2}+\frac{1}{2} = 1$

$\Gamma^1_{12} = \frac{\partial^2\tilde{x}^k}{\partial x^1\partial x^2}\cdot\frac{\partial x^1}{\partial\tilde{x}^k}$

$= \left(e^{x^1+x^2}\right)\left(\frac{1}{2e^{x^1+x^2}}\right) + \left(-e^{x^1-x^2}\right)\left(\frac{1}{2e^{x^1-x^2}}\right)$

$= \frac{1}{2}-\frac{1}{2} = 0 = \Gamma^1_{21}$

$\Gamma^1_{22} = \frac{\partial^2\tilde{x}^k}{\partial x^2\partial x^2}\cdot\frac{\partial x^1}{\partial\tilde{x}^k}$

$= e^{x^1+x^2}\cdot\frac{1}{2}e^{-(x^1+x^2)} + e^{x^1-x^2}\cdot\frac{1}{2}e^{-(x^1-x^2)}$

$= \frac{1}{2}+\frac{1}{2} = 1$

and similarly:

$\Gamma^2_{11} = 0$, $\Gamma^2_{12} = 1 = \Gamma^2_{21}$

$\Gamma^2_{22} = 0$

Covariant Differentiation:

Here we will derive the formula for covariant derivative of a vector. Consider the transformation law:

$$\tilde{T}_i = T_r \frac{\partial x^r}{\partial \tilde{x}^i}$$

Its partial differentiation yields:

$$\frac{\partial \tilde{T}_i}{\partial \tilde{x}^k} = \frac{\partial T_r}{\partial \tilde{x}^k}\frac{\partial x^r}{\partial \tilde{x}^i} + T_r \frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i} \longrightarrow (\star)$$

now we find an expression for $\frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i}$ in terms of Christoffel symbols of second kind.

Recall the transformation law for Christoffel Symbols of second kind:

$$\tilde{\Gamma}^s_{ik} = \Gamma^r_{ut}\frac{\partial \tilde{x}^s}{\partial x^r}\frac{\partial x^u}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k} + \frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i}\frac{\partial \tilde{x}^s}{\partial x^r}$$

$$\Rightarrow \frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i} = \tilde{\Gamma}^s_{ik}\frac{\partial x^r}{\partial \tilde{x}^s} - \Gamma^v_{ut}\frac{\partial \tilde{x}^s}{\partial x^v}\cdot\frac{\partial x^r}{\partial \tilde{x}^s}\frac{\partial x^u}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k}$$

$$\Rightarrow \frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i} = \tilde{\Gamma}^s_{ik}\frac{\partial x^r}{\partial \tilde{x}^s} - \Gamma^v_{ut}\delta^r_v\frac{\partial x^u}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k}$$

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$$\Rightarrow \frac{\partial^2 x^r}{\partial \tilde{x}^k \partial \tilde{x}^i} = \tilde{\Gamma}^s_{ik}\frac{\partial x^r}{\partial \tilde{x}^s} - \Gamma^r_{st}\frac{\partial x^s}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k} \quad \checkmark$$

$$\Rightarrow \frac{\partial \tilde{T}_i}{\partial \tilde{x}^k} = \frac{\partial T_r}{\partial \tilde{x}^k}\frac{\partial x^r}{\partial \tilde{x}^i} + T_r\left[\tilde{\Gamma}^s_{ik}\frac{\partial x^r}{\partial \tilde{x}^s} - \Gamma^r_{st}\frac{\partial x^s}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k}\right]$$

$$\Rightarrow \frac{\partial \tilde{T}_i}{\partial \tilde{x}^k} - \tilde{\Gamma}^s_{ik}\left(T_r\frac{\partial x^r}{\partial \tilde{x}^s}\right) = \frac{\partial T_r}{\partial x^s}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^k} - \Gamma^r_{st}T_r\frac{\partial x^s}{\partial \tilde{x}^i}\frac{\partial x^t}{\partial \tilde{x}^k}$$

$$\Rightarrow \frac{\partial \tilde{T}_i}{\partial \tilde{x}^k} - \tilde{\Gamma}^s_{ik}\tilde{T}_s = \frac{\partial T_r}{\partial x^s}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^k} - \Gamma^t_{rs}T_t\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^k} \qquad \begin{cases} r \to t \\ s \to r \\ t \to s \end{cases}$$

$$\Rightarrow \frac{\partial \tilde{T}_i}{\partial \tilde{x}^k} - \tilde{\Gamma}^s_{ik}\tilde{T}_s = \left(\frac{\partial T_r}{\partial x^s} - \Gamma^t_{rs}T_t\right)\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial \tilde{x}^s}{\partial \tilde{x}^k}$$

$\Rightarrow$ When the components of $\frac{\partial T_i}{\partial x^k}$ are corrected by subtracting certain linear combinations of T_i itself, the result is a tensor.

Definition:

In any coordinate system x^i, the covariant derivative with respect to x^k of a covariant vector T_i is the tensor

$$T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik} T_t$$

* here the second index 'k' emphasize that the derivative is w.r.t kth coordinate.
* The covariant derivative and partial derivative coincide when g_{ij} are constants.

Definition:

In any coordinate system x^i, the covariant derivative with respect to x^k of a contravariant vector T^i is the tensor

$$T^i_{;k} = \frac{\partial T^i}{\partial x^k} + \Gamma^i_{jk} T^t$$

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Covariant Derivative of any tensor:

In any coordinate system x^i, the covariant derivative with respect to x^k of a tensor $T^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q}$ is defined as:

$$T^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q ; k} = \frac{\partial T^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q}}{\partial x^k} + \Gamma^{i_1}_{tk} T^{t i_2 i_3 \cdots i_p}_{j_1 j_2 \cdots j_q} + \cdots$$

$$- \Gamma^t_{j_1 k} T^{i_1 i_2 \cdots i_p}_{t j_2 \cdots j_q} - \cdots - \Gamma^t_{j_q k} T^{i_1 i_2 \cdots i_p}_{j_1 j_2 \cdots j_q}$$

* $T^\alpha{}_{\beta;\gamma} = T^\alpha{}_{\beta,\gamma} + \Gamma^\alpha_{\gamma\delta} T^\delta_\beta - \Gamma^\delta_{\beta\gamma} T^\alpha_\delta$

$$T_{\alpha\beta;\gamma} = T_{\alpha\beta,\gamma} - \Gamma^\delta_{\alpha\gamma} T_{\delta\beta} - \Gamma^\delta_{\beta\gamma} T_{\alpha\delta}$$

$$T^{\alpha\beta}{}_{;\gamma} = T^{\alpha\beta}{}_{,\gamma} + \Gamma^\alpha_{\delta\gamma} T^{\delta\beta} + \Gamma^\beta_{\gamma\delta} T^{\alpha\delta}$$

* The covariant derivative of a Scalar function is just the partial derivative.

$$\varphi_{;\alpha} = \varphi_{,\alpha}$$

Question: Consider polar coordinates and find the covariant derivative $V^{\alpha}{}_{;\alpha}$ of $V^{\alpha} = r^2\cos\theta\,\hat{e}_r - \sin\theta\,\hat{e}_\theta$.

Solution: Using the formula we have:

$$V^{\alpha}{}_{;\alpha} = V^{\alpha}{}_{,\alpha} + \Gamma^{\alpha}_{\beta\alpha}V^{\beta}$$

$$V^{\alpha}{}_{;\alpha} = V^{1}{}_{;1} + V^{2}{}_{;2}$$

$$V^{1}{}_{;1} = V^{1}{}_{,1} + \Gamma^{1}_{\beta 1}V^{\beta}$$

$$V^{2}{}_{;2} = V^{2}{}_{,2} + \Gamma^{2}_{\beta 2}V^{\beta}$$

Consider,

$$V^{1}{}_{;1} = V^{1}{}_{,1} + \Gamma^{1}_{\beta 1}V^{\beta}$$

$$= (r^2\cos\theta)_{,r} + \Gamma^{1}_{11}V^{1} + \Gamma^{1}_{21}V^{2}$$

now using the Christoffel symbols given below:

$$\Gamma^{1}_{11} = 0, \quad \Gamma^{1}_{21} = 0$$

$$\Gamma^{2}_{12} = \Gamma^{2}_{21} = \frac{1}{r} \quad \text{and} \quad \Gamma^{1}_{22} = -r$$

$$\Rightarrow V^{1}{}_{;1} = 2r\cos\theta$$

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$$V^{2}{}_{;2} = V^{2}{}_{,2} + \Gamma^{2}_{\beta 2}V^{\beta}$$

$$= (-\sin\theta)_{,\theta} + \Gamma^{2}_{12}V^{1} + \overset{\nearrow 0}{\Gamma^{2}_{22}}V^{2}$$

$$= -\cos\theta + \frac{1}{r}(r^2\cos\theta)$$

$$= -\cos\theta + r\cos\theta$$

$$\Rightarrow V^{\alpha}{}_{;\alpha} = 2r\cos\theta + r\cos\theta - \cos\theta$$

$$V^{\alpha}{}_{;\alpha} = 3r\cos\theta - \cos\theta$$

$$= (3r-1)\cos\theta$$

Question: Using the spherical polar coordinates $(r,\theta,\phi) \to (x^1,x^2,x^3)$ and the christoffel symbols

$$\Gamma^{2}_{21} = \Gamma^{2}_{12} = \frac{1}{r}, \quad \Gamma^{3}_{31} = \Gamma^{3}_{13} = \frac{1}{r}$$

$$\Gamma^{3}_{32} = \Gamma^{3}_{23} = \cot\theta, \quad \Gamma^{1}_{22} = -r,$$

$$\Gamma^{1}_{33} = -r\sin^2\theta, \quad \Gamma^{2}_{33} = -\sin\theta\cos\theta$$

find the covariant derivative $V^{\alpha}{}_{;\beta}$ for $\beta = 1,2,3$. $\alpha = 1,2,3$

where $V^{\alpha} = (r, r^2\cos\theta, \sin\phi)$

Solution:

$$V^{\alpha}_{;\beta} = V^{\alpha}_{,\beta} + \Gamma^{\alpha}_{t\beta} V^{t}$$

$$V^{1}_{;1} = V^{1}_{,1} + \Gamma^{1}_{t1} V^{t} = V^{1}_{,1} + 0$$

$$= V^{1}_{,1} = (r)_{,1} = 1$$

$$V^{1}_{;2} = V^{1}_{,2} + \Gamma^{1}_{t2} V^{t}$$

$$V^{1}_{;2} = V^{1}_{,2} + \Gamma^{1}_{22} V^{2} = 0 + (-r)(r^2\cos\theta) = -r^3\cos\theta$$

$$V^{1}_{;3} = V^{1}_{,3} + \Gamma^{1}_{t3} V^{t} = \overset{\nearrow 0}{V^{1}_{,3}} + \Gamma^{1}_{33} V^{3}$$

$$V^{1}_{;3} = (-r\sin^2\theta)(\sin\phi) = -r\sin^2\theta\sin\phi$$

$$V^{2}_{;1} = V^{2}_{,1} + \Gamma^{2}_{t1} V^{t} = V^{2}_{,1} + \Gamma^{2}_{21} V^{2}$$

$$V^{2}_{;1} = (r^2\cos\theta)_{,1} + \left(\frac{1}{r}\right)(r^2\cos\theta)$$

$$V^{2}_{;1} = 2r\cos\theta + r\cos\theta = 3r\cos\theta$$

$$V^{2}_{;2} = V^{2}_{,2} + \Gamma^{2}_{t2} V^{t} = V^{2}_{,2} + \Gamma^{2}_{12} V^{1}$$

$$= (r^2\cos\theta)_{,2} + \left(\frac{1}{r}\right)(r)$$

$$V^{2}_{;2} = -r^2\sin\theta + 1$$

$$V^{2}_{;3} = V^{2}_{,3} + \Gamma^{2}_{3t} V^{t} = \overset{\nearrow 0}{V^{2}_{,3}} + \Gamma^{2}_{33} V^{3}$$

$$= (-\sin\theta\cos\theta)\sin\phi$$

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$$V^{3}_{;1} = V^{3}_{,1} + \Gamma^{3}_{1t} V^{t}$$

$$= \overset{\nearrow 0}{V^{3}_{,1}} + \Gamma^{3}_{13} V^{3} = \frac{1}{r}\sin\phi$$

$$V^{3}_{;2} = V^{3}_{,2} + \Gamma^{3}_{2t} V^{t}$$

$$= \overset{\nearrow 0}{V^{3}_{,2}} + \Gamma^{3}_{23} V^{3} = \cot\theta\sin\phi$$

$$V^{3}_{;3} = V^{3}_{,3} + \Gamma^{3}_{3t} V^{t} = V^{3}_{,3} + \Gamma^{3}_{31} V^{1} + \Gamma^{3}_{32} V^{2}$$

$$= (\sin\phi)_{,3} + \frac{1}{r}(r) + \cot\theta\,(r^2\cos\theta)$$

$$= \cos\phi + 1 + \frac{r^2\cos^2\theta}{\sin\theta}$$

Question: The covariant derivative of Kronecker delta is zero.

Solution: $\delta^{i}_{j;k} = ?$

$$\delta^{i}_{j;k} = \overset{\nearrow 0}{\delta^{i}_{j,k}} + \Gamma^{i}_{lk}\delta^{l}_{j} - \Gamma^{l}_{jk}\delta^{i}_{l}$$

$$= \Gamma^{i}_{jk} - \Gamma^{i}_{jk}$$

$$= 0$$

Question: Show that $g_{\alpha\beta;\gamma} = 0$

Solution:

$$g_{\alpha\beta;\gamma} = g_{\alpha\beta,\gamma} - \Gamma^{s}_{\alpha\gamma}\, g_{s\beta} - \Gamma^{s}_{\beta\gamma}\, g_{\alpha s}$$

using the formulas:

$$\Gamma^{s}_{\alpha\gamma} = \frac{1}{2} g^{st}\left(g_{t\gamma,\alpha} + g_{t\alpha,\gamma} - g_{\alpha\gamma,t}\right)$$

$$\Gamma^{s}_{\beta\gamma} = \frac{1}{2} g^{st}\left(g_{t\gamma,\beta} + g_{t\beta,\gamma} - g_{\beta\gamma,t}\right)$$

Implies:

$$g_{\alpha\beta;\gamma} = g_{\alpha\beta,\gamma} - g_{s\beta}\frac{g^{st}}{2}\left(g_{t\gamma,\alpha} + g_{t\alpha,\gamma} - g_{\alpha\gamma,t}\right)$$

$$- g_{\alpha s}\frac{g^{st}}{2}\left(g_{t\gamma,\beta} + g_{t\beta,\gamma} - g_{\beta\gamma,t}\right)$$

$$= g_{\alpha\beta,\gamma} - \frac{1}{2}\delta^{t}_{\beta}\left(g_{t\gamma,\alpha} + g_{t\alpha,\gamma} - g_{\alpha\gamma,t}\right)$$

$$- \frac{\delta^{t}_{\alpha}}{2}\left(g_{t\gamma,\beta} + g_{t\beta,\gamma} - g_{\beta\gamma,t}\right)$$

$\because g_{\alpha s} g^{st} = \delta^{t}_{\alpha}$, $g_{s\beta}\, g^{st} = \delta^{t}_{\beta}$

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$$\Rightarrow g_{\alpha\beta;\gamma} = g_{\alpha\beta,\gamma} - \frac{1}{2}g_{\beta\gamma,\alpha} - \frac{1}{2}g_{\beta\alpha,\gamma} + \frac{1}{2}g_{\alpha\gamma,\beta}$$

$$- \frac{1}{2}g_{\alpha\gamma,\beta} - \frac{1}{2}g_{\alpha\beta,\gamma} + \frac{1}{2}g_{\beta\gamma,\alpha}$$

$$= g_{\alpha\beta,\gamma} - g_{\alpha\beta,\gamma}$$

$$= 0$$

$$\Rightarrow g_{\alpha\beta;\gamma} = 0$$

Question: If A_{ij} be a Skew Symmetric tensor of rank two Then Show that

$$A_{ij;k} + A_{jk;i} + A_{ki;j} = A_{ij,k} + A_{jk,i} + A_{ki,j}$$

Solution: consider the L.H.S and expand the formulas of covariant derivatives.

$$A_{ij;k} = A_{ij,k} - \Gamma^{l}_{ik} A_{lj} - \Gamma^{l}_{jk} A_{il}$$

$$A_{jk;i} = A_{jk,i} - \Gamma^{l}_{ji} A_{lk} - \Gamma^{l}_{ki} A_{jl}$$

$$A_{ki;j} = A_{ki,j} - \Gamma^{l}_{kj} A_{li} - \Gamma^{l}_{ij} A_{kl}$$

Adding all these we get:

$$A_{ij;k} + A_{jk;i} + A_{ki;j} = A_{ij,k} + A_{jk,i} + A_{ki,j} - (A_{lj} + A_{jl})\Gamma^{l}_{ik} - (A_{il} + A_{li})\Gamma^{l}_{jk} - (A_{lk} + A_{kl})\Gamma^{l}_{ij}$$

Since A_{ij} is skew symmetric, we have

$$A_{ij;k} + A_{jk;i} + A_{ki;j} = A_{ij,k} + A_{jk,i} + A_{ki,j}$$

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Gradient, Divergence and Curl:

1) Gradient:

If ϕ be a scalar function $\phi = \phi(x^i)$, then the gradient of ϕ is denoted by

$$\nabla\phi = \text{grad}\,\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$$

$$\phi_{;i} = \phi_{,i}$$

Divergence:

The divergence of the contravariant vector A^i is defined by (in component form)

$$\text{div}(A^i) = A^i_{\ ;i} = A^i_{\ ,i} + \Gamma^{i}_{ik} A^k$$

and similarly:

$$\text{div}(A_i) = A_{i;i} = A_{i,i} - \Gamma^{k}_{ii} A_k$$

Question: Prove that

$$div\, A^i = \frac{1}{\sqrt{g_{\alpha\alpha}}} \frac{\partial(\sqrt{g_{\alpha\alpha}} A^k)}{\partial x^k}$$

Solution:

$$div\, A^i = A^i_{;i} = A^i_{,i} + \Gamma^{ii}_{ki} A^k$$

where

$$\Gamma^{ii}_{ki} = \frac{1}{2} g^{is} \left[g_{si,k} + g_{sk,i} - g_{ki,s} \right]$$

we have proved that if g_{ij} is a diagonal metric then

$$\Gamma^{ii}_{ki} = \frac{\partial}{\partial x^k}\left(\frac{1}{2}\ln|g_{ii}|\right) = \frac{\partial}{\partial x^k}\left(\ln\sqrt{g_{ii}}\right) = \frac{1}{\sqrt{g_{ii}}}\frac{\partial\sqrt{g_{ii}}}{\partial x^k}$$

so this implies

$$\Rightarrow div\, A^i = A^i_{,i} + \frac{1}{\sqrt{g_{ii}}}\frac{\partial\sqrt{g_{ii}}}{\partial x^k} A^k$$

$$= \frac{1}{\sqrt{g_{ii}}}\left(\sqrt{g_{ii}}\, A^k_{,k} + A^k \frac{\partial\sqrt{g_{ii}}}{\partial x^k}\right)$$

$$\Rightarrow div\, A^i = \frac{1}{\sqrt{g_{ii}}}\frac{\partial}{\partial x^k}\left(\sqrt{g_{ii}}\, A^k\right)$$

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Curl: The curl is defined as:

$$\left[\nabla \times \vec{A}\right]_i = \varepsilon_{ijk} A_{k,j}$$

so in terms of covariant derivative we have

$$\left[\nabla \times \vec{A}\right]_i = \varepsilon_{ijk} A_{k;j}$$

where

$$A_{k;j} = A_{k,j} - \Gamma^{s}_{kj} A_s$$

Laplacian:

The Laplacian operator ∇^2 is defined as:

$$\nabla^2\phi = \nabla.\nabla\phi \quad i.e \quad div\,(grad\,\phi)$$

Since

$$\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$$

Implies

$$\nabla.\nabla\phi = \left(\frac{\partial\phi}{\partial x^i}\right)_{;i}$$

Absolute Differentiation Along a curve:

As $T^i_{;j}$ is a tensor, the inner product of $T^i_{;j}$ with another tensor is also a tensor. Suppose that the other tensor is $\frac{dx^j}{dt}$, that is the tangent vector of the curve $C: x^i = x^i(t)$. Then the inner product $T^i_{;r}\frac{dx^r}{dt}$ is a tensor of the same type and order as the original tensor T^i.

This tensor is known as the absolute derivative of T^i along the curve C, with components written as

$$\frac{\delta T^i}{\delta t} = \frac{dT^i}{dt} + \Gamma^i_{rs}T^s\frac{dx^r}{dt} \; ; \; T^i = T^i(x^i(t))$$

It is clear that, in coordinate systems in which the g_{ij} are constant, absolute differentiation reduces to ordinary differentiation.

$$T^i_{;r} = \frac{\partial T^i}{\partial x^r} + \Gamma^i_{rs}T^s \Rightarrow T^i_{;r}\frac{dx^r}{dt} = \frac{\partial T^i}{\partial x^r}\cdot\frac{dx^r}{dt} + \Gamma^i_{rs}T^s\frac{dx^r}{dt}$$

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Theorem: (Uniqueness of the absolute derivative) The only tensor derivable from a given tensor T^i that coincides with the ordinary derivative $\frac{dT^i}{dt}$ along some curve in a rectangular coordinate system is the absolute derivative of T^i along that curve.

Rules for Tensor Differentiation:

The basic rules for differentiation from Calculus carry over to covariant and absolute differentiation of tensors.

For covariant differentiation: For tensors T and S we have:

1) Sum: $(T+S)_{;k} = T_{;k} + S_{;k}$

2) outer product: $[TS]_{;k} = [T_{;k}S] + [TS_{;k}]$

3) Inner product: $(TS)_{;k} = T_{;k}S + TS_{;k}$

For Absolute differentiation:

1) Sum: $\frac{\delta}{\delta t}(T+S) = \frac{\delta T}{\delta t} + \frac{\delta S}{\delta t}$

2) Outer product: $\frac{\delta}{\delta t}[TS] = \left[\frac{\delta T}{\delta t} S\right] + \left[T \frac{\delta S}{\delta t}\right]$

3) Inner product: $\frac{\delta}{\delta t}(TS) = \frac{\delta T}{\delta t} S + T \frac{\delta S}{\delta t}$

Question: Prove that absolute differentiation $\frac{\delta T^i}{\delta t}$ is the result of inner product of $T^i_{;j}$ with the tangent vector $\frac{dx^i}{dt}$ of the curve.

Proof:

$$T^i_{;j} = \frac{\partial T^i}{\partial x^j} + \Gamma^i_{\lambda j} T^\lambda$$

Inner product:

$$T^i_{;j} \frac{dx^j}{dt} = \left(\frac{\partial T^i}{\partial x^j} + \Gamma^i_{\lambda j} T^\lambda\right) \frac{dx^j}{dt} = \frac{\partial T^i}{\partial x^j} \frac{dx^j}{dt} + \Gamma^i_{\lambda j} T^\lambda \frac{dx^j}{dt}$$

$$= \frac{dT^i}{dt} + \Gamma^i_{\lambda j} T^\lambda \frac{dx^j}{dt} = \frac{\delta T^i}{\delta t}$$

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The Torsion Tensor:

The torsion tensor is defined as:

$$T^\alpha_{\beta\gamma} = \Gamma^\alpha_{\beta\gamma} - \Gamma^\alpha_{\gamma\beta}$$

Even though Christoffel Symbols are not tensors their difference is a tensor.

In torsion free spaces $\Gamma^\alpha_{\beta\gamma} = \Gamma^\alpha_{\gamma\beta}$

However, in some theories the torsion tensor does not vanish.

Acceleration in General Coordinates:

In rectangular coordinates, the acceleration vector of a particle is the time derivative of its velocity vector, or the second time derivative of its position function $\vec{x} = \vec{x}(x^i(t))$.

$$\vec{a} = (a^i) = \left(\frac{d}{dt}\frac{dx^i}{dt}\right) = \left(\frac{d^2x^i}{dt^2}\right)$$

The length of this vector at time t is the instantaneous acceleration of the particle.

$$a = \sqrt{\delta_{ij} a^i a^j}$$

The generalization of $\frac{d}{dt}\left(\frac{dx^i}{dt}\right)$

$$\frac{\delta}{\delta t}\left(\frac{dx^i}{dt}\right) = \frac{d}{dt}\left(\frac{dx^i}{dt}\right) + \Gamma^i_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}$$

Hence,

$$\vec{a} = (a^i) = \left(\frac{d^2x^i}{dt} + \Gamma^i_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}\right)$$

and $a = \sqrt{|g_{ij}a^i a^j|}$

Question: A particle is in motion along the circular arc given parametrically in spherical coordinates by $x^1 = b$, $x^2 = \frac{\pi}{4}$, $x^3 = \omega t$. Find its acceleration using the formula

$$a = \sqrt{|g_{ij}a^i a^j|}$$

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Solution: The Christoffel symbols for spherical coordinate system and parametric equations $x^1 = b$, $x^2 = \pi/4$, $x^3 = \omega t$ become:

$$\Gamma^1_{22} = -x^1 = -b, \quad \Gamma^1_{33} = -x^1\sin^2 x^2 = -b\sin^2\frac{\pi}{4} = -\frac{b}{2}$$

$$\Gamma^1_{12} = \Gamma^1_{21} = \frac{1}{x^1} = \frac{1}{b}, \quad \Gamma^2_{33} = -\sin x^2\cos x^2 = -\sin\frac{\pi}{4}\cos\frac{\pi}{4} = -\frac{1}{2}$$

$$\Gamma^3_{13} = \Gamma^3_{13} = \frac{1}{x^1} = \frac{1}{b}, \quad \Gamma^3_{23} = \Gamma^3_{32} = \cot x^2 = \cot\frac{\pi}{4} = 1$$

The components of acceleration are:

$$a^1 = \frac{d^2x^1}{dt^2}^{\to 0} + \Gamma^1_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}$$

$$= 0 + \Gamma^1_{22}\frac{dx^2}{dt}\frac{dx^2}{dt}^{\to 0} + \Gamma^1_{33}\frac{dx^3}{dt}\frac{dx^3}{dt}$$

$$= -\frac{b}{2}(\omega)(\omega) = -\frac{b\omega^2}{2}$$

$$a^2 = \frac{d^2x^2}{dt^2} + \Gamma^2_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}$$

$$= \Gamma^2_{33}\frac{dx^3}{dt}\frac{dx^3}{dt} = -\frac{\omega^2}{2}$$

$$a^3 = \frac{d^2x^3}{dt^2} + \Gamma^3_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}$$

$$= \Gamma^3_{13}\frac{dx^1}{dt}\frac{dx^2}{dt} + \Gamma^3_{23}\frac{dx^2}{dt}\frac{dx^3}{dt}$$

$$= 0$$

Metric components are:

$$g_{11} = 1 \;, g_{22} = (x^1)^2 = b^2$$

$$g_{33} = (x^1)^2 \sin^2(x^2) = b^2 \sin^2\frac{\pi}{4} = \frac{b^2}{2}$$

The acceleration is given by:

$$a = \sqrt{g_{ij}\, a^i a^j}$$

$$= \sqrt{g_{11}a^1a^1 + g_{22}a^2a^2 + g_{33}a^3a^3}$$

$$= \left((1)\left(-\frac{bw^2}{2}\right)^2 + b^2\left(-\frac{w^2}{2}\right)^2 + \frac{b^2}{2}(0)\right)^{1/2}$$

$$= \left(\frac{b^2w^4}{4} + \frac{b^2w^4}{4}\right)^{1/2} = \left(\frac{b^2w^4}{2}\right)^{1/2}$$

$$= \frac{bw^2}{\sqrt{2}}$$

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Curvature in General Coordinates:

In Euclidean geometry an important role is played by the curvature. The curvature of a curve $C: x^i = x^i(t)$ is defined as the norm of the second derivative of $(x^i(s))$:

$$K(s) = \sqrt{\delta_{ij}\frac{d^2x^i}{ds^2}\frac{d^2x^j}{ds^2}}$$

where

$s(t) = \int_a^t \sqrt{\delta_{ij}\left(\frac{dx^i}{dt}\right)\left(\frac{dx^j}{dt}\right)}\,dt$ gives the arc length parameter.

The generalized formula for curvature is:

$$K(s) = \sqrt{|g_{ij}\, b^i b^j|}$$

where

$$(b^i) = \left(\frac{\delta}{\delta s}\frac{dx^i}{ds}\right) = \left(\frac{d^2x^i}{ds^2} + \Gamma^i_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds}\right)$$

and

$$s(t) = \int_a^t \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

Question: Using the formula

$$k(s) = \left(g_{ij}\, b^i b^j\right)^{1/2}$$

find the curvature of the circle $x^1 = b$, $x^2 = t$ in polar coordinates.

Solution: For polar coordinates: $(g_{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & (x^1)^2 \end{bmatrix}$

and christoffel symbols are:

$\Gamma^1_{22} = -x^1$ and

$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$, remaining all are zero.

now:

$$k(s) = \left(g_{ij}\, b^i b^j\right)^{1/2} = \left(g_{11} b^1 b^1 + g_{22} b^2 b^2\right)^{1/2}$$

$$b^i = \frac{d^2x^i}{ds^2} + \Gamma^i_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds}$$

where $\frac{ds}{dt} = \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}$

$$\Rightarrow \frac{ds}{dt} = \left[g_{11} \frac{dx^1}{dt}^{\to 0} \frac{dx^1}{dt}^{\to 0} + g_{22} \frac{dx^2}{dt} \frac{dx^2}{dt}\right]^{1/2}$$

$$\frac{ds}{dt} = \left[(x^1)^2 (1)(1)\right]^{1/2} = x^1$$

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Now:

$$b^1 = \frac{d^2x^1}{ds^2} + \Gamma^1_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = \frac{d^2x^1}{ds^2} + \Gamma^1_{22} \frac{dx^2}{ds} \frac{dx^2}{ds}$$

$$\frac{dx^2}{ds} = \frac{dx^2}{dt} \cdot \frac{dt}{ds} = \frac{(1)}{x^1} = 1/x^1$$

$$\frac{dx^1}{ds} = \frac{dx^1}{dt}^{\to 0} \cdot \frac{dt}{ds} = 0 \Rightarrow \frac{d^2x^1}{ds^2} = 0$$

$$\Rightarrow b^1 = 0 + (-x^1)\left(\frac{1}{x^1}\right)^2 = -\frac{1}{x^1}$$

$$b^2 = \frac{d^2x^2}{ds^2} + \Gamma^2_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = \frac{d^2x^2}{ds^2} + 2\Gamma^2_{12} \frac{dx^1}{ds} \frac{dx^2}{ds}$$

as $\frac{dx^1}{ds} = 0$

$$\frac{dx^2}{ds} = \frac{1}{x^1} \Rightarrow \frac{d}{ds}\left(\frac{dx^2}{ds}\right) = \frac{d}{dt}\left(\frac{1}{x^1}\right) \cdot \frac{dt}{ds}$$

$$= 0$$

$$\Rightarrow b^2 = 0$$

Hence:

$$k(s) = \left(g_{11} b^1 b^1 + g_{22} b^2 b^2\right)^{1/2}$$

$$= \left((1)\left(\frac{-1}{x^1}\right)^2\right)^{1/2}$$

$$= \frac{1}{x^1}$$

where $x^1 = b$

$$\Rightarrow \quad k(s) = \frac{1}{b}$$

Geodesics:

An important application of curvature $k(s)$ in curvilinear coordinates is the following.

Suppose that we seek those curves for which $k = 0$, i.e., the straight lines or geodesics.

$$k=0 \Rightarrow b^i = 0 \Rightarrow \frac{d^2x^i}{ds^2} + \Gamma^i_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = 0$$

The solution of this system of second order differential equations will define the geodesics $x^i = x^i(s)$

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Question:

Verify that $x^1 = a \sec x^2$ is a geodesic for the Euclidean metric in polar coordinates. (In rectangular coordinates the curve is $x = a$).

Solution: Given $x^1 = a \sec x^2$, let $x^2 = t$

then the parametic equations of curve are:

$$x^1 = a \sec t$$
$$x^2 = t \quad ; \left(-\pi/2 < t < \pi/2\right)$$

we have the formula for geodesic:

$$\frac{d^2x^i}{ds^2} + \Gamma^i_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = 0$$

First we find how t is related to s:

$$\frac{ds}{dt} = \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}$$

In polar coordinates:

$$(g_{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & (x^1)^2 \end{bmatrix}$$

$$\Rightarrow \frac{ds}{dt} = \left[\left(\frac{dx^1}{dt}\right)^2 + a^2 \sec^2 t \left(\frac{dx^2}{dt}\right)^2\right]^{1/2}$$

$$\frac{dx^1}{dt} = a \sec t \tan t \quad \text{and} \quad \frac{dx^2}{dt} = 1$$

$$\frac{ds}{dt} = \left(a^2\sec^2 t\tan^2 t + a^2\sec^2 t\right)^{1/2}$$

$$= a\sec t\,(1+\tan^2 t)^{1/2}$$

$$\frac{ds}{dt} = a\sec t\,(\sec^2 t)^{1/2} = a\sec^2 t$$

and $\frac{dt}{ds} = \frac{\cos^2 t}{a}$ ✓

Now: $\frac{dx'}{ds} = \frac{dx'}{dt}\cdot\frac{dt}{ds} = a\sec t\tan t\cdot\frac{\cos^2 t}{a}$

$= \sin t$

and $\frac{d^2x'}{ds^2} = \frac{d}{ds}(\sin t) = \frac{d}{dt}(\sin t)\frac{dt}{ds}$

$\Rightarrow \frac{d^2x'}{ds^2} = \cos t\,\frac{\cos^2 t}{a} = \frac{\cos^3 t}{a}$

and $\frac{dx^2}{ds} = \frac{d}{dt}(x^2)\frac{dt}{ds} = (1)\frac{\cos^2 t}{a} = \frac{\cos^2 t}{a}$

$\frac{d^2x^2}{ds^2} = \frac{d}{ds}\left(\frac{\cos^2 t}{a}\right) = \frac{d}{dt}\left(\frac{\cos^2 t}{a}\right)\cdot\frac{dt}{ds}$

$= \frac{1}{a}(-2\cos t\sin t)\,\frac{\cos^2 t}{a}$

$= -\frac{2}{a^2}\sin t\cos^3 t$

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So, $\frac{d^2x^1}{ds^2} + \Gamma^1_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds} = 0$

The non-zero Christoffel symbols are:

$\Gamma^1_{22} = -x^1$, $\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$

$\Rightarrow \frac{\cos^2 t}{a} + \Gamma^1_{22}\frac{dx^2}{ds}\frac{dx^2}{ds} = 0$

$\frac{\cos^2 t}{a} + (-a\sec t)\left(\frac{\cos^2 t}{a}\right)^2 = 0$

$\Rightarrow \frac{\cos^2 t}{a}\left\{1 - \frac{a}{\cos t}\cdot\frac{\cos t}{a}\right\} = 0 \Rightarrow 0 = 0$ satisfied.

Now,

$\frac{d^2x^2}{ds^2} + \Gamma^2_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds} = 0$

$\Rightarrow -\frac{2}{a^2}\sin t\cos^3 t + 2\Gamma^2_{12}\frac{dx^1}{ds}\frac{dx^2}{ds} = 0$

$\Rightarrow \frac{-2}{a^2}\sin t\cos^3 t + \frac{2}{a\sec t}\left(\sin t\cdot\frac{\cos^2 t}{a}\right) = 0$

$\Rightarrow \frac{-2}{a^2}\sin t\cos^3 t + \frac{2}{a^2}\sin t\cos^3 t = 0$

$\Rightarrow$ geodesic equation is satisfied

$\Rightarrow x^1 = a\sec x^2$ is a geodesic for the Euclidean metric.

Question: Write the geodesic equations on the surface of a sphere.

Solution: Here the non-zero Christoffel symbols are:

$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$$

$$\Gamma^3_{31} = \Gamma^3_{13} = \frac{1}{x^1}, \quad \Gamma^3_{32} = \Gamma^3_{23} = \cot x^2$$

$$\Gamma^1_{22} = -x^1, \quad \Gamma^1_{33} = -x^1 \sin^2 x^2$$

$$\Gamma^2_{33} = -\sin x^2 \cos x^2$$

using the formula:

$$\frac{d^2x^i}{ds^2} + \Gamma^i_{pq} \frac{dx^p}{ds} \cdot \frac{dx^q}{ds} = 0$$

for $i=1$:

$$\frac{d^2x^1}{ds^2} + \Gamma^1_{22}\left(\frac{dx^2}{ds}\right)^2 + \Gamma^1_{33}\left(\frac{dx^3}{ds}\right)^2 = 0$$

$$\Rightarrow \frac{d^2x^1}{ds^2} - x^1\left(\frac{dx^2}{ds}\right)^2 - x^1 \sin^2 x^2 \left(\frac{dx^3}{ds}\right)^2 = 0$$

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for $i=2$:

$$\frac{d^2x^2}{ds^2} + \Gamma^2_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = 0$$

$$\Rightarrow \frac{d^2x^2}{ds^2} + 2\Gamma^2_{12} \frac{dx^1}{ds} \frac{dx^2}{ds} + \Gamma^2_{33}\left(\frac{dx^3}{ds}\right)^2 = 0$$

$$\Rightarrow \frac{d^2x^2}{ds^2} + \frac{2}{x^1} \frac{dx^1}{ds} \frac{dx^2}{ds} - \sin x^2 \cos x^2 \left(\frac{dx^3}{ds}\right)^2 = 0$$

for $i=3$:

$$\frac{d^2x^3}{ds^2} + \Gamma^3_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = 0$$

$$\frac{d^2x^3}{ds^2} + 2\Gamma^3_{13} \frac{dx^1}{ds} \frac{dx^3}{ds} + 2\Gamma^3_{23} \frac{dx^2}{ds} \frac{dx^3}{ds} = 0$$

$$\frac{d^2x^3}{ds^2} + \frac{2}{x^1} \frac{dx^1}{ds} \frac{dx^3}{ds} + 2\cot x^2 \frac{dx^2}{ds} \frac{dx^3}{ds} = 0$$

Remark: If the radius of the sphere is constant then

$(x^1, x^2, x^3) \to (r, \theta, \phi)$

$x^1 = r = a$ (some constant)

$\Rightarrow$ all derivatives of x^1 will vanish and equations are reduced to:

$$\frac{d^2x^2}{ds^2} - (\sin x^2 \cos x^2)\left(\frac{dx^3}{ds}\right)^2 = 0$$

$$\frac{d^2x^3}{ds^2} + 2\cot x^2 \frac{dx^2}{ds}\frac{dx^3}{ds} = 0$$

Question: If the metric for (x^i) system is

$$(g_{ij}) = \begin{bmatrix} (x^1)^2 & 0 \\ 0 & 1 \end{bmatrix}$$

a) Write the differential equations of the geodesics in terms of the dependent variables $u = (x^1)^2$ and $v = x^2$

b) Integrate these equations and eliminate the arc-length parameter from the solution.

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Solution: (a) Given $u = (x^1)^2$ and $v = x^2$

and $(g_{ij}) = \begin{bmatrix} (x^1)^2 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow (g^{ij}) = \begin{bmatrix} \frac{1}{(x^1)^2} & 0 \\ 0 & 1 \end{bmatrix}$

The geodesic equation is:

$$\frac{d^2x^i}{ds^2} + \Gamma^i_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds} = 0 \quad ; i = 1, 2$$

So, first we will find the Christoffel symbols using the formula:

$$\Gamma^i_{jk} = \frac{g^{ir}}{2}\left[-g_{jk,r} + g_{kr,j} + g_{rj,k}\right]$$

$$\Gamma^1_{11} = \frac{g^{11}}{2}\left[-g_{11,1} + g_{11,1} + g_{11,1}\right]$$

$$= \frac{1}{2(x^1)^2}(2x^1) = \frac{1}{x^1}$$

$$\Gamma^1_{21} = \frac{g^{11}}{2}\left[-g_{21,1}^{\to 0} + g_{11,2}^{\to 0} + g_{12,1}^{\to 0}\right] = 0 = \Gamma^1_{12}$$

$$\Gamma^1_{22} = \frac{g^{11}}{2}\left[-g_{22,1}^{\to 0} + g_{21,2}^{\to 0} + g_{12,2}^{\to 0}\right] = 0$$

$$\Gamma^2_{11} = \frac{g^{22}}{2}\left[-g_{11,2} + g_{12,2} + g_{21,1}\right] = 0$$

$$\Gamma^2_{21} = \frac{g^{22}}{2}\left[-g_{21,2} + g_{22,1} + g_{21,2}\right] = 0 = \Gamma^2_{12}$$

$$\Gamma^2_{22} = \frac{g^{22}}{2}\left[-g_{22,2} + g_{22,2} + g_{22,2}\right] = 0$$

$\Rightarrow$ The only non-zero Christoffel symbol is $\Gamma^1_{11} = \frac{1}{x^1}$.

Now for $i = 1$, geodesic equation gives:

$$\frac{d^2x^1}{ds^2} + \Gamma^1_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds} = 0$$

$$\Rightarrow \frac{d^2x^1}{ds^2} + \Gamma^1_{11}\frac{dx^1}{ds}\frac{dx^1}{ds} = 0$$

$$\Rightarrow \frac{d^2x^1}{ds^2} + \frac{1}{x^1}\left(\frac{dx^1}{ds}\right)^2 = 0$$

Given $u = (x^1)^2 \Rightarrow x^1 = \sqrt{u}$

$$\Rightarrow \frac{dx^1}{ds} = \frac{1}{2\sqrt{u}}\frac{du}{ds}$$

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$$\Rightarrow \frac{d^2x^1}{ds^2} = \frac{d}{ds}\left(\frac{1}{2\sqrt{u}}\frac{du}{ds}\right)$$

$$\Rightarrow \frac{d^2x^1}{ds^2} = \frac{1}{2\sqrt{u}}\frac{d^2u}{ds^2} - \frac{1}{4u^{3/2}}\left(\frac{du}{ds}\right)^2$$

So: $\frac{d^2x^1}{ds^2} + \frac{1}{x^1}\left(\frac{dx^1}{ds}\right)^2 = 0$

$$\Rightarrow \frac{1}{2\sqrt{u}}\frac{d^2u}{ds^2} - \frac{1}{4u\sqrt{u}}\left(\frac{du}{ds}\right)^2 + \frac{1}{\sqrt{u}}\frac{1}{4u}\left(\frac{du}{ds}\right)^2 = 0$$

$$\Rightarrow \frac{d^2u}{ds^2} = 0$$

Similarly, for $i = 2$:

$$\frac{d^2x^2}{ds^2} + \Gamma^{2\,\neq 0}_{pq}\frac{dx^p}{ds}\frac{dx^q}{ds} = 0$$

$$\Rightarrow \frac{d^2x^2}{ds^2} = 0 \quad \text{given} \quad x^2 = v$$

$$\Rightarrow \frac{d^2v}{ds^2} = 0$$

Implies Geodesic equations are:

$$\frac{d^2u}{ds^2} = 0 \quad \text{and} \quad \frac{d^2v}{ds^2} = 0$$

(b) Now we will integrate both equations and eliminat the arc length parameter s.

Consider, $\frac{d^2u}{ds^2} = 0 \Rightarrow \frac{du}{ds} = c_1 \Rightarrow u = c_1 s + c_2$

and Similarly: $\frac{d^2v}{ds^2} = 0$

$v = c_3 s + c_4$

$\Rightarrow s = \frac{1}{c_1}(u - c_2)$

and $s = \frac{1}{c_3}(v - c_4)$

comparison yields:

$\frac{1}{c_1}(u - c_2) = \frac{1}{c_3}(v - c_4)$

$\Rightarrow \frac{c_3}{c_1}u - \frac{c_2 c_3}{c_1} = v - c_4$

$\Rightarrow v = \frac{c_3}{c_1}u - \frac{c_2 c_3}{c_1} + c_4$

$\Rightarrow v = pu + q$ where $p = c_3/c_1$ and $q = c_4 - \frac{c_2 c_3}{c_1}$

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Question: Find the absolute differentiation of $T^i = (\cos\theta, \frac{\sin\theta}{r})$ along the curve C with parametric equations: $x^1 = b$, $x^2 = t$ in polar coordinates.

Solution: here $(x^1, x^2) \to (r, \theta)$

$\Rightarrow x^1 = r = b$ and $x^2 = \theta = t$

and $T^i = (\cos\theta, \frac{\sin\theta}{r}) = (\cos t, \frac{\sin t}{b})$

The non-zero Christoffel Symbols for polar coordinates are:

$\Gamma^1_{22} = -x^1$, $\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$

$\Rightarrow \Gamma^1_{22} = -b$ and $\Gamma^2_{12} = 1/b$

Now using the formula for absolute derivative:

$\frac{\delta T^i}{\delta t} = \frac{d}{dt}(T^i) + \Gamma^i_{jk} T^j \frac{dx^k}{dt}$

$$\Rightarrow \frac{\delta T^1}{\delta t} = \frac{dT^1}{dt} + \Gamma^1_{jk} T^j \frac{dx^k}{dt}$$

$$= \frac{d}{dt}(\cos t) + \Gamma^1_{22} T^2 \frac{dx^2}{dt}$$

$$= -\sin t + (-b)\left(\frac{\sin t}{b}\right)\frac{dt}{dt}$$

$$= -\sin t - \sin t \,(1)$$

$$\frac{\delta T^1}{\delta t} = -2\sin t$$

Similarly:

$$\frac{\delta T^2}{\delta t} = \frac{dT^2}{dt} + \Gamma^2_{jk} T^j \frac{dx^k}{dt}$$

$$= \frac{d}{dt}\left(\frac{\sin t}{b}\right) + \Gamma^2_{12} T^1 \frac{dx^2}{dt} + \Gamma^2_{21} T^2 \frac{dx^1}{dt}$$

$$= \frac{1}{b}\cos t + \left(\frac{1}{b}\right)\cos t \frac{dt}{dt} + \left(\frac{1}{b}\right)\frac{\sin t}{b}\frac{d}{dt}(b) \quad (b \to 0)$$

$$= \frac{\cos t}{b} + \frac{\cos t}{b}$$

$$\frac{\delta T^2}{\delta t} = \frac{2}{b}\cos t \Rightarrow \frac{\delta T^i}{\delta t} = \left(-2\sin t, \frac{2}{b}\cos t\right)$$

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Question: Find the divergence of

$$A^i = \left(\frac{1}{x^1}, x^1 \cos x^2\right)$$

in polar coordinates using Christoffel symbols

$$\Gamma^1_{22} = -x^1, \quad \Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$$

Solution: consider the formula:

$$\text{div}A^i = A^i_{;i} = A^i_{,i} + \Gamma^i_{ji} A^j$$

also $\text{div}A^i = A^i_{;i} = A^1_{;1} + A^2_{;2}$

$$\Rightarrow A^1_{;1} = A^1_{,1} + \Gamma^1_{j1} A^j$$

$$= \left(\frac{1}{r}\right)_{,r} + \Gamma^1_{j1} A^j \quad (\Gamma^1_{j1} \to 0) = -\frac{1}{r^2}$$

$$A^2_{;2} = A^2_{,2} + \Gamma^2_{j2} A^j$$

$$= (r\cos\theta)_{,2} + \Gamma^2_{12} A^1$$

$$= -r\sin\theta + \frac{1}{r}\left(\frac{1}{r}\right) = -r\sin\theta + \frac{1}{r^2}$$

$$\Rightarrow \text{div}A^i = A^i_{;i} = -\frac{1}{r^2} - r\sin\theta + \frac{1}{r^2} = -r\sin\theta$$

Question: Find the curl of $\vec{A} = (A_i) = \left(1, x^1, \frac{\cos x^2}{x^1}\right)$ in spherical polar coordinates

Solution: Here we use the following formulas:

$$(\nabla \times \vec{A})_i = \varepsilon_{ijk} A_{k;j}$$

$$(\nabla \times \vec{A})_j = \varepsilon_{jki} A_{i;k}$$

$$(\nabla \times \vec{A})_k = \varepsilon_{kij} A_{j;i}$$

$$(\nabla \times \vec{A})_i = \varepsilon_{ijk} A_{k;j}$$

$$(\nabla \times \vec{A})_1 = \varepsilon_{1jk}\left[A_{k,j} - \Gamma^{r}_{kj} A_r\right]$$

$$= \varepsilon_{123}\left[A_{3,2} - \Gamma^{r}_{23} A_r\right]$$

$$+ \varepsilon_{132}\left[A_{2,3} - \Gamma^{r}_{32} A_r\right]$$

$$= (+1)\left[x^1{}_{,3} - \Gamma^{3}_{23} A_3\right] + (-1)\left[\left(\frac{\cos x^2}{x^1}\right)_{,2} - \Gamma^{3}_{32} A_3\right]$$

$$= 0 + (\cot x^2)\frac{\cos x^2}{x^1} + \left[-\frac{\sin x^2}{x^1} - \cot x^2 \frac{\cos x^2}{x^1}\right]$$

$$= \frac{\cos^2 x^2}{x^1 \sin x^2} - \frac{\sin x^2}{x^1} - \frac{\cos^2 x^2}{x^1 \sin x^2} = -\frac{\sin x^2}{x^1}$$

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$$(\nabla \times \vec{A})_j = \varepsilon_{jki} A_{i;k}$$

$$(\nabla \times \vec{A})_2 = \varepsilon_{2ki} A_{i;k}$$

$$= \varepsilon_{231} A_{1;3} + \varepsilon_{213} A_{3;1}$$

$$= (1)\left[A_{1,3} - \Gamma^{r}_{13} A_r\right] + (-1)\left[A_{3,1} - \Gamma^{r}_{31} A_r\right]$$

$$= 0 - \Gamma^{3}_{13} A_3 - \left(\frac{\cos x^2}{x^1}\right)_{,1} + \Gamma^{3}_{31} A_3$$

$$= -\frac{1}{x^1}\left(\frac{\cos x^2}{x^1}\right) + \frac{\cos x^2}{(x^1)^2} + \frac{1}{x^1}\left(\frac{\cos x^2}{x^1}\right)$$

$$(\nabla \times \vec{A})_2 = \frac{\cos x^2}{(x^1)^2}$$

$$(\nabla \times \vec{A})_k = \varepsilon_{kij} A_{j;i}$$

$$\Rightarrow (\nabla \times \vec{A})_3 = \varepsilon_{3ij} A_{j;i}$$

$$= \varepsilon_{321} A_{1;2} + \varepsilon_{312} A_{2;1}$$

$$= (1)\left[A_{1,2} - \Gamma^{r}_{12} A_r\right] + (-1)\left[A_{2,1} - \Gamma^{r}_{12} A_r\right]$$

$$= 0 - \Gamma^{2}_{12} A_2 - (x^1)_{,1} + \Gamma^{2}_{12} A_2$$

$$= -1$$

$$\Rightarrow \nabla \times \vec{A} = \left(\frac{\sin x^2}{x^1}\right)\hat{e}_r + \frac{\cos x^2}{(x^1)^2}\hat{e}_\theta + (-1)\hat{e}_\phi$$

$$\nabla \times \vec{A} = \frac{\sin\theta}{r}\hat{e}_r + \frac{\cos\theta}{r^2}\hat{e}_\theta - \hat{e}_\phi$$

Question: Find $\nabla^2\phi$ for $\phi = \frac{x^1 x^2}{2}$ in polar coordinates.

Solution: First consider:

$$\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i = \frac{\partial\phi}{\partial x^1}\hat{e}_1 + \frac{\partial\phi}{\partial x^2}\hat{e}_2$$

$$\Rightarrow \nabla\phi = \frac{\partial}{\partial x^1}\left(\frac{x^1 x^2}{2}\right)\hat{e}_1 + \frac{\partial}{\partial x^2}\left(\frac{x^1 x^2}{2}\right)\hat{e}_2$$

$$\nabla\phi = \frac{x^2}{2}\hat{e}_1 + \frac{x^1}{2}\hat{e}_2$$

$$\Rightarrow \nabla\phi = \left(\frac{x^2}{2}, \frac{x^1}{2}\right) = \left(\frac{\theta}{2}, \frac{r}{2}\right)$$

now:

$$\nabla^2\phi = \nabla\cdot\nabla\phi = \left(\frac{\partial\phi}{\partial x^i}\right)_{;i}$$

$$= \left(\frac{\partial\phi}{\partial x^1}\right)_{;1} + \left(\frac{\partial\phi}{\partial x^2}\right)_{;2}$$

($\nabla\phi$ is a covariant vector)

$$= \nabla\phi^1{}_{;1} + \nabla\phi^2{}_{;2}$$

$$= \nabla\phi^1{}_{,1} - \Gamma^1_{j1}\nabla\phi^j + \nabla\phi^2{}_{,2} - \Gamma^2_{j2}\nabla\phi^j$$

$$= \left(\frac{\theta}{2}\right)_{,1}^{\to 0} - \Gamma^1_{j1}\nabla\phi^j + \left(\frac{r}{2}\right)_{,2}^{\to 0} - \Gamma^2_{12}\nabla\phi^1$$

$$= -\left(\frac{1}{x^1}\right)\left(\frac{x^2}{2}\right) = -\frac{1}{2}\frac{x^2}{x^1} = -\frac{\theta}{2r}$$

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Applications of Tensors in Special Relativity:

In 1905, Albert Einstein proposed the special theory of relativity based on three simple postulates:

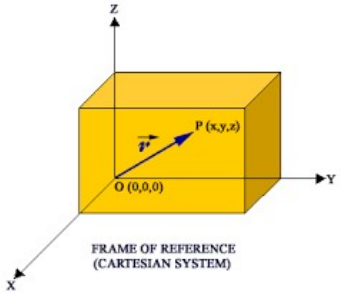
Postulate 1: The principle of relativity.
The laws of physics are the same in all inertial reference frames.

Postulate 2: The speed of light is invariant.
All observers in inertial frames will measure the same speed of light, regardless of their state of motion.

Postulate 3: Uniform motion is invariant.
A particle at rest or with constant velocity in one inertial frame will be at rest or have constant velocity in all inertial frames.

Frame of reference

The frame of reference is defined as any coordinate system with respect to which one can study the motion of another object. For example a bus, train, the Earth, the road etc.



Inertial frame of reference

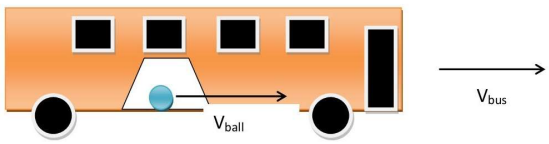
Inertia – resistance of an object to change its state of motion

Inertial Frame of Reference

- Non accelerating
- Newton's 1st law and other laws of physics are valid

For example:

Inside a bus moving at constant velocity with a ball inside



Here we will discuss some concepts of Special Relativity where tensors are being used.

Event Space:

Here the concept of space and time is combined. Thus each event is assigned four coordinates (t,x,y,z) where t is the time of the event and (x,y,z) is the location of the event in ordinary rectangular coordinates. Such coordinates

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are called spacetime coordinates.

Definition: An event space is an $\mathbb{R}^4$ space whose points are events.

Event coordinates: $(x^i) = (x^0, x^1, x^2, x^3)$

where $x^0 = ct$ is the temporal coordinate, and $(x^1, x^2, x^3) = (x, y, z)$ the rectangular positional coordinates of an event.

Let $E_1(x_1)$ and $E_2(x_2)$ are two events then:

* They are identical if $x_1^i = x_2^i$ for all i
* They are simultaneous if $x_1^0 = x_2^0$
* They are copositional if $x_1^i = x_2^i$ for $i = 1,2,3$
* The spatial distance b/w E_1 and E_2 is $S_d = \sqrt{(dx^1)^2 + (dx^2)^2 + (dx^3)^2}$ where $dx^i = x_2^i - x_1^i$ for $i = 1,2,3$.

Relativistic Length:

For an arbitrary event $E(x^i)$, the relativistic distance from origin to $E(x^i)$ is the real number $s \geq 0$ such that

$$s^2 = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$$

More generally, the length of interval or relativistic distance between $E_1(x_1^i)$ and $E_2(x_2^i)$ is:

$$(ds)^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

Interval Types:

$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$

The interval between $E_1(x_1^i)$ and $E_2(x_2^i)$ is

(1) Spacelike if $(ds)^2 < 0$

$\Rightarrow (dx^1)^2 + (dx^2)^2 + (dx^3)^2 > (dx^0)^2$

(2) lightlike if $(ds)^2 = 0$

$\Rightarrow (dx^0)^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2$

(3) timelike if $(ds)^2 > 0$

$\Rightarrow (dx^0)^2 > (dx^1)^2 + (dx^2)^2 + (dx^3)^2$

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Spacelike Interval:
If the interval between two events is spacelike then there is some frame of reference where the events occur at the same time, and there is no frame of reference where the events occur at the same place. The value of a spacelike interval is actually the proper distance between the two events—that is the distance as measured in a frame of reference where the two events occur at the same time.

Lightlike Interval:
We say that the interval is light-like or null if the interval is zero. If the interval is light-like (or null) then a ray of light can go from one event to the other. If the interval between two events is light-like then there is no frame of reference where the events occur at either the same time or at the same place.

Timelike Interval:
If the interval between two events is timelike then there is some frame of reference where the events occur at the same place, and there is no frame of reference where the events occur at the same time. And the event that occurs first in one frame of reference, occurs first in all frames of reference. The value of a timelike interval determines the proper time between the two events—that is the time between the events as measured in a frame of reference where they occur at the same position.

Exercises:

1) Calculate ds^2 for the event pairs:

a) $E_1(5,1,-2,0)$ and $E_2(0,3,1,-3)$

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$
$$= (0-5)^2 - (3-1)^2 - (1+2)^2 - (-3)^2$$
$$= 25 - 4 - 9 - 9$$
$$ds^2 = 3$$

$\Rightarrow$ Timelike interval

Also determine the interval type.

b) $E_1(7,2,4,4)$ and $E_2(4,1,2,6)$

$$ds^2 = (3)^2 - (1)^2 - (2)^2 - (2)^2$$
$$= 9 - 1 - 4 - 4 = 0$$

$\Rightarrow ds^2 = 0 \Rightarrow$ light like interval.

c) $E_1(5,1,3,3)$ and $E_2(2,-1,1,1)$

$$ds^2 = (3)^2 - (2)^2 - (2)^2 - (2)^2$$
$$= 9 - 4 - 4 - 4 = -3$$

$\Rightarrow$ Space like interval

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Question: Show that

a) Simultaneous events have a spacelike interval.

Solution: Simultaneous events means

$$x_1^0 = x_2^0 \Rightarrow dx^0 = 0$$

for $E_1(x_1^i), E_2(x_2^i)$; $i = 0,1,2,3$.

$$\Rightarrow ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)$$
$$\Rightarrow ds^2 = 0 - \left[(dx^1)^2 + (dx^2)^2 + (dx^3)^2\right] < 0$$

$\Rightarrow$ Spacelike interval.

b) Copositional events have a timelike interval.

Solution: Copositional events means

$$x_1^i = x_2^i \text{ for } i = 1,2,3$$
$$\Rightarrow dx^i = 0 \text{ for } i = 1,2,3$$

So,

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$
$$\Rightarrow ds^2 = (dx^0)^2 > 0$$

$\Rightarrow$ timelike interval.

Minkowski Spacetime:

Minkowski space or Minkowski spacetime is named after the mathematician Hermann Minkowski.

In its setting the three ordinary dimensions of space are combined with a single dimension of time to form a four-dimensional spacetime.

Minkowski space is often contrasted with Euclidean space. While a Euclidean space has only spacelike dimensions, a Minkowski space also has one timelike dimension.
Therefore the symmetry group of a Euclidean space is the Euclidean group and for a Minkowski space it is the Poincaré group.

Locally flat spacetime:

The use of the Minkowski space to describe physical systems over finite distances applies only in the Newtonian limit of systems without significant gravitation.

In the case of significant gravitation, spacetime becomes curved and one must abandon special relativity in favor of the full theory of general relativity.

In the presence of gravity, spacetime is described by a curved 4-dimensional manifold for which the tangent space to any point is a 4-dimensional Minkowski space. Thus, the structure of Minkowski space is still essential in the description of general relativity.

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Mathematical Definition

Minkowski spacetime is a 4-dimensional coordinate system in which the axes are $(ct, x, y, z) \rightarrow (x^0, x^1, x^2, x^3)$

The differential for arc length in Minkowski spacetime is given by

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

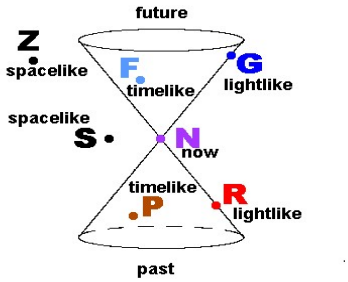
and in matrix notation the Minkowski spacetime is:

$$g_{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

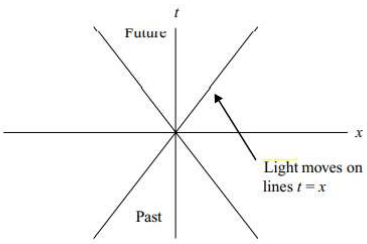
This metric is denoted by $\eta_{\mu\nu}$ and its inverse $\eta^{\mu\nu}$ is also same.

Light Cones and Spacetime Diagrams

It is often helpful to visualize spacetime by considering a flash of light emitted at the origin. Such a flash of light is described by a spherical wavefront. However, our minds cannot visualize four dimensions and it's not possible to draw it on paper. So we do the next best thing and suppress one or more of the spatial dimensions.



Let's start with the simplest of all cases, where we suppress two spatial dimensions. Doing so gives us a simple spacetime diagram. In a spacetime diagram, the vertical axis represents time while one or two horizontal axes represent space. It is convenient to work in units where $c = 1$. The upper half plane where $t > 0$ represents events to the future of the origin. Past events are found in the lower half plane where $t < 0$.



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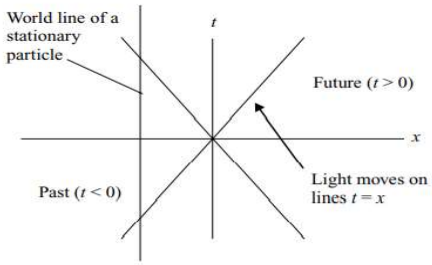
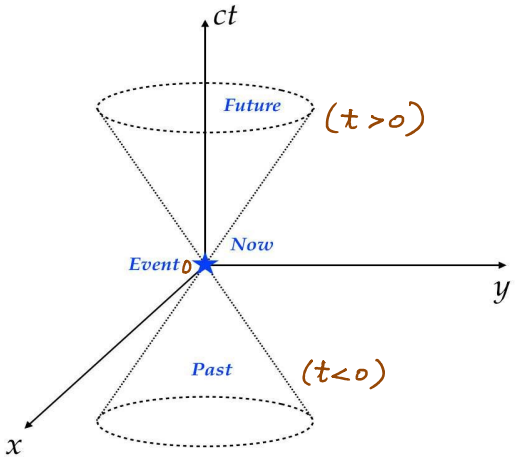
The motion of light in such a diagram is then described by lines that make a 45◦angle with the x-axis, i.e., lines that satisfy $t^2 = x^2$. The motion of a particle through spacetime as depicted in a spacetime diagram is called a worldline.



The simplest of all particle motion is a particle just sitting somewhere. To indicate the worldline of a stationary particle on a spacetime diagram, we simply draw a straight vertical line that passes through the x-axis at the spatial location of the particle. This makes sense because the particle is located at some position x that does not change, but time keeps marching forward.

Special relativity tells us that a particle cannot move faster than the speed of light. On a spacetime diagram, this is indicated by the fact that particle motion is restricted to occur only inside the light cone. The region inside the light cone is called timelike. Regions outside the light cone, which are casually unrelated to the event E, are called spacelike.

The following figure shows the projection of the light cone onto the hyperplane $x^3 = 0$.



Light Cone: A flash of light at position $(0,0,0)$ and time $t=0$ sends out an expanding spherical wave front, with equation

$$x^2 + y^2 + z^2 = c^2 t^2 \quad \text{or}$$

$$(x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 = 0$$

this is the equation of the light cone in event space, relative to the inertial frame (x^i).

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In any other inertial frame $(\tilde{x}^i)$ the equation of the light cone is exactly the same, (since all observers measure the velocity of light as c).

$$(\tilde{x}^0)^2 - (\tilde{x}^1)^2 - (\tilde{x}^2)^2 - (\tilde{x}^3)^2 = 0$$

Note: Relativistic distance is an invariant across all inertial frames. This is due to postulate of special relativity, i.e.,

$$ds^2 = d\tilde{s}^2.$$

The relativistic distance b/w two events is also called interval.

Lorentz Matrices and Transformations:

The invariance of the equation of the light cone can be expressed as:

$$ds^2 = 0 = d\tilde{s}^2$$

$$g_{ij}\, dx^i dx^j = 0 = g_{ij}\, d\tilde{x}^i d\tilde{x}^j$$

where g_{ij} is the Minkowski metric.

$\Rightarrow g_{00} = 1$, $g_{11} = g_{22} = g_{33} = -1$ and $g_{ij} = 0$ for $i \neq j$.

also $d\tilde{x}^i = a^i_j\, dx^j$

$$\Rightarrow g_{ij}\, dx^i dx^j = g_{ij}\, a^i_r a^j_s\, dx^r dx^s$$

$$\Rightarrow g_{ij}\, \frac{dx^i}{dx^r}\frac{dx^j}{dx^s} = g_{ij}\, a^i_r a^j_s$$

$$\Rightarrow g_{ij}\, \delta^i_r \delta^j_s = g_{ij}\, a^i_r a^j_s$$

$$\Rightarrow g_{rs} = g_{ij}\, a^i_r a^j_s$$

or in matrix form:

$$A^t G A = G$$

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or:

$$(a^0_0)^2 - (a^1_0)^2 - (a^2_0)^2 - (a^3_0)^2 = 1$$

$$(a^0_j)^2 - (a^1_j)^2 - (a^2_j)^2 - (a^3_j)^2 = -1 \quad (j = 1,2,3)$$

$$a^0_i\, a^0_j - a^1_i\, a^1_j - a^2_i\, a^2_j - a^3_i\, a^3_j = 0 \quad (i \neq j)$$

Definition: Any 4x4 matrix (or corresponding linear transformation) that preserves the quadratic form $A^t G A$ is called Lorentz.

★ The set of Lorentz matrices form a group under matrix multiplication called the Lorentz group.

Simple Lorentz Transformations:

The simple Lorentz transformations are

$$\tilde{x}^0 = \frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}}, \quad \tilde{x}^1 = -\frac{\beta x^0 + x^1}{\sqrt{1-\beta^2}}, \quad \tilde{x}^2 = x^2, \quad \tilde{x}^3 = x^3$$

OR in matrix form:

$$A = \begin{bmatrix} 1/\sqrt{1-\beta^2} & -\beta/\sqrt{1-\beta^2} & 0 & 0 \\ -\beta/\sqrt{1-\beta^2} & 1/\sqrt{1-\beta^2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $\beta = v/c$ is relative velocity.

Any 4x4 matrix of the form:

$$A = \begin{bmatrix} a & b & 0 & 0 \\ b & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{where} \quad a^2 - b^2 = 1$$

is called Simple Lorentz and the relative velocity is recovered as:

$$\beta = -\frac{b}{a}$$

The inverse of a simple Lorentz matrix is:

$$A^{-1} = \begin{bmatrix} a & -b & 0 & 0 \\ -b & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which is itself a simple Lorentz matrix, corresponding to $-\beta$.
i.e: If the velocity of $\tilde{O}$ (frame of reference) w.r.t O is v, then the velocity of O with respect to $\tilde{O}$ is $-v$.

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Consequences of Simple Lorentz Transformation:

Length Contraction:

For any fixed x^0, Simple Lorentz transformation gives:

$$\Delta\tilde{x}^1 = \frac{1}{\sqrt{1-\beta^2}}\Delta x^1 \quad \text{or} \quad \Delta x^1 = \sqrt{1-\beta^2}\,\Delta\tilde{x}^1 < \Delta\tilde{x}^1$$

If the frame $\tilde{O}$ is moving at a uniform velocity v relative to O, distance in $\tilde{O}$ appear to Observer O to run slow by the factor $\sqrt{1-\beta^2}$.

Two events: $(\tilde{x}^0_1, \tilde{x}^1_1, \tilde{x}^2_1, \tilde{x}^3_1) \to A$

$(\tilde{x}^0_2, \tilde{x}^1_2, \tilde{x}^2_2, \tilde{x}^3_2) \to B$

x^0 is fixed, Implies:

$$\tilde{x}^1_1 = \frac{-\beta x^0_1 + x^1_1}{\sqrt{1-\beta^2}}, \quad \tilde{x}^1_2 = \frac{-\beta x^0_2 + x^1_2}{\sqrt{1-\beta^2}}$$

$$\Rightarrow \Delta\tilde{x}^1 = \frac{1}{\sqrt{1-\beta^2}}\left(-\beta\,\Delta x^0 + \Delta x^1\right)$$

$$\Rightarrow \Delta\tilde{x}^1 = \frac{1}{\sqrt{1-\beta^2}}\,\Delta x^1$$

Time Dilation:

For any fixed $\vec{x}'$,

$$\Delta x^0 = \frac{1}{\sqrt{1-\beta^2}}\Delta\vec{x}^0 \quad \text{or} \quad \Delta t = \frac{1}{\sqrt{1-\beta^2}}\Delta\vec{t} > \Delta\vec{t}$$

If the frame $\bar{O}$ is moving at a uniform velocity v relative to O, the clock of Observer $\bar{O}$ appears to observer O to run slow by the factor $\sqrt{1-\beta^2}$.

consider inverse Lorentz transformation as:

$$x^0_1 = \frac{\vec{x}^0_1 + \beta\vec{x}'_1}{\sqrt{1-\beta^2}}, \quad x^0_2 = \frac{\vec{x}^0_2 - \beta\vec{x}'_2}{\sqrt{1-\beta^2}}$$

$$\Rightarrow \Delta x^0 = \frac{\Delta\vec{x}^0 + \beta\Delta\vec{x}'^0}{\sqrt{1-\beta^2}}$$

$$\Rightarrow \Delta x^0 = \frac{1}{\sqrt{1-\beta^2}}\Delta\vec{x}^0$$

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Question: Verify that the following matrix is Lorentz:

$$A = \begin{bmatrix} \sqrt{3} & \sqrt{2} & 0 & 0 \\ 1 & \sqrt{6}/2 & 1/2 & 1/2 \\ 1 & \sqrt{6}/2 & -1/2 & -1/2 \\ 0 & 0 & -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

$$g_{rs} = g_{ij}\, a^i_r a^j_s$$

Solution: To show that the above matrix is Lorentz, we have to show that:

$$(a^0_0)^2 - (a^1_0)^2 - (a^2_0)^2 - (a^3_0)^2 = 1 \longrightarrow (A)$$

$$(a^0_j)^2 - (a^1_j)^2 - (a^2_j)^2 - (a^3_j)^2 = -1, \; j = 1,2,3 \longrightarrow (B)$$

$$a^0_i a^0_j - a^1_i a^1_j - a^2_i a^2_j - a^3_i a^3_j = 0, \; i \neq j \longrightarrow (C)$$

For A:

$$(\sqrt{3})^2 - (1)^2 - (1)^2 - (0)^2 = 3 - 2 = 1$$

For B:

$$(\sqrt{2})^2 - \left(\frac{\sqrt{6}}{2}\right)^2 - \left(\frac{\sqrt{6}}{2}\right)^2 + (0)^2 = 2 - \frac{3}{2} - \frac{3}{2} = -1$$

$$(0)^2 - \left(\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right)^2 - \left(-\frac{\sqrt{2}}{2}\right)^2 = 0 - \frac{1}{4} - \frac{1}{4} - \frac{1}{2} = -1$$

$$(0)^2 - \left(\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right)^2 - \left(\frac{\sqrt{2}}{2}\right)^2 = 0 - \frac{1}{4} - \frac{1}{4} - \frac{1}{2} = -1$$

For c:

$$a^0_1 a^0_2 - a^1_1 a^1_2 - a^2_1 a^2_2 - a^3_1 a^3_2 = 0$$

$$(\sqrt{2})(0) - \left(\frac{\sqrt{6}}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{\sqrt{6}}{2}\right)\left(-\frac{1}{2}\right) - (0)\left(-\frac{\sqrt{2}}{2}\right) = 0$$

$$a^0_0 a^0_1 - a^1_0 a^1_1 - a^2_0 a^2_1 - a^3_0 a^3_1 = 0$$

$$(\sqrt{3})(\sqrt{2}) - (1)\left(\frac{\sqrt{6}}{2}\right) - (1)\left(\frac{\sqrt{6}}{2}\right) - 0 = 0$$

and similarly:

for $i=0, j=2$:

$$(\sqrt{3})(0) - (1)\left(\frac{1}{2}\right) - (1)\left(-\frac{1}{2}\right) - (0)\left(\frac{\sqrt{2}}{2}\right) = 0$$

for $i=0, j=3$:

$$(\sqrt{3})(0) - (1)\left(\frac{1}{2}\right) - (1)\left(-\frac{1}{2}\right) - (0)\left(\frac{\sqrt{2}}{2}\right) = 0$$

for $i=1, j=3$:

$$(\sqrt{2})(0) - \left(\frac{\sqrt{6}}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{\sqrt{6}}{2}\right)\left(-\frac{1}{2}\right) - (0)\left(\frac{\sqrt{2}}{2}\right) = 0$$

for $i=2, j=3$:

$$0 - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = -\frac{1}{4} - \frac{1}{4} + \frac{1}{2} = 0$$

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Question: Prove that for simple Lorentz transformations

$$ds^2 = d\tilde{s}^2$$

Solution: Consider:

$$\tilde{x}^0 = \frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}}, \quad \tilde{x}^1 = -\frac{\beta x^0 + x^1}{\sqrt{1-\beta^2}}, \quad \tilde{x}^2 = x^2, \quad \tilde{x}^3 = x^3$$

also, $ds^2 = g_{ij}\, dx^i dx^j$

where g_{ij} is Minkowski spacetime.

$$\Rightarrow d\tilde{s}^2 = \tilde{g}_{ij}\, d\tilde{x}^i d\tilde{x}^j$$

here $g_{ij} = \tilde{g}_{ij}$

$$\Rightarrow d\tilde{s}^2 = g_{00}(d\tilde{x}^0)^2 + g_{11}(d\tilde{x}^1)^2 + g_{22}(d\tilde{x}^2)^2 + g_{33}(d\tilde{x}^3)^2$$

$$d\tilde{s}^2 = (1)\left[d\left(\frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}}\right)\right]^2 - \left[d\left(\frac{-\beta x^0 + x^1}{\sqrt{1-\beta^2}}\right)\right]^2 - (dx^2)^2 - (dx^3)^2$$

$$= \frac{1}{1-\beta^2}\left[(dx^0)^2 + \beta^2(dx^1)^2 - 2\beta dx^0 dx^1 - \beta^2(dx^0)^2 - (dx^1)^2 + 2\beta dx^0 dx^1\right] - (dx^2)^2 - (dx^3)^2$$

$$\Rightarrow (d\tilde{s})^2 = \frac{1}{1-\beta^2}(dx^0)^2(1-\beta^2) - \frac{1}{1-\beta^2}(dx^1)^2(1-\beta^2) - (dx^2)^2 - (dx^3)^2$$

$$\Rightarrow (d\tilde{s})^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

$$= (ds)^2$$

Four vectors

In special relativity, we work with a unified entity called spacetime rather than viewing space as an arena with time flowing in the background.

As a result, a vector is going to have a time component in addition to the spatial components we are used to. This is called a four vector.

In general, (V^i) is a four vector if it transforms according to the law $\tilde{V}^i = a^i_j V^j$, where (a^i_j) is the Lorentz matrix.

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We usually use the following notation:

$(V^i) = (V^0, \vec{V})$ where

$V^0 = V_t$ is reffered to as the time component of the vector and

$\vec{V} = (V^1, V^2, V^3) = (V_x, V_y, V_z)$ are the usual space components.

Four velocity and acceleration:

In the inertial frame $(x^i) = (ct, x, y, z)$, let a particle describe the class C^2-curve:

$$(x^i) = (ct, r(t)) = (ct, x(t), y(t), z(t))$$
$$\downarrow x^0 \quad \downarrow x^1 \quad \downarrow x^2 \quad \downarrow x^3$$

Then we have the formulas:

$$v_i = \frac{dx^i}{dt} = (c, \vec{V})$$

where $\vec{v} = \frac{d\vec{r}}{dt}$ and $\|\vec{V}\| = \sqrt{v_x^2 + v_y^2 + v_z^2}$

$$(a_i) = \left(\frac{d^2 x^i}{dt^2}\right) = (0, \vec{a})$$

where $\vec{a} = \frac{d\vec{v}}{dt}$ and $\hat{a} = \|\vec{a}\| = \sqrt{a_x^2 + a_y^2 + a_z^2}$.

If $(\vec{v}_i)$ and $(\vec{a}_i)$ are the like quantities in $(\vec{x}_i)$, then using the simple Lorentz transformations we have:

$$\tilde{V}_0 = c = V_0 \quad , \quad \tilde{V}_y = \frac{V_y \sqrt{1-\beta^2}}{1 - V_x}$$

$$\tilde{V}_x = \frac{V_x - V}{1 - V_x V/c^2} \quad , \quad \tilde{V}_z = \frac{V_z \sqrt{1-\beta^2}}{1 - V_x V/c^2}$$

Derivation of Lorentz transformation of Velocity:

consider the simple Lorentz transformation

$$\tilde{x}^0 = \frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}} \quad , \quad \tilde{x}^1 = \frac{-\beta x^0 + x^1}{\sqrt{1-\beta^2}} \quad , \quad \tilde{x}^2 = x^2, \quad \tilde{x}^3 = x^3$$

let $\gamma = (1-\beta^2)^{-1/2}$ and $x^0 = ct$

$$\Rightarrow c\tilde{t} = \gamma(ct - \beta x)$$

$$\tilde{x} = \gamma(-\beta ct + x)$$

$$\tilde{y} = y$$

$$\tilde{z} = z$$

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Now differentiate the first equation w.r.t $\tilde{t}$ and use the chain rule:

$$c = \gamma(c - \beta V_x)\frac{dt}{d\tilde{t}} \quad \text{or} \quad \frac{d}{d\tilde{t}} = \frac{d}{dt}\frac{dt}{d\tilde{t}}$$

$$\frac{dt}{d\tilde{t}} = \frac{c}{\gamma(c - \beta V_x)} = \frac{c}{\gamma c\left(1 - \frac{\beta}{c}V_x\right)}$$

Also $\beta = V/c$

$$\Rightarrow \frac{dt}{d\tilde{t}} = \frac{1}{\gamma\left(1 - \frac{V V_x}{c^2}\right)}$$

$$\tilde{x} = \gamma(-\beta ct + x)$$

$$\Rightarrow \frac{d\tilde{x}}{d\tilde{t}} = \frac{d}{dt}\gamma(-\beta ct + x)\frac{dt}{d\tilde{t}}$$

$$= \gamma(-\beta c + V_x)\frac{dt}{d\tilde{t}}$$

$$\Rightarrow \tilde{V}_x = \frac{\gamma(-\beta c + V_x)}{\gamma\left(1 - \frac{V V_x}{c^2}\right)} \qquad (\because \beta c = V)$$

$$= \frac{V_x - V}{1 - \frac{V V_x}{c^2}}$$

$$\Rightarrow \tilde{V}_x = \frac{V_x - V}{1 - \frac{VV_x}{c^2}} \Rightarrow \tilde{V}_x = \frac{c^2(V_x - V)}{c^2 - VV_x}$$

$$\Rightarrow c^2\tilde{V}_x - VV_x\tilde{V}_x = c^2V_x - c^2V$$

$$\Rightarrow c^2(\tilde{V}_x + V) = V_x(c^2 + V\tilde{V}_x)$$

$$\Rightarrow V_x = \frac{\tilde{V}_x + V}{1 + \frac{V\tilde{V}_x}{c^2}}$$

Now consider

$$\tilde{y} = y$$

$$\Rightarrow \frac{d\tilde{y}}{d\tilde{t}} = \frac{dy}{dt}\cdot\frac{dt}{d\tilde{t}} \Rightarrow \tilde{V}_y = V_y\frac{dt}{d\tilde{t}}$$

$$\Rightarrow \tilde{V}_y = V_y\frac{1}{\gamma\left(1 - \frac{VV_x}{c^2}\right)} \quad ; \quad \gamma = (1-\beta^2)^{-1/2}$$

$$\Rightarrow \tilde{V}_y = \frac{V_y\sqrt{1-\beta^2}}{1 - \frac{VV_x}{c^2}}$$

and similarly: $\tilde{V}_z = \dfrac{V_z\sqrt{1-\beta^2}}{1 - \frac{VV_x}{c^2}}$

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Lorentz Transformations of Acceleration:

By differentiating the velocity components we have:

$$\tilde{a}_x = \frac{d\tilde{V}_x}{d\tilde{t}} = \frac{d\tilde{V}_x}{dt}\cdot\frac{dt}{d\tilde{t}}$$

$$= \frac{(a_x - 0)(1 - V_xV/c^2) - (V_x - V)\left(0 - \frac{a_xV}{c^2}\right)}{\left(1 - \frac{V_xV}{c^2}\right)^2}\cdot\frac{1}{\gamma\left(1 - \frac{V_xV}{c^2}\right)}$$

$$= \frac{a_x - a_xV_xV/c^2 + V_xVa_x/c^2 - a_xV^2/c^2}{\gamma\left(1 - \frac{V_xV}{c^2}\right)^3}$$

$$\Rightarrow \tilde{a}_x = \frac{a_x\left(1 - \frac{V^2}{c^2}\right)}{\gamma\left(1 - \frac{VV_x}{c^2}\right)^3}$$

also $\gamma = \dfrac{1}{\sqrt{1-\beta^2}}$ or $\gamma = \dfrac{1}{\sqrt{1 - V^2/c^2}}$

$$\Rightarrow \tilde{a}_x = \frac{a_x(1-\beta^2)^{3/2}}{\left(1 - \frac{VV_x}{c^2}\right)^3}$$

Similarly:

$$\tilde{a}_y = \frac{a_y + (a_x v_y - v_x a_y)\,v/c^2}{(1 - v_x v/c^2)^3}\,(1-\beta^2)$$

and

$$\tilde{a}_z = \frac{a_z + (a_x v_z - v_x a_z)\,v/c^2}{(1 - v_x v/c^2)^3}\,(1-\beta^2)$$

Proper Time: $\sqrt{g_{00}\frac{dx^0}{du}\frac{dx^0}{du} + g_{11}\frac{dx^1}{du}\frac{dx^1}{du} + g_{22}\frac{dx^2}{du}\frac{dx^2}{du} + g_{33}\frac{dx^3}{du}\frac{dx^3}{du}}$

Let us reparameterize the curve, choosing the quantity: $\sqrt{c^2 - v_x^2 - v_y^2 - v_z^2} = c\sqrt{1 - \hat{v}^2/c^2}$

$$\tau = \frac{s}{c} = \frac{1}{c}\int_{t_0}^{t} \sqrt{g_{ij}\frac{dx^i}{du}\frac{dx^j}{du}}\,du$$

or $\frac{d\tau}{dt} = \sqrt{1 - \hat{v}^2/c^2}$

where $\hat{v} < c$. The new parameter τ is known as the proper time for the particle.

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consider:

$$\tau = \frac{1}{c}\int_{t_0}^{t} \sqrt{g_{ij}\frac{dx^i}{du}\frac{dx^j}{du}}\,du$$

$$\Rightarrow \frac{d\tau}{dt} = \frac{1}{c}\sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}$$ where g_{ij} is the Minkowski metric.

$$\Rightarrow \frac{d\tau}{dt} = \frac{1}{c}\left[\left(\frac{dx^0}{dt}\right)^2 - \left(\frac{dx^1}{dt}\right)^2 - \left(\frac{dx^2}{dt}\right)^2 - \left(\frac{dx^3}{dt}\right)^2\right]^{1/2}$$

$$= \frac{1}{c}\left[c^2 - \hat{v}^2\right]^{1/2} = \left[1 - \hat{v}^2/c^2\right]^{1/2}$$

Proper Velocity and Acceleration:

When τ-derivatives replace t-derivatives, the corresponding velocity and acceleration become proper velocity and acceleration. That is we have:

$$u^i = \frac{dx^i}{d\tau}, \quad b^i = \frac{du^i}{d\tau} = \frac{d^2x^i}{d\tau^2}$$

Using simple Lorentz transformations we have:

$$\tilde{u}^0 = \frac{u^0 - \beta u^1}{\sqrt{1-\beta^2}}, \quad \tilde{u}^1 = \frac{-\beta u^0 + u^1}{\sqrt{1-\beta^2}}, \quad \tilde{u}^2 = u^2, \quad \tilde{u}^3 = u^3$$

$\tilde{x}^0 = \frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}}$

and

$$\tilde{b}^0 = \frac{b^0 - \beta b^1}{\sqrt{1-\beta^2}}, \quad \tilde{b}^1 = \frac{-\beta b^0 + b^1}{\sqrt{1-\beta^2}}, \quad \tilde{b}^2 = b^2, \quad \tilde{b}^3 = b^3$$

Also: $u_i u^i = c^2$

Question: Derive $u_i u^i = c^2$.

Solution:

consider: $u_i u^i = g_{ij} u^i u^j$

$$= (u^0)^2 - (u^1)^2 - (u^2)^2 - (u^3)^2$$

$$= \left[(v_t)^2 - (v_x)^2 - (v_y)^2 - (v_z)^2\right]\left(\frac{dt}{d\tau}\right)^2$$

$$= \left[c^2 - \hat{v}^2\right]\frac{1}{1 - \hat{v}^2/c^2} = c^2$$

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Question: Find the expressions

$$u^i = \frac{dx^i}{d\tau} \quad \text{and} \quad b^i = \frac{du^i}{d\tau}$$

Solution: $u^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt} \cdot \frac{dt}{d\tau}$

$$= \frac{v_i}{\sqrt{1 - \hat{v}^2/c^2}}$$

$$b^i = \frac{du^i}{d\tau} = \frac{d}{dt}\left(\frac{v_i}{\sqrt{1-\hat{v}^2/c^2}}\right) \cdot \frac{dt}{d\tau}$$

$$= \frac{a_i}{1 - \hat{v}^2/c^2} + \frac{(\vec{v}\cdot\vec{a})\, v_i}{c^2\left(1 - \hat{v}^2/c^2\right)^2}$$

Instantaneous Rest Frame:

An instantaneous rest frame for the particle is an inertial frame that translates in such a manner that its origin coincides with P_1 at $t = t_1$, where $P_1 = \vec{r}(t_1)$ is instantaneous position of the particle at $t = t_1$.

★ The particle's motion (with respect to frame O) is uniformly accelerated if its spatial acceleration relative to an instantaneous rest frame $\tilde{O}$ does not vary, i.e.,

$$\hat{a} = \sqrt{a_x^2 + a_y^2 + a_z^2} = \alpha$$

Relativistic Mass and Force:

1) Rest Mass and Relativistic Mass:

The rest mass of a particle is its mass as measured from Newtonian mechanics in any instantaneous rest frame for that particle.

The relativistic mass of a particle with spatial velocity $\vec{v}$ is:

$$\hat{m} = \frac{m}{\sqrt{1 - \hat{v}^2/c^2}}$$

where m is the rest mass of the particle and $\hat{m}$ is relativistic mass.

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Relativistic Momentum and Force:

The 4-momentum of special Relativity is defined by:

$$(P^i) = (P^0, \vec{P}) = (\hat{m} v_i)$$

and the Lorentz force $(F_0, \vec{F})$ is defined as the time derivative of the 4-momentum:

$$F_0 = \frac{dP^0}{dt} = \frac{d}{dt}(\hat{m} v_0) = \frac{d}{dt}\left(\frac{mc}{\sqrt{1 - \hat{v}^2/c^2}}\right)$$

$$\vec{F} = \frac{d}{dt}\left(\frac{m\vec{v}}{\sqrt{1 - \hat{v}^2/c^2}}\right)$$

In terms of proper time τ. The 4-force also called Minkowski force is defined as:

$$(K^i) = \left(\frac{dP^i}{d\tau}\right)$$

$$\Rightarrow K^i = \frac{dP^i}{dt} \cdot \frac{dt}{d\tau} = F_i \frac{dt}{d\tau} = \frac{F_i}{\sqrt{1 - \hat{v}^2/c^2}}$$

The Riemann Tensor:

The Riemann tensor emerges from an analysis of a simple question. Starting with a covariant vector (V_i) and taking the covariant derivative with respect to x^j and then with respect to x^k produces the third order tensor $((V_i)_{;j})_{;k} = V_{i;jk}$

The Question is:
Does the order of differentiation matters?
OR does $V_{i;jk} = V_{i;kj}$ hold in general?

By definition of the covariant derivative:

$$V_{i;jk} = (V_{i;j})_{;k} = \frac{\partial}{\partial x^k}(V_{i;j}) - \Gamma^r_{ik}(V_{r;j}) - \Gamma^r_{jk}(V_{i;r})$$

Also, $V_{i;j} = \frac{\partial V_i}{\partial x^j} - \Gamma^s_{ij} V_s$

$$\Rightarrow V_{i;jk} = \frac{\partial^2 V_i}{\partial x^k \partial x^j} - \frac{\partial \Gamma^s_{ij}}{\partial x^k} V_s - \Gamma^s_{ij}\frac{\partial V_s}{\partial x^k} - \Gamma^r_{ik}\frac{\partial V_r}{\partial x^j} + \Gamma^r_{ik}\Gamma^s_{rj} V_s - \Gamma^r_{jk}\frac{\partial V_i}{\partial x^r} + \Gamma^r_{jk}\Gamma^s_{ir} V_s \longrightarrow (A)$$

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Interchanging j and k yields:

$$V_{i;kj} = \frac{\partial^2 V_i}{\partial x^j \partial x^k} - \frac{\partial \Gamma^s_{ik}}{\partial x^j} V_s - \Gamma^s_{ik}\frac{\partial V_s}{\partial x^j} - \Gamma^r_{ij}\frac{\partial V_r}{\partial x^k} + \Gamma^r_{ij}\Gamma^s_{rk} V_s - \Gamma^r_{kj}\frac{\partial V_i}{\partial x^r} + \Gamma^r_{kj}\Gamma^s_{ir} V_s \longrightarrow (B)$$

Subtracting (B) from (A), we have:

$$V_{i;jk} - V_{i;kj} = -\frac{\partial \Gamma^s_{ij}}{\partial x^k} V_s + \Gamma^r_{ik}\Gamma^s_{rj} V_s + \frac{\partial \Gamma^s_{ik}}{\partial x^j} V_s - \Gamma^r_{ij}\Gamma^s_{rk} V_s$$

$$\Rightarrow V_{i;jk} - V_{i;kj} = \left[\frac{\partial \Gamma^s_{ik}}{\partial x^j} - \frac{\partial \Gamma^s_{ij}}{\partial x^k} + \Gamma^r_{ik}\Gamma^s_{rj} - \Gamma^r_{ij}\Gamma^s_{rk}\right] V_s$$

$$V_{i;jk} - V_{i;kj} = R^s_{ijk} V_s \longrightarrow (*)$$

where $R^s_{ijk} = \frac{\partial \Gamma^s_{ik}}{\partial x^j} - \frac{\partial \Gamma^s_{ij}}{\partial x^k} + \Gamma^r_{ik}\Gamma^s_{rj} - \Gamma^r_{ij}\Gamma^s_{rk}$

The Quotient theorem immediately implies:

"The n^4 components defined by Equation (*) are those of a forth-order tensor, contravariant of order one and covariant of order three."

R^i_{jkl} is called the Riemann or Riemann-Christoffel tensor of the second kind. Lowering the contravariant index gives

$$R_{ijkl} = g_{is} R^s_{jkl}$$

is the Riemann tensor of first kind.

Note: We may now say that covariant differentiation is order-dependent unless the metric is such as to make Riemann tensor vanish.

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Question: Show that at any point where the Christoffel symbols vanish,

$$R^i_{jkl} + R^i_{klj} + R^i_{ljk} = 0$$

Solution: Given that the christoffel symbols vanish, the expression for Riemman tensor becomes:

$$R^i_{jkl} = \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l},$$

$$R^i_{klj} = \frac{\partial \Gamma^i_{kj}}{\partial x^l} - \frac{\partial \Gamma^i_{kl}}{\partial x^j}$$

$$R^i_{ljk} = \frac{\partial \Gamma^i_{lk}}{\partial x^j} - \frac{\partial \Gamma^i_{lj}}{\partial x^k}$$

and adding all these:

$$R^i_{jkl} + R^i_{klj} + R^i_{ljk} = \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l} + \frac{\partial \Gamma^i_{kj}}{\partial x^l} - \frac{\partial \Gamma^i_{kl}}{\partial x^j} + \frac{\partial \Gamma^i_{lk}}{\partial x^j} - \frac{\partial \Gamma^i_{lj}}{\partial x^k}$$

$$= 0$$

because christoffel symbols are symmetric w.r.t lower index.

Properties of the Riemann Tensor:

Interchange of k and l in the definition of R^i_{jkl} shows that

$$R^i_{jkl} = -R^i_{jlk}$$

where:

$$R^i_{jkl} = \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l} + \Gamma^r_{jl}\Gamma^i_{rk} - \Gamma^r_{jk}\Gamma^i_{rl}$$

and

$$R^i_{jlk} = \frac{\partial \Gamma^i_{jk}}{\partial x^l} - \frac{\partial \Gamma^i_{jl}}{\partial x^k} + \Gamma^r_{jk}\Gamma^i_{rl} - \Gamma^r_{jl}\Gamma^i_{rk}$$

also $R_{ijkl} = -R_{ijlk}$

Below are the symmetry properties of Riemann tensor:

1) $R^i_{jkl} = -R^i_{jlk}$

2) $R_{ijkl} = -R_{jikl}$ (first skew symmetry)

3) $R_{ijkl} = -R_{ijlk}$ (second skew symmetry)

4) $R_{ijkl} = R_{klij}$

5) $R_{ijkl} + R_{iklj} + R_{iljk} = 0$ (Bianchi's Identity)

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Number of Independent Components:

We count the separate types of potentially nonzero components, using the above symmetry properties.

The first two properties imply that R_{aacd} and R_{abcc} (not summed on a or c) are zero. In the following list, the repeated indices are not summed over.

$R_{11cd} = -R_{11cd} \Rightarrow R_{11cd} = 0$

(A) Type R_{abab}, $a<b$: → 1212, 1313, 1414

$$n_A = {}^nC_2 = \frac{n(n-1)}{2!}$$

(B) Type R_{abac}, $b<c$: → R_{1213}, R_{1314}

$$n_B = 3\,{}^nC_3 = \frac{n(n-1)(n-2)}{2}$$

(C) Type R_{abcd} or R_{acbd}, all different indices, $a<b<c<d$

$$n_C = 2\,{}^nC_4 = \frac{n(n-1)(n-2)(n-3)}{12}$$

Theorem: There are a total of $\frac{n^2(n^2-1)}{12}$ components of the Riemann tensor R_{ijkl} that are not identically zero and that are independent.

Proof: The symmetries of the Riemann tensor imply that

$$n_A = \frac{n(n-1)}{2}, \quad n_B = \frac{n(n-1)(n-2)}{2}$$

and $$n_C = \frac{n(n-1)(n-2)(n-3)}{12}$$

now adding all these

No of non-zero independent components of the Riemann tensor $= n_A + n_B + n_C$

$$= \frac{n(n-1)}{2}\left[1 + (n-2) + \frac{(n-2)(n-3)}{6}\right]$$

$$= \frac{n(n-1)}{2}\left[\frac{6(n-1) + (n-2)(n-3)}{6}\right]$$

$$= \frac{n(n-1)}{12}\left[6n - 6 + n^2 - 5n + 6\right]$$

$$= \frac{n(n-1)(n^2+n)}{12} = \frac{n^2(n^2-1)}{12}.$$

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Corollary:

In a two dimensional space, the only components of the Riemann tensor not identically zero are

$$R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$$

Question: List the independent, potentially nonzero components of R_{ijkl} for $n=5$.

Solution:

Type A: $R_{1212}, R_{1313}, R_{1414}, R_{1515},$
$R_{2323}, R_{2424}, R_{2525}$
$R_{3434}, R_{3535},$
R_{4545}

Type B: $R_{1213}, R_{1214}, R_{1215}, R_{1314}, R_{1315}, R_{1415}$

$R_{2123}, R_{2124}, R_{2125}, R_{2324}, R_{2325}, R_{2425}$

$R_{3132}, R_{3134}, R_{3135}, R_{3234}, R_{3235}, R_{3435}$

$R_{4142}, R_{4143}, R_{4145}, R_{4243}, R_{4245}, R_{4345}$

$R_{5152}, R_{5153}, R_{5154}, R_{5253}, R_{5254}, R_{5354}$

Type C: $R_{1234}, R_{1235}, R_{1245}, R_{1345}, R_{2345}$

$R_{1324}, R_{1325}, R_{1425}, R_{1435}, R_{2435}$

Implies:

Type A: 10 components

Type B: 30 components

Type C: 10 components

$\Rightarrow$ Total 50 components

and also $\frac{n^2(n^2-1)}{12} = \frac{25(24)}{12} = 50$

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Question: Compute the components of the Riemann tensor for a unit 2-sphere, where

$$ds^2 = d\theta^2 + \sin^2\theta\, d\phi^2$$

and the non-zero christoffel symbols are:

$$\Gamma^1_{22} = -\sin\theta\cos\theta$$

$$\Gamma^2_{21} = \Gamma^2_{12} = \cot\theta$$

$x^1, x^2, x^3 \to r, \theta, \phi$ but $r = 1$ $\Rightarrow x^1, x^2 \to \theta, \phi$

Solution: Using the fact that the non-zero components of the Riemann tensor in two dimensions are given by $R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$

$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^s_{22}\Gamma^1_{s1} - \Gamma^s_{21}\Gamma^1_{s2}$$

$$= \frac{\partial}{\partial\theta}(-\sin\theta\cos\theta) - 0 + \Gamma^1_{22}\Gamma^1_{11} + \Gamma^2_{22}\Gamma^1_{21} - \Gamma^1_{21}\Gamma^1_{12} - \Gamma^2_{21}\Gamma^1_{22}$$

$$= -\cos^2\theta + \sin^2\theta - \cot\theta(-\sin\theta\cos\theta)$$

$$= \sin^2\theta - \cos^2\theta + \cos^2\theta$$

$$= \sin^2\theta$$

Question: For the metric of spherical coordinates,

$$ds^2 = (dx^1)^2 + (x^1 dx^2)^2 + (x^1 \sin x^2 dx^3)^2$$

list and calculate the non-zero components R_{ijkl}, if any.

Solution: There are $\frac{n^2(n^2-1)}{12} = \frac{9(8)}{12} = 6$

$\Rightarrow$ There are six non-zero components:

A) R_{1212}, R_{1313}, R_{2323}

B) R_{1213}, R_{1232}, R_{1323}

As $R_{ijkl} = g_{ii} R^{i}_{jkl}$, so we find the mixed components.

The non-zero Christoffel symbols are:

$$\Gamma^1_{22} = -r, \quad \Gamma^1_{33} = -r\sin^2\theta$$

$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{r}, \quad \Gamma^2_{33} = -\sin\theta\cos\theta$$

$$\Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{r}, \quad \Gamma^3_{23} = \Gamma^3_{32} = \cot\theta$$

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$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^r_{22}\Gamma^1_{r1} - \Gamma^r_{21}\Gamma^1_{r2}$$

$$= \frac{\partial}{\partial r}(-r) - \Gamma^2_{21}\Gamma^1_{22} = (-1) - \frac{1}{r}(-r)$$

$$R^1_{212} = -1 + 1 = 0$$

$$R^1_{313} = \frac{\partial \Gamma^1_{33}}{\partial x^1} - \frac{\partial \Gamma^1_{31}}{\partial x^1} + \Gamma^r_{33}\Gamma^1_{r1} - \Gamma^r_{31}\Gamma^1_{r3}$$

$$= -\sin^2\theta - \Gamma^3_{13}\Gamma^1_{33}$$

$$= -\sin^2\theta - \left(\frac{1}{r}\right)(-r\sin^2\theta) = 0$$

Similarly:

$$R^2_{323} = 0$$

$$R^1_{213} = 0$$

$$R^1_{232} = 0$$

$$R^1_{323} = 0$$

Therefore $R_{ijkl} = 0$ for all i, j, k, l.

The Ricci Tensor and Ricci Scalar:

The Riemann tensor can be used to derive two more quantities that are used to define the Einstein tensor. The first of these is the Ricci tensor, which is calculated from the Riemann tensor by contraction on the first and third indices.

$$R_{ab} = R^{c}{}_{acb}$$

The Ricci tensor is symmetric, so

$$R_{ab} = R_{ba}$$

Using contraction on the Ricci tensor, we obtain the Ricci scalar:

$$R = g^{ab} R_{ab} = R^{a}{}_{a}$$

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Question: Show that $R = 2$ for the unit 2-Sphere. Given

$$R^{1}_{212} = \sin^2\theta, \quad R_{1212} = \sin^2\theta$$

Solution:

The Symmetry conditions of the Riemann tensor implies that:

$$R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$$

The metric of 2-sphere is:

$$ds^2 = d\theta^2 + \sin^2\theta\, d\phi^2$$

So,

$$R^{2}_{121} = g^{22} R_{2121} = \frac{1}{\sin^2\theta} \sin^2\theta = 1$$

$$R^{1}_{221} = g^{11} R_{1221} = -\sin^2\theta$$

$$R^{2}_{112} = g^{22} R_{2112} = \left(\frac{1}{\sin^2\theta}\right)\left(-\sin^2\theta\right) = -1$$

The components of the Ricci tensor are given by:

$$R_{11} = R^{\alpha}_{1\alpha 1} = R^{1}_{111} + R^{2}_{121} = 0 + 1 = 1$$

$$R_{22} = R^{\alpha}_{2\alpha 2} = R^{1}_{212} + R^{2}_{222} = \sin^2\theta$$

$$R_{12} = R^{\alpha}_{1\alpha 2} = R^{1}_{112} + R^{2}_{122} = 0$$

$$\Rightarrow R_{12} = R_{21} = 0$$

So, $R = g^{\alpha\beta} R_{\alpha\beta} = g^{11} R_{11} + g^{22} R_{22}$

$$R = 1 + \frac{1}{\sin^2\theta} \sin^2\theta = 1 + 1 = 2$$

Question: For the metric

$$ds^2 = (dx^1)^2 + 2x^1 (dx^2)^2 + 2x^2 (dx^3)^2$$

calculate R_{ij}, R, Given:

$$R^{1}_{212} = \frac{1}{2x^1}, \quad R^{2}_{323} = \frac{1}{4x^1x^2}, \quad R^{3}_{132} = \frac{1}{4x^1x^2}$$

$$R_{1212} = \frac{1}{2x^1}, \quad R_{2323} = \frac{1}{2x^2}, \quad R_{3132} = \frac{1}{2x^1}$$

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Solution: Using the formula:

$$R_{ij} = R^{k}_{ikj} = R^{1}_{i1j} + R^{2}_{i2j} + R^{3}_{i3j}$$

$$= g^{11} R_{1i1j} + g^{22} R_{2i2j} + g^{33} R_{3i3j}$$

Implies:

$$R_{11} = g^{11} \overbrace{R_{1111}}^{0} + g^{22} R_{2121} + g^{33} \overbrace{R_{3131}}^{0}$$

$$= g^{22} R_{2121}$$

where $g^{22} = \frac{1}{2x^1}$ and $R_{2121} = \frac{1}{2x^1}$

$$\Rightarrow R_{11} = \frac{1}{2x^1}\left(\frac{1}{2x^1}\right) = \frac{1}{4(x^1)^2}$$

$$R_{22} = g^{11} R_{1212} + g^{33} R_{3232}$$

$$= (1)\left(\frac{1}{2x^1}\right) + \frac{1}{2x^2}\left(\frac{1}{2x^2}\right)$$

$$= \frac{1}{2x^1} + \frac{1}{4(x^2)^2}$$

$$R_{33} = g^{22} R_{2323} = \frac{1}{2x^1}\left(\frac{1}{2x^2}\right)$$

$$= \frac{1}{4x^1x^2}$$

$$R_{12} = g^{33} R_{3132} = \frac{1}{2x^2}\left(\frac{1}{2x^1}\right)$$

$$R_{12} = \frac{1}{4x^1x^2} = R_{21}$$

Hence

$$R = g^{11}R_{11} + g^{22}R_{22} + g^{33}R_{33}$$

$$= (1)\left(\frac{1}{4(x^1)^2}\right) + \frac{1}{2x^1}\left(\frac{1}{2x^1} + \frac{1}{4(x^2)^2}\right)$$

$$+ \frac{1}{2x^2}\left(\frac{1}{4x^1x^2}\right)$$

$$\Rightarrow R = \frac{x^1 + 2(x^2)^2}{(2x^1x^2)^2}$$

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Physical Interpretation:

Curvature describes how a geometric object such as a curve deviates from a straight line or a surface from a flat plane.

Curvature can be expressed simply as a scalar that represents the magnitude of this deviation. Curvature can also be described as a vector that takes into account the direction of the curve along with the magnitude.

For more complex objects such as surfaces or n-dimensional spaces, a more complex object is needed to describe the curvature. **One such object, is the Riemann curvature tensor.**

Intrinsic objects are those which can be obtained by measuring distances and computing the derivatives of those distances in some space.

The **metric tensor** is an example of an intrinsic object. The metric tensor describes how to compute distances and lengths of curves in a given space.

Riemann generalized this idea and extended it to spaces of n-dimensions. Because the metric tensor is an intrinsic object, subsequent objects that can be described in terms of the metric tensor and its derivatives are also intrinsic.

One object that can be derived from the metric tensor is the Christoffel symbol. The Christoffel symbol describes the variation in basis vectors from one point to another in curvilinear coordinate systems. The Christoffel symbols measure the rate of change of the covariant basis with respect to the coordinate variables.

The Riemann tensor, Ricci tensor, and Ricci scalar are all derived from the metric tensor and are therefore intrinsic measures of curvature.

A **manifold** is a space that can be curved but is locally flat. That is, near each point the space resembles Euclidean space. One example of a curved manifold is the surface of a sphere.

The **Riemann te**nsor is the most common tool used to describe the curvature of a Riemannian manifold.

The Riemann curvature tensor is a tool used to describe the curvature of n-dimensional spaces such as Riemannian manifolds in the field of differential geometry. The Riemann tensor plays an important role in the theories of general relativity and gravity as well as the curvature of spacetime.

Contraction, of the Riemann tensor produces the **Ricci tensor**. The Ricci tensor provides a way measure the degree to which a space differs from Euclidean space.

Contraction of the Ricci tensor produces the scalar curvature or **Ricci scalar**. The Ricci scalar is the simplest curvature invariant of a manifold.

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Riemnnian Curvature:

The Riemannian or sectional curvature relative to a given metric (g_{ij}), a Riemannian manifold and two linearly independent tangent vectors at the same point, (U^i) and (V^i) is defined as:

$$K = K(x^i; U^i, V^i) = \frac{R_{ijkl}\, U^i V^j U^k V^l}{G_{pqrs}\, U^p V^q U^r V^s}$$

where, $G_{pqrs} = g_{pr} g_{qs} - g_{ps} g_{qr}$

This curvature can be associated to Gaussian curvature of the surface.

This sort of curvature depends not only on position, but also on a pair of directions selected at each point.

By contrast, the curvature k of a curve depends only on the points along the curve.

In the calculations of K it is helpful that G_{ijkl} possess exactly the same symmetries as the R_{ijkl}.

$R_{ijkl}\,U^i V^j U^k V^l \longrightarrow$ is an invariant

$G_{pqrs}\,U^p V^q U^r V^s \longrightarrow$ is also an invariant

It follows that K is an invariant and it serves to generalize the Gaussian curvature of a surface to higher dimensions.

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Question: Using the fact that in a two-dimensional space the only non-zero components of Riemann tensor are

$R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$

simplify the expression of Riemannian Curvature.

Solution:

$$K = \frac{R_{ijkl}U^iV^jU^kV^l}{G_{pqrs}U^pV^qU^rV^s} = \frac{R_{1212}U^1V^2U^1V^2 + R_{2121}U^2V^1U^2V^1 - R_{1221}U^1V^2U^2V^1 - R_{2112}U^2V^1U^1V^2}{}$$

$$= \frac{R_{1212}\left[(U^1)^2(V^2)^2 - 2U^1V^2U^2V^1 + (U^2)^2(V^1)^2\right]}{G_{1212}\left[(U^1)^2(V^2)^2 - 2U^1V^2U^2V^1 + (U^2)^2(V^1)^2\right]}$$

$$= \frac{R_{1212}}{G_{1212}} = \frac{R_{1212}}{g_{11}g_{22} - g_{12}^2}$$

Example: Calculate K for the metric

$$ds^2 = (x^1)^{-2}(dx^1)^2 - (x^1)^{-2}(dx^2)^2,$$ using the result

$$K = \frac{R_{1212}}{g_{11}g_{22} - g_{12}^2}$$

and $\Gamma^1_{11} = \frac{-1}{x^1}$, $\Gamma^1_{22} = -\frac{1}{x^1}$

$$\Gamma^2_{12} = \Gamma^2_{21} = -\frac{1}{x^1}$$

Solution:

$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^k_{22}\Gamma^1_{k1} - \Gamma^k_{21}\Gamma^1_{k2}$$

$$= \frac{\partial}{\partial x^1}\left(-\frac{1}{x^1}\right) - \Gamma^2_{21}\Gamma^1_{22} + \Gamma^1_{22}\Gamma^1_{11}$$

$$= \frac{1}{(x^1)^2} - \left(\frac{-1}{x^1}\right)\left(\frac{-1}{x^1}\right) - \frac{1}{x^1}\left(\frac{-1}{x^1}\right)$$

$$= \frac{1}{(x^1)^2}$$

$$\Rightarrow R_{1212} = -g_{11}R^1_{212} = (x^1)^{-2}(x^1)^{-2} = -(x^1)^{-4}$$

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Now,

$$K = \frac{R_{1212}}{g_{11}g_{22} - g_{12}^2}$$

$$= \frac{-\frac{1}{(x^1)^2}\left(\frac{1}{x^1}\right)^2}{-\left(\frac{1}{x^1}\right)^2\left(\frac{1}{x^1}\right)^2 - 0}$$

$$\boxed{K = -1}$$

Riemannian Space: A Riemannian space is the space R^n coordinatized by (x^i), together with a fundamental form or Riemannian metric, $g_{ij}\,dx^i dx^j$, where $g = (g_{ij})$ obeys conditions for a metric.

Zero Curvature and the Euclidean metric.

$ds^2 = dr^2 + r^2 d\theta^2$, $ds^2 = dx^2 + dy^2 \mid \begin{matrix} x^1, x^2 \to \tilde{x}^1, \tilde{x}^2 \\ r, \theta \to x, y \end{matrix}$

A Riemannian metric $g = (g_{ij})$, specified in a coordinate system (x^i), is the Euclidean metric if, under some permissible coordinate transformation, $(\tilde{g}_{ij}) = \delta_{ij}$.

Now a coordinate system $(\tilde{x}^i)$ in which $(\tilde{g}_{ij}) = \delta_{ij}$ is a rectangular coordinate system. Then $\tilde{K} = 0$. Since all christoffel symbols vanish in $(\tilde{x}^i)$.

But Riemannian curvature is an invariant, so that $K = 0$ in the original coordinates (x^i) as well. Hence

Theorem:

"A Riemannian metric (g_{ij}) is the Euclidean metric if and only if the Riemannian Curvature K is zero at all points."

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Proof: Here the necessity part of the statement is: If the metric is Euclidean then this implies $K = 0$, and the Sufficiency portion is if $K = 0$ then the metric is Euclidean metric.

To prove the first part consider that the metric is Euclidean then there exist a transformation b/w (x^i) system and the rectangular coordinate system $(\tilde{x}^i)$. Also, all christoffel symbols vanish in rectangular coordinates implying $\tilde{K} = 0$, but $\tilde{K}$ is an invariant $\Rightarrow K = 0$.

To prove the Sufficiency part consider $K = 0$. Also to prove the corresponding metric is Euclidean we establish a system of first order PDEqs b/w $(x^j$ and $(\tilde{x}^i)$ system.

Here we set up a system of first order partial differential equations for $(\tilde{x}^i)$ as a function of (x^j).

So consider, $G = J^t J$

or $\dfrac{\partial \tilde{x}^k}{\partial x^i}\dfrac{\partial x^k}{\partial \tilde{x}^j} = \tilde{g}_{ij}(x^i)$

also the transformation law of Γ^i_{jk}

$$\tilde{\Gamma}^i_{jk} = \Gamma^h_{st}\frac{\partial \tilde{x}^i}{\partial x^h}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k} + \frac{\partial^2 x^h}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial \tilde{x}^i}{\partial x^h}$$

Interchanging tilde and non-tilde coordinates and replacing $\tilde{\Gamma}^i_{jk} = 0$

$$\Rightarrow \Gamma^i_{jk} = \frac{\partial^2 x^h}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial \tilde{x}^i}{\partial x^h}$$

$$\Rightarrow \frac{\partial^2 x^h}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial \tilde{x}^i}{\partial x^h} = \Gamma^i_{jk}$$

$$\Rightarrow \frac{\partial^2 x^h}{\partial \tilde{x}^j \partial \tilde{x}^k} = \Gamma^i_{jk}\frac{\partial x^i}{\partial \tilde{x}^h}$$

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let $w = \tilde{x}^i$ and $u_i = \dfrac{\partial \tilde{x}^i}{\partial x^h}$

$$\Rightarrow \frac{\partial w}{\partial x^i} = u_i$$

$$\frac{\partial u_i}{\partial x^j} = \Gamma^h_{ij} u_h$$

Flat Riemannian Spaces:

A Riemannian Space is flat if and only if $K = 0$ at all points.

Theorem: If $K = 0$, then $R = 0$.

Proof: If $K = 0$, then the space is flat and hence the $\tilde{g}_{ij}$ are constant for some coordinate system $(\tilde{x}^i)$. It follows that all $\tilde{\Gamma}^i_{jk}$, $\tilde{R}^i_{jkl}$, $\tilde{R}_{ij}$ vanish and hence $\tilde{R} = \tilde{R}^i_i = 0$.

and since the Ricci curvature is invariant,

$R = 0$.

* The converse of this theorem does not hold i.e., if $R=0$ then it does not implies $K=0$.

Question: Show that R^3 under the metric

$$ds^2 = [(x^1)^2 + (x^2)^2](dx^1)^2 + [(x^1)^2 + (x^2)^2](dx^2)^2 + (dx^3)^2$$

is Euclidean. Given that $\Gamma^i_{jk} = 0$ whenever i, j or k equals to 3, and

$$\Gamma^1_{11} = \frac{x^1}{w}, \quad \Gamma^1_{12} = \Gamma^1_{21} = \frac{x^2}{w}, \quad \Gamma^1_{22} = -\frac{x^1}{w}$$

$$\Gamma^2_{11} = -\frac{x^2}{w}, \quad \Gamma^2_{21} = \Gamma^2_{12} = \frac{x^1}{w}, \quad \Gamma^2_{22} = \frac{x^2}{w}$$

where $w = (x^1)^2 + (x^2)^2$

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Solution: The Six independent components of the Riemann tensor are:

R_{1212}, R_{2323} (→ 0), R_{1313} (→ 0), R_{1213} (→ 0), R_{2123} (→ 0), R_{3132} (→ 0)

All components of the Riemann tensor involving index 3 goes to zero. So the only possibly non zero component is R_{1212}.

So,

$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^1_{22}\Gamma^1_{11} + \Gamma^2_{22}\Gamma^1_{21} - \Gamma^1_{21}\Gamma^1_{12} - \Gamma^2_{21}\Gamma^1_{22}$$

$$\Rightarrow R^1_{212} = -\frac{w + x^1(2x^1)}{w^2} - \frac{w - x^2(2x^2)}{w^2} + \left(-\frac{x^1}{w}\right)\left(\frac{x^1}{w}\right) + \frac{x^2}{w}\left(\frac{x^2}{w}\right) - \frac{x^2}{w}\left(\frac{x^2}{w}\right) - \frac{x^1}{w}\left(-\frac{x^1}{w}\right)$$

$$= 0$$

As all the components of Riemann tensor are zero, we have

$K = 0$

$\Rightarrow$ The metric is Euclidean.

Question: Verify that $ds^2 = (dx^1)^2 + (x^1)^2 (dx^2)^2$ represent the Euclidean metric.

Solution: This is actually polar metric:

$$ds^2 = dr^2 + r^2 d\theta^2 \quad (x^1, x^2 \to r, \theta)$$

with christoffel symbols:

$$\Gamma^1_{22} = -x^1 \,, \quad \Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1} \quad \text{(remaining all zero)}$$

The independent components of Riemann tensor are: $\frac{n^2(n^2-1)}{12} = \frac{4(3)}{12} = 1$

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i.e., R_{1212}

$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^r_{22}\Gamma^1_{r1} - \Gamma^r_{21}\Gamma^1_{r2}$$

$$= \frac{\partial}{\partial x^1}(-x^1) - \frac{\partial}{\partial x^2}(0) + 0 - \Gamma^2_{21}\Gamma^1_{22}$$

$$= -1 - (-x^1)\left(\tfrac{1}{x^1}\right) = -1 + 1$$

$$R^1_{212} = 0$$

$\Rightarrow K = 0$

$\Rightarrow$ metric is Euclidean.

Question: Determine whether the following metric is Euclidean or not.

$$ds^2 = (dx^1)^2 + (x^3)^2 (dx^2)^2 + (dx^3)^2$$

Solution: First we will find the nonzero christoffel Symbols which are:

$$\Gamma^2_{23} = \Gamma^2_{32} = \frac{1}{x^3} \,, \quad \Gamma^3_{22} = -x^3$$

The independent components of the Riemann tensor are:

R_{1212}, R_{1313}, R_{2323}, R_{1312}, R_{1323}, R_{2321}

the only possibly non zero component is R_{2323}.

$$R^2_{323} = \frac{\partial \Gamma^{12}_{33}}{\partial x^2} - \frac{\partial \Gamma^{12}_{32}}{\partial x^3} + \Gamma^{\alpha}_{33}\Gamma^{2}_{\alpha 2} - \Gamma^{\alpha}_{32}\Gamma^{2}_{\alpha 3}$$

$$= -\left(-\frac{1}{(x^3)^2}\right) - \left(\frac{1}{x^3}\right)\left(\frac{1}{x^3}\right) = 0$$

$\Rightarrow K = 0 \Rightarrow$ metric is Euclidean.

Question: Consider the Riemannian metric

$$ds^2 = (dx^1)^2 + 4(x^2)^2(dx^2)^2 + 4(x^3)^2(dx^3)^2 - 4(x^4)^2(dx^4)^2$$

a) Calculate the Riemanian curvature

b) Find a solution of $\frac{\partial u_i}{\partial x^j} = \Gamma^{\alpha}_{ij} u_\alpha$, $\frac{\partial w}{\partial x^i} = u_i$

where,

$$\Gamma^2_{22} = \frac{1}{x^2}, \quad \Gamma^3_{33} = \frac{1}{x^3}, \quad \Gamma^4_{44} = \frac{1}{x^4}$$

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Solution: $K = \dfrac{R_{ijkl}\, U^i V^j U^k V^l}{G_{pqrs}\, U^p V^q U^r V^s}$

$$R^i_{jkl} = \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l} + \Gamma^{\alpha}_{jl}\Gamma^{i}_{\alpha k} - \Gamma^{\alpha}_{jk}\Gamma^{i}_{\alpha l}$$

here $\Gamma^i_{jk} = 0$ unless $i = j = k$, Implies

$R^i_{jkl} = 0 \Rightarrow R_{ijkl} = 0$

and hence $K = 0$

b) Consider, $\frac{\partial u_i}{\partial x^j} = \Gamma^{\alpha}_{ij} u_\alpha$

$$\Rightarrow \frac{\partial u_1}{\partial x^1} = 0, \quad \frac{\partial u_2}{\partial x^2} = \frac{u_2}{x^2}$$

$$\frac{\partial u_3}{\partial x^3} = \frac{u_3}{x^3}, \quad \frac{\partial u_4}{\partial x^4} = \frac{u_4}{x^4}$$

and $\frac{\partial u_i}{\partial x^j} = 0$ for $i \neq j$

$$\frac{\partial u_2}{u_2} = \frac{\partial x^2}{x^2}$$

$$\Rightarrow \ln u_2 = \ln x^2 + \ln f_2$$

$$\Rightarrow e^{\ln u_2} = e^{\ln x^2 + \ln f_2}$$

$$\Rightarrow u_2 = x^2 f_2(x^1, x^3, x^4)$$

Integration yields:

$$u_1 = f_1(x^2, x^3, x^4)$$

$$u_2 = x^2 f_2(x^1, x^3, x^4)$$

$$u_3 = x^3 f_3(x^1, x^2, x^4)$$

and $u_4 = x^4 f_4(x^1, x^2, x^3)$

where f_i, $i = 1, 2, 3, 4$ are arbitrary functions.

and $\frac{\partial w}{\partial x^i} = u_i$

$$\Rightarrow \int \partial w = \int u_i \, \partial x^i$$

$$\Rightarrow w = \int u_1 \partial x^1 + u_2 \partial x^2 + u_3 \partial x^3 + u_4 \partial x^4$$

consider $f_i = c_i$ = constant, we have

$$w = \int c_1 \partial x^1 + c_2 x^2 \partial x^2 + c_3 x^3 \partial x^3 + c_4 x^4 \partial x^4$$

$$w = c_1 x^1 + c_2 \frac{(x^2)^2}{2} + c_3 \frac{(x^3)^2}{2} + c_4 \frac{(x^4)^2}{2} + c_5$$

$$\Rightarrow w = a_1 x^1 + a_2 (x^2)^2 + a_3 (x^3)^2 + a_4 (x^4)^2 + a_5$$

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and as $w = \tilde{x}^k$

$$\Rightarrow \tilde{x}^k = a^k_1 x^1 + a^k_2 (x^2)^2 + a^k_3 (x^3)^2 + a^k_4 (x^4)^2 + a^k_5$$

where a^k_i are constants.

Normal Coordinates:

It is possible to introduce local, quasi rectangular coordinates in Riemannian space the use of which greatly simply the proofs of certain complicated tenson identities.

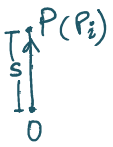
Let '0' denote an arbitrary point of R^n, and (P^i) an arbitrary unit vector at 0.

Consider the differential Equations for geodesics,

$$\frac{d^2x^i}{ds^2} + \Gamma^i_{jk} \frac{dx^j}{ds} \frac{dx^k}{ds} = 0$$

along with the initial conditions:

$$\left.\frac{dx^i}{ds}\right|_{s=0} = p^i$$



For a given (p^i) the previous system has a unique solution. Moreover, for each point P in some neighbourhood $\mathcal{N}$ of 0, there is a unique choice of direction (p^i) at 0 such that the solution curve $x^i = x^i(s)$ (i.e., a geodesic) passes through P.

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Accordingly, for each P in $\mathcal{N}$, define the coordinates of P as:

$$y^i = s p^i$$

where s is the distance along the geodesic from 0 to P. The numbers (y^i) are called the normal coordinates or the geodesic coordinates or the Riemannian coordinates of point P.

Theorem: At the origin of a Riemannian coordinate system (y^i), all $\frac{\partial g_{ij}}{\partial y^k}$, $\frac{\partial g^{ij}}{\partial y^k}$, Γ_{ijk} and Γ^i_{jk} are zero.

Proof: If (y^i) are normal coordinates, then the geodesic through 0 and any point P in some neighbourhood $\mathcal{N}$ of 0 has the parametric form: $p^i = (p^1, p^2, \ldots p^n)$, $y^i = (y^1, y^2, \ldots y^n)$

$y^1 = sp^1$, $y^2 = sp^2 \ldots y^n = sp^n$ $\quad y^i = sp^i$ ($p^i \to$ constant)

$$\Rightarrow \frac{dy^i}{ds} = p^i \quad \text{and} \quad \frac{d^2y^i}{ds^2} = 0$$

Also (y^i) coordinates must satisfy the geodesic equation:

$$\frac{d^2y^i}{ds^2} + \Gamma^i_{jk}\frac{dy^j}{ds}\frac{dy^k}{ds} = 0$$

$\frac{d^2y^i}{ds^2} \to 0$

$$\Rightarrow \Gamma^i_{jk} p^j p^k = 0 \quad (\text{for all } (p^i))$$

now p^i is constant.

$$\Rightarrow \Gamma^i_{jk} = 0$$

also $\Gamma_{ijk} = g_{kr}\Gamma^r_{ij}$

$$\Rightarrow \Gamma_{ijk} = 0$$

now consider:

$$\frac{\partial g_{ik}}{\partial y^j} = \Gamma_{ijk} + \Gamma_{jki}$$

$$\Rightarrow \frac{\partial g_{ik}}{\partial y^i} = 0$$

and

$$g^{ij} g_{jr} = \delta^i_r$$

$$\Rightarrow g^{ij}\frac{\partial g_{jr}}{\partial y^k} + \frac{\partial g^{ij}}{\partial y^k} g_{jr} = 0$$

$$\Rightarrow 0 + \frac{\partial g^{ij}}{\partial y^k} g_{jr} = 0$$

$$g_{jr} \neq 0$$

$$\Rightarrow \frac{\partial g^{ij}}{\partial y^k} = 0$$

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Question: Prove Bianchi's first identity.

$$R_{ijkl} + R_{iklj} + R_{iljk} = 0 \quad (\text{using normal coordinates})$$

Solution: consider the Riemannian coordinates.

Then

$$R_{ijkl} = \frac{\partial \Gamma_{jli}}{\partial x^k} - \frac{\partial \Gamma_{jki}}{\partial x^l} + \Gamma_{ilr}\Gamma^r_{jk} - \Gamma_{ikr}\Gamma^r_{jl}$$

becomes:

$$R_{ijkl} = \frac{\partial \Gamma_{jli}}{\partial x^k} - \frac{\partial \Gamma_{jki}}{\partial x^l}$$

at origin of Riemannian coordinates.

$\Rightarrow R_{ijk\ell} = \Gamma_{j\ell i,k} - \Gamma_{jki,\ell}$

and Similarly:

$R_{ik\ell j} = \Gamma_{kji,\ell} - \Gamma_{k\ell i,j}$

$R_{i\ell jk} = \Gamma_{\ell ki,j} - \Gamma_{\ell ji,k}$

$\because \Gamma_{ijk}$ is Symmetric in first two indices

$\Rightarrow R_{ijk\ell} + R_{ik\ell j} + R_{i\ell jk} = 0$

Question: Prove Bianchi's second identity.

$R_{ijk\ell,u} + R_{ij\ell u,k} + R_{iju k,\ell} = 0$

Solution: Consider,

$$R^i_{jk\ell} = \frac{\partial \Gamma^i_{j\ell}}{\partial y^k} - \frac{\partial \Gamma^i_{jk}}{\partial y^\ell} + \Gamma^r_{j\ell}\Gamma^i_{rk} - \Gamma^r_{jk}\Gamma^i_{r\ell}$$

$$R^i_{jk\ell,u} = \frac{\partial}{\partial y^u}\left[\frac{\partial \Gamma^i_{j\ell}}{\partial y^k} - \frac{\partial \Gamma^i_{jk}}{\partial y^\ell} + \Gamma^r_{j\ell}\Gamma^i_{rk} - \Gamma^r_{jk}\Gamma^i_{r\ell}\right]$$

$R^i_{jk\ell,u} = \Gamma^i_{j\ell,ku} - \Gamma^i_{jk,\ell u}$ at '0' of Riemannian coordinates.

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$\Rightarrow R^i_{j\ell u,k} = \Gamma^i_{ju,k\ell} - \Gamma^i_{j\ell,\ell u}$

and

$R^i_{juk,\ell} = \Gamma^i_{jk,\ell u} - \Gamma^i_{ju,\ell k}$

Implies:

$R^i_{jk\ell,u} + R^i_{j\ell u,k} + R^i_{juk,\ell} = 0$

$\Rightarrow g^{hi}\left[R_{ijk\ell,u} + R_{ij\ell u,k} + R_{iju k,\ell}\right] = 0$

$\Rightarrow R_{ijk\ell,u} + R_{ij\ell u,k} + R_{iju k,\ell} = 0$

The Einstein Tensor:

The Einstein tensor is defined in terms of the Ricci tensor R_{ij} and the curvature invariant R as:

$$G^i_j = R^i_j - \frac{1}{2}\delta^i_j R$$

This is a mixed tensor of order 2.

Theorem: For any Riemannian metric, the divergence of the Einstein tensor is zero at all points.

Proof: We have to show that

$$G^{r}_{i;r} = 0 \qquad A^{i}_{;i} = 0$$

consider,

$$R_{ijkl;u} + R_{ijlu;k} + R_{iluk;l} = 0$$

Multiply both sides by $g^{il}\, g^{jk}$

$$\Rightarrow g^{il} g^{jk} R_{ijkl;u} + g^{il} g^{jk} R_{ijlu;k} + g^{il} g^{jk} R_{iluk;l} = 0$$

$$\Rightarrow g^{il} g^{jk} R_{ijkl;u} - g^{il} g^{jk} R_{ijul;k} - g^{il} g^{jk} R_{jiuk;l} = 0$$

$$\Rightarrow g^{jk} R^{l}_{jkl;u} - g^{jk} R^{l}_{jul;k} - g^{il} R^{k}_{iuk;l} = 0$$

$$\Rightarrow R^{l}_{l;u} - R^{k}_{u;k} - R^{l}_{u;l} = 0$$

$$R^{k}_{k;u} - R^{k}_{u;k} - R^{k}_{u;k} = 0$$

$$2\left(\frac{1}{2} R_{;u} - R^{k}_{u;k}\right) = 0$$

$$\left(R^{k}_{u} - \frac{1}{2}\delta^{k}_{u} R\right)_{;k} = 0$$

$$\Rightarrow G^{k}_{u\,;k} = 0$$

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Question: Show that the Einstein tensor G_{ij} obtained by lowering the index i in G^{i}_{j} is symmetric.

Solution: $G_{ij} = g_{ik} G^{k}_{j}$

$$G_{ij} = g_{ik}\left(R^{k}_{j} - \frac{1}{2}\delta^{k}_{j} R\right)$$

$$G_{ij} = R_{ij} - \frac{1}{2} g_{ij} R$$

(R_{ij}: symmetric; g_{ij}: symmetric)

$\Rightarrow G_{ij}$ is symmetric.

Question: Show that Einstein invariant $G = G^i_i$, vanishes if the space is flat.

Solution:

$$G^i_j = R^i_j - \frac{1}{2}\delta^i_j R$$

$$\Rightarrow G = G^i_i = R^i_i - \frac{1}{2}R$$

$$= R - \frac{1}{2}R = \frac{1}{2}R$$

$$\Rightarrow G = \frac{1}{2}R$$

using the result that if space is flat $K = 0$ and this implies $R = 0$.

$\Rightarrow G = 0$ if space is flat.

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Energy Momentum Tensor:

- The quantity T^{ij}; $i, j = 0,1,2,3$ is the so-called energy-momentum-stress tensor or in a shorter version the energy-momentum tensor or the stress-energy tensor.
- It is a rank-2 symmetric tensor encoding all the information about energy density, momentum density, stress, pressure etc.
- In general relativity, the stress-energy or energy-momentum tensor T^{ij} acts as the source of the gravitational field.

$$T^{\alpha\beta} = \begin{pmatrix} \underset{\text{energy density}}{T^{00}} & \underset{\text{energy flux}}{T^{0i}} \\ \underset{\text{mom. density}}{T^{i0}} & \underset{\text{stress tensor}}{T^{ij} \to i,j=1,2,3} \end{pmatrix}.$$

The ten components of this tensor have the following interpretation:

- T^{00} is the local energy density, including any potential contribution from forces between particles and their kinetic energy.

The T^{00} component represents energy density. To see why, consider the momentum four vector $p^\mu = (E, p^i)$. Using this definition, we see that in this case T^{00} is the p^0 component of the momentum four vector. This is energy density.

* Energy density is the amount of energy in a system per unit volume.

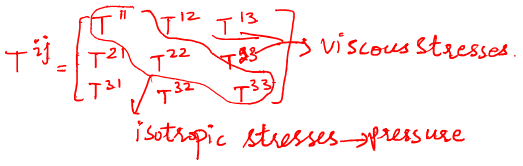
- T^{0i} is the energy flux in the ith direction. This includes not only the bulk motion but also any other processes giving rise to transfers of energy, as for instance heat conduction.
- T^{i0} is the density of the momentum component in the ith direction, i.e. the 3−momentum density. As the previous case, it also takes into account the changes in momentum associated to heat conduction.

Momentum density is momentum per unit volume. This is the flow of momentum crossing a surface of constant time. As energy-momentum tensor is symmetric $T^{0i} = T^{i0}$ represents the energy flow across the surface.

Energy flux is the rate of transfer of energy through a surface.

the rate of transfer of momentum across a unit area.

- T^{ij} is the 3-momentum flux or stress tensor, i.e the rate of flow of the ith momentum component per unit area in the plane orthogonal to the jth -direction. The component T^{ii} encodes the isotropic pressure in the ith direction while the components T^{ij} with $i \neq j$ refer to the viscous stresses of the fluid.



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Covariant Derivative of Energy Momentum Tensor:

The covariant derivative of energy momentum tensor is given by:

$$\nabla_j T^{ij} = 0$$

The covariant derivative of energy momentum tensor is actually the generalization of the conservation of energy and momentum in flat space.

This relation is called the local conservation of energy-momentum because it reduces to the conservation law $\frac{\partial T^{ij}}{\partial x^j} = 0$ (in flat spacetime) in a local inertial frame. However, it is not a conservation law like in flat space, as the energy of matter is not conserved in the presence of dynamic spacetime curvature but changes in response to it.

Energy Momentum Tensor of Dust:

One of the simplest energy-momentum tensors is the dust energy momentum tensor. This type of matter field consists of noninteracting matter.

However, consider that the dust particles carry energy and momentum. The energy and momentum of the moving dust particles give rise to a gravitational field.

In this case, there are only two quantities that can be used to describe the matter field in the problem, the energy density and how fast (and in what direction) the dust is moving.

The simplest way to obtain the first quantity, the energy density, is to jump over to the co-moving frame. If you're in the comoving frame, then you're moving along with the dust particles. In that case there is a number of dust particles per unit volume n, and each dust particle has energy m. So the energy density is given by $\rho = mn$.

The second item of interest is none other than the velocity four vector u. This of course will give us the momentum carried by the dust. Generally speaking, to get the stress-energy tensor for dust, we put this together with the energy density. So for dust, the stress-energy tensor is given by

$$T^{ij}=\rho u^i u^j$$

Where u^i is the proper velocity.

The Components of Energy Momentum Tensor of Dust in Minkowski spacetime (Flat space)

Here we discuss two cases: when frame is comoving and non comoving.

Case I:

A frame moving with the flow of the dust fluid is called comoving frame. For a co-moving observer, the four velocity reduces to

$$u^i = (c,0,0,0)$$

In this case, the stress-energy tensor takes on the remarkably simple form

$T^{ij} = \rho u^i u^j \rightarrow T^{00} = \rho u^0 u^0 = \rho c^2$

$$T^{ij}=\begin{bmatrix} c^2\rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Here u^i is the proper velocity.

So Recall

$v^i = \frac{dx^i}{dt} = \frac{d}{dt}(ct,x,y,z) = (c, v^x, v^y, v^z)$

and $\frac{d\tau}{dt} = \sqrt{1-v^2/c^2} \Rightarrow \frac{dt}{d\tau} = \frac{1}{\sqrt{1-v^2/c^2}} = \gamma$

Implies proper velocity is:

$u^i = \frac{d}{d\tau}(ct,x,y,z) = \frac{dt}{d\tau}\cdot\frac{d}{dt}(ct,x,y,z)$

$= \frac{dt}{d\tau}(c, v^x, v^y, v^z)$

$u^i = \gamma(c, v^x, v^y, v^z)$

Now in a comoving frame: $V=0$

$\Rightarrow \gamma = \frac{1}{\sqrt{1-v^2/c^2}} = 1$

$\Rightarrow u^i = (c, 0, 0, 0)$

Case II:
Now consider the case of a stationary observer seeing the dust particles go by with four velocity u. In that case, we have

$$u = (\gamma c, \gamma v^x, \gamma v^y, \gamma v^z)$$

where u^i are the ordinary components of three velocity and $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$. In this case the stress-energy tensor is

$$T^{ij} = \rho\gamma^2 \begin{bmatrix} c & cv^x & cv^y & cv^z \\ cv^x & (v^x)^2 & v^x v^y & v^x v^z \\ cv^y & u^y u^x & (v^y)^2 & v^y v^z \\ cv^z & v^z v^x & v^z v^y & (v^z)^2 \end{bmatrix}$$

Question: Show that the conservation equations for the energy-momentum tensor in case of dust lead to the equation of continuity of a fluid. (in flat space)

Solution: In a flat space the conservation equation is

$$\frac{\partial T^{ij}}{\partial x^j} = 0$$

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for $i = 0$ or $i = t$

$$\frac{\partial T^{tj}}{\partial x^j} = \frac{\partial T^{tt}}{c\partial t} + \frac{\partial T^{tx}}{\partial x} + \frac{\partial T^{ty}}{\partial y} + \frac{\partial T^{tz}}{\partial z}$$

$$= \frac{\partial}{c\partial t}(\gamma^2 \rho c^2) + \frac{\partial}{\partial x}(\gamma^2 \rho c v^x) + \frac{\partial}{\partial y}(c\gamma^2 \rho v^y) + \frac{\partial}{\partial z}(c\gamma^2 \rho v^z)$$

$$= c\left[\frac{\partial}{\partial t}(\gamma^2\rho) + \frac{\partial}{\partial x}(\gamma^2 \rho v^x) + \frac{\partial}{\partial y}(\gamma^2 \rho v^y) + \frac{\partial}{\partial z}(\gamma^2 \rho v^z)\right]$$

$\Rightarrow \frac{\partial}{\partial t}(\gamma^2\rho) + \nabla.(\gamma^2 \rho \vec{u}) = 0$, This is the equation of continuity.

$\gamma^2 \rho \vec{u} = \gamma^2 \rho (v^x, v^y, v^z) \rightarrow \nabla.(\gamma^2 \rho \vec{u})$

Question: Find trace of energy momentum tensor defined by $x^0 = ct$

$$T = T^i_i = T^{ij} g_{ij}$$

when $g_{ij} = \eta_{ij} = (cdt)^2 - dx^2 - dy^2 - dz^2$

and $T^{ij} = \rho u^i u^j$ in a comoving frame.

Solution: here only

$$T^{00} = T^{tt} = c^2\rho$$ is non zero

in a comoving frame.

$$T = T^{ij} g_{ij} = T^{ij} \eta_{ij}$$

$$= T^{00}\eta_{00} + T^{11}\eta_{11} + T^{22}\eta_{22} + T^{33}\eta_{33}$$

$$= c^2\rho(1) + 0 + 0 + 0$$

$$T = c^2\rho$$

Question: Find the trace of $T^{ij} = \rho u^i u^j$ with $g_{ij} = \eta_{ij}$ for a non comoving frame.

Solution:

$$T = T^{ij}\eta_{ij}$$

$$= T^{00}\eta_{00} + T^{11}\eta_{11} + T^{22}\eta_{22} + T^{33}\eta_{33}$$

$$= \rho\gamma^2\left[c^2 + (v^x)^2(-1) + (v^y)^2(-1) + (v^z)^2(-1)\right]$$

$$\Rightarrow T = \rho\gamma^2\left[c^2 - (v^x)^2 - (v^y)^2 - (v^z)^2\right]$$

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Question: Find the components of $T^{ij} = \rho u^i u^j$ when $x^0 = ct$, $(dx^0)^2 \overset{(1)}{=} c^2dt^2$

$g_{ij} = c^2dt^2 - a^2(t)(dx^2 + dy^2 + dz^2)$

considering a comoving frame and

$u^i = \frac{1}{\sqrt{g_{00}}}(c,0,0,0)$

Solution: $u^i = \frac{1}{\sqrt{g_{00}}}(c,0,0,0)$ here $(g_{00} = 1)$

$$\Rightarrow u^i = \frac{1}{\sqrt{1}}(c,0,0,0) = (c,0,0,0)$$

$$\Rightarrow T^{00} = \rho u^0 u^0$$

$$\Rightarrow T^{00} = \rho(c)(c) = \rho c^2$$

remaing all components are zero.

Also $T = T^{ij}g_{ij} = T^{00}g_{00} = \rho c^2$

Energy-Momentum Tensor of Perfect Fluid

A perfect fluid is a fluid that has no heat conduction or viscosity. Such fluid is characterized by its mass density ρ and the pressure P. The stress-energy tensor that describes a perfect fluid is

$$T^{ij} = \left(\rho + \frac{P}{c^2}\right) u^i u^j - P g^{ij}$$

The Components of Energy Momentum Tensor of perfect fluid in Minkowski spacetime (Flat space)

Here we discuss two cases: when frame is comoving and non comoving.

Case I:

A frame moving with the flow of the perfect fluid is called comoving frame.

For a co-moving observer, the four velocity reduces to

$$u^i = (c, 0, 0, 0)$$

In this case, the stress-energy tensor takes on the remarkably simple form

$$T^{ij} = \begin{bmatrix} c^2\rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{bmatrix}$$

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Consider,

$T^{ij} = (\rho + \frac{P}{c^2}) u^i u^j - P g^{ij}$

$u^i = (c, 0, 0, 0)$

$\Rightarrow T^{00} = (\rho + P/c^2) u^0 u^0 - P g^{00}$

$\Rightarrow T^{00} = (\rho + \frac{P}{c^2}) c^2 - P = \rho c^2$

$T^{11} = 0 - P(-1) = P$

$T^{22} = 0 - P(-1) = P$

$T^{33} = 0 - P(-1) = P$

Case II:

Now consider the case of a stationary observer seeing the perfect fluid particles go by with four velocity u^i. In that case, we have

$$u^i = (\gamma c, \gamma v^x, \gamma v^y, \gamma v^z)$$

where v^i are the ordinary components of three velocity and $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$. In this case the stress-energy tensor is

$T^{ij} = (\rho + P/c^2) u^i u^j - P g^{ij}$

$\Rightarrow T^{00} = (\rho + P/c^2) \gamma^2 c^2 - P(1)$

$= \rho c^2 \gamma^2 + P\gamma^2 - P$

$T^{00} = \rho c^2 \gamma^2 + (\gamma^2 - 1) P$

$$T^{01} = (\rho + p/c^2)\, u^0 u^1$$
$$= (\rho + p/c^2)(\gamma c)(\gamma v^x)$$
$$= \gamma^2 c v^x (\rho + p/c^2)$$

$$T^{02} = \gamma^2 c v^y (\rho + p/c^2)$$

$$T^{03} = \gamma^2 c v^z (\rho + p/c^2)$$

$$T^{11} = (\rho + p/c^2)\gamma^2 (v^x)^2 + p$$

$$T^{12} = (\rho + p/c^2)\, u^1 u^2$$
$$= \gamma^2 v^x v^y (\rho + p/c^2)$$

$$T^{13} = (\rho + p/c^2)\gamma^2 v^x v^z$$

$$T^{22} = (\rho + p/c^2)\gamma^2 (v^y)^2 + p$$

$$T^{23} = (\rho + p/c^2)\gamma^2 v^y v^z$$

$$T^{33} = (\rho + p/c^2)\gamma^2 (v^z)^2 + p$$

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Question: Find the trace of Energy-momentum tensor of perfect fluid in Minkowski space and comoving frame.

Solution: $T = T^{ij} g_{ij}$ (where $g_{ij} = \eta_{ij}$)

$g_{00} = 1$, $g_{11} = -1 = g_{22} = g_{33}$

$$\Rightarrow T = T^{00} g_{00} + T^{11} g_{11} + T^{22} g_{22} + T^{33} g_{33}$$
$$= \rho c^2 (1) + p(-1) + p(-1) + p(-1)$$
$$= \rho c^2 - 3p$$

Question: Find the trace of perfect fluid energy-momentum tensor in flat space and non-comoving frame.

Solution: $T = T^{ij} g_{ij}$

$$\Rightarrow T = T^{00} g_{00} + T^{11} g_{11} + T^{22} g_{22} + T^{33} g_{33}$$

here $g_{00} = 1$, $g_{11} = -1 = g_{22} = g_{33}$

and $T^{00} = (\rho + p/c^2)\gamma^2 c^2 - p$

$T^{11} = (\rho + p/c^2)\gamma^2 (v^x)^2 + p$

$T^{22} = (\rho + p/c^2)\gamma^2 (v^y)^2 + p$

$T^{33} = (\rho + p/c^2)\gamma^2 (v^z)^2 + p$

$$\Rightarrow T = (\rho + p/c^2)\gamma^2 \{c^2 - (v^x)^2 - (v^y)^2 - (v^z)^2\} - 4p$$

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Question: Find the components of

$$T^{ij} = (\rho + p/c^2) u^i u^j - p g^{ij}$$

where

$$g_{ij} = c^2 dt^2 - a^2(t)(dx^2 + dy^2 + dz^2)$$

and

$$u^i = \frac{1}{\sqrt{g_{00}}}(c, 0, 0, 0)$$

Solution:

$$u^i = \frac{1}{\sqrt{1}}(c, 0, 0, 0) = (c, 0, 0, 0)$$

$$T^{00} = (\rho + p/c^2) u^0 u^0 - p g^{00}$$

$$= (\rho + p/c^2) c^2 - p(1)$$

$$= \rho c^2 + p - p$$

$$\Rightarrow T^{00} = \rho c^2$$

$$T^{11} = (\rho + p/c^2)\overset{0}{u^1}\overset{0}{u^1} - p g^{11} \quad g^{11} = 1/g_{11}$$

$$= 0 - p(-1/a^2)$$

$$T^{11} = p/a^2$$

Similarly $T^{22} = p/a^2$ and $T^{33} = p/a^2$

and $T = \rho c^2 (1) + p/a^2(-a^2) + p/a^2(-a^2) + p/a^2(-a^2)$

$$T = \rho c^2 - 3p$$

Question: Discuss the conservation equation for perfect fluid in flat space and comoving frame.

Solution: The conservation equation in flat space is:

$$\frac{\partial T^{ij}}{\partial x^j} = 0$$

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for $i = 0$, we have:

$$\frac{\partial T^{0j}}{\partial x^j} = \frac{\partial T^{00}}{\partial x^0} + \frac{\partial T^{01}}{\partial x^1} + \frac{\partial T^{02}}{\partial x^2} + \frac{\partial T^{03}}{\partial x^3}$$

$$= \frac{\partial (\rho c^2)}{c\partial t} + 0 + 0 + 0 = 0$$

$$\Rightarrow c\frac{\partial \rho}{\partial t} = 0$$

$$\Rightarrow \frac{\partial \rho}{\partial t} = 0 \Rightarrow \rho = \text{constant}$$

Question: Find the conservation equation for perfect fluid in flat space and non-comoving frame, for $i = 0$.

Solution: Using the conservation equation $\frac{\partial T^{ij}}{\partial x^j} = 0$

for $i = 0$:

$$\frac{\partial T^{00}}{\partial x^0} + \frac{\partial T^{01}}{\partial x^1} + \frac{\partial T^{02}}{\partial x^2} + \frac{\partial T^{03}}{\partial x^3} = 0 \rightarrow (*)$$

where

$$T^{00} = (\rho + p/c^2)\gamma^2 c^2 - p$$

$$T^{01} = \gamma^2 c v^x (\rho + p/c^2)$$

$$T^{02} = \gamma^2 c v^y (\rho + p/c^2)$$

$$T^{03} = \gamma^2 c v^z (\rho + p/c^2)$$

$x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$

Eq. (4) yields:

$$\frac{1}{c}\frac{\partial}{\partial t}\left[(\rho + p/c^2)\gamma^2 c^2 - p\right] + \frac{\partial}{\partial x}\left[(\rho + p/c^2)\gamma^2 c v^x\right]$$

$$+ \frac{\partial}{\partial y}\left[(\rho + p/c^2)\gamma^2 c v^y\right] + \frac{\partial}{\partial z}\left[(\rho + p/c^2)\gamma^2 c v^z\right] = 0$$

$$\Rightarrow \frac{\partial}{\partial t}\left[(\rho + p/c^2)\gamma^2 - p/c^2\right] + \nabla \cdot \left[(\rho + p/c^2)\gamma^2 \vec{v}\right] = 0$$

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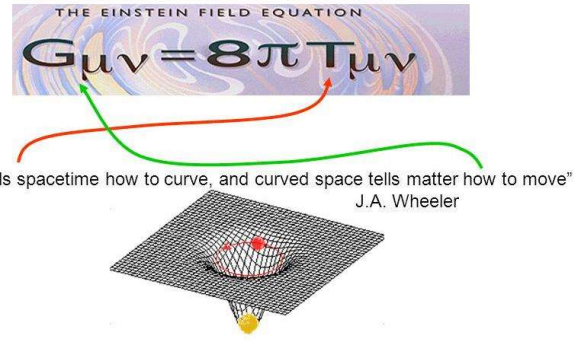
THE EINSTEIN FIELD EQUATIONS

- To take into account spacetime curvature, tensor calculus was necessary in Einstein's General Relativity.



"matter tells spacetime how to curve, and curved space tells matter how to move"
J.A. Wheeler

Einstein's Formulation of the Field Equation:

Einstein did not derive the field equation, i.e., the law of gravitation, but formulated it from a number of considerations which we will discuss here.

1) At first, the field equation must be a tensor equation.

2) secondly, under appropriate conditions the new law of gravitation must yield an equation of the form

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}=4\pi G\rho \Big\}\text{Poisson law}$$

So, the field equations must not contain higher derivatives of $g_{\mu\nu}$ than the Second. So that the ten metric functions $g_{\mu\nu}$ of Einstein theory can correspond to a single function ψ of Newton's theory.

3) Thirdly, the right hand side of Poisson law should be related to $T_{\mu\nu}$. Obeying the conservation law

$$T^{\alpha\beta}_{\ ;\beta}=0 \quad \text{or} \quad \frac{\partial T^{\alpha\beta}}{\partial x^\beta}=0 \text{ (in flat space)}$$

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4) Finally,

$$\left(R^{\alpha\beta}-\frac{1}{2}g^{\alpha\beta}R\right)_{;\beta}=0$$

Inspired by these consideration, Einstein write down the general relativistic analogue of the Poisson equation as:

$$R_{\alpha\beta}-\frac{1}{2}Rg_{\alpha\beta}=-\kappa T_{\alpha\beta}$$

where $\kappa=8\pi G/c^4$

Question: Write down the field equations for the cosmological model $ds^2=dt^2-a^2(t)(dx^2+dy^2+dz^2)$, in case of dust fluid, using

$$R_{00}=\frac{3\ddot{a}}{a}$$

$$R_{11}=-(a\ddot{a}+2\dot{a}^2)=R_{22}=R_{33}$$

$$\Rightarrow R = R_{ij} g^{ij}$$

$$= R_{00} g^{00} + R_{11} g^{11} + R_{22} g^{22} + R_{33} g^{33}$$

$$= \frac{3\ddot{a}}{a}(1) + \frac{3}{a^2}(a\ddot{a} + 2\dot{a}^2)$$

$$= \frac{6\ddot{a}}{a} + \frac{6\dot{a}^2}{a^2}$$

$$= \frac{6[\dot{a}^2 + a\ddot{a}]}{a^2}$$

$$\Rightarrow G_{00} = R_{00} - \frac{1}{2} R g_{00}$$

$$= \frac{3\ddot{a}}{a} - \frac{1}{2}(1)\left[\frac{3\ddot{a}}{a} + \frac{3\dot{a}^2}{a^3} + \frac{6\dot{a}^2}{a^4}\right]$$

$$= \frac{3}{2}\frac{\ddot{a}}{a} + 3\frac{\ddot{a}}{a^3} + 6\frac{\dot{a}^2}{a^4}$$

Similarly

$$G_{11} = R_{11} - \frac{1}{2} R g_{11}$$

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Now $T^{ij} = \rho u^i u^j$

$u^i = (1,0,0,0)$

$\Rightarrow T^{00} = \rho$, $T^{11} = T^{22} = T^{33} = 0$

$\Rightarrow T_{00} = g_{00} g_{00} T^{00}$

$T_{00} = (1)(1)\rho = \rho$

So the field equations:

$G_{\mu\nu} = \kappa T_{\mu\nu}$ yield:

$G_{00} = \kappa T_{00}$ $\quad T_{00} = g_{00} g_{00} T^{00}$

$\Rightarrow -\frac{3\dot{a}^2}{a^2} = \kappa\rho$ $\quad = (1)(1)\rho = \rho$

$T^{11} = 0 \Rightarrow T_{11} = 0$

and $G_{11} = \kappa T_{11}$

$\Rightarrow 2a\ddot{a} + \dot{a}^2 = \kappa(0)$

$\Rightarrow 2a\ddot{a} + \dot{a}^2 = 0$

Electrodynamics

It is the branch of Physics which deals with rapidly changing electric and magnetic fields.

Here we will discuss the applications of tensors or use of tensors in the well known Maxwell field equations of electrodynamics.

Maxwell Field Equations;

Maxwell's equations are four equations that together form a complete description of the production and interrelation of electric and magnetic fields.

In the 19th century, the physicist James Clerk Maxwell based his description of electromagnetic fields on these four equations, which express experimental laws.

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Maxwell Field Equations

electric field → $\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$ ← charge density; ε_0 → permittivity of vacuum

$$\nabla \cdot \mathbf{B} = 0$$

Magnetic flux

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$

μ_0 → permeability of vacuum; $\mathbf{j}$ → electric current

$\frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c$ → speed of light

Maxwell Field Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$ → Gauss' law of electricity

$$\nabla \cdot \mathbf{B} = 0$$ → Gauss' law of Magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$ → Faraday's law of Induction

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$ → Ampere's law

These are differential equations for the electric field $\vec{E}$ and magnetic field $\vec{B}$ in the presence of a charge function ρ and an electrical current $\vec{j}$. The quantities ε_0 and μ_0 are physical constants called the permittivity and permeability of vacuum, respectively.

Where electric field is the physical field that surrounds electrically charged particles and exerts force to all other charged particles in the field.

A magnetic field describes the magnetic influence on moving electric charges, electric currents and magnetic materials.

Here we will discuss how different electromagnetic quantities are transformed in relativity under Lorentz transformations.

Charge Density:

Charge density is defined as the measure of electric charge per unit volume.

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Consider two frames of reference S and S′ such that frame S′ is moving with velocity $\vec{v}$ away from frame S. Now if charge density is defined as

$$\rho_0 = \frac{Ne}{l_0^3}$$

in the frame S′.

- $N \rightarrow$ No of electrons
- $e \rightarrow$ charge
- $l_0 \rightarrow$ length of one side of cube

Then in frame S the charge density appears to be

$$\rho = \frac{Ne}{l_0^3\sqrt{1-v^2/c^2}}$$

→ this factor appears due to length contraction.

$$\Rightarrow \rho = \frac{\rho_0}{\sqrt{1-v^2/c^2}} \quad \left(\because \rho_0 = \frac{Ne}{l^3}\right)$$

Current Density: It is the amount of current flowing across the given region.

There are two frames of reference here, S and S'. where S' frame is moving away from S with a velocity $\vec{V}$. Then there is no current for frame S'.

However, for frame S we have:

$$\vec{j} = \rho \vec{v}$$

$$\Rightarrow \vec{j} = \frac{\rho_0 \vec{v}}{\sqrt{1 - v^2/c^2}}$$

or in component form:

$$j_x = \frac{\rho_0 v_x}{\sqrt{1 - v^2/c^2}}, \quad j_y = \frac{\rho_0 v_y}{\sqrt{1 - v^2/c^2}}, \quad j_z = \frac{\rho_0 v_z}{\sqrt{1 - v^2/c^2}}$$

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Mass:

If m_0 is the mass of the charge density ρ_0 according to frame S', then mass according to S is:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

Consider, $m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$ and $\rho = \frac{\rho_0}{\sqrt{1 - v^2/c^2}}$

$\frac{m}{m_0} = \frac{\rho}{\rho_0} \Rightarrow \frac{\rho_0}{m_0} = \frac{\rho}{m}$

$$\Rightarrow \frac{\rho}{m} = \frac{\rho_0}{m_0} \Rightarrow \rho = \left(\frac{\rho_0}{m_0}\right) m$$

and $\vec{j} = \rho \vec{u}$, $\rho = \left(\frac{\rho_0}{m_0}\right) m$

$$\Rightarrow \vec{j} = \left(\frac{\rho_0}{m_0}\right) m \vec{u} \Rightarrow \vec{j} = \left(\frac{\rho_0}{m_0}\right) \vec{p}$$

Four current:

It is a four vector whose three space components are those of the ordinary current density and its time component is the charge density.

Mathematically, $J^\mu = (\rho c, \vec{j})$

$J_\mu = g_{\mu\nu} J^\nu$

$= g_{00} J^0 + g_{11} J^1 + g_{22} J^2 + g_{33} J^3$

$= \rho c - J^1 - J^2 - J^3$

$= (\rho c, -\vec{j})$

where $\vec{j} = j_x \hat{i} + j_y \hat{j} + j_z \hat{k}$

$\rho \rightarrow$ charge density

or $J_\mu = (\rho c, -\vec{j})$ [in Minkowski space]

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Conservation equation for four current density is:

$\nabla_\alpha J^\alpha = 0$ or $J^\alpha{}_{;\alpha} = 0$

In Special relativity:

$J^\alpha{}_{,\alpha} = 0 \Rightarrow J^0{}_{,0} + J^1{}_{,1} + J^2{}_{,2} + J^3{}_{,3} = 0$

$\Rightarrow \frac{1}{q} \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$

$\Rightarrow \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$ ✓✓

Electromagnetic four potential:

The four potential has time and space components defined as

$A^\mu = (\frac{\phi}{c}, \vec{A})$

$\vec{A} \rightarrow$ vector potential

$\phi \rightarrow$ scalar potential

The scalar and vector potential are related to the potential energy U and potential momentum $\vec{Q}$ of a particle having charge q by:

$$U = q\phi$$

$$\vec{Q} = q\vec{A}$$

The magnetic field $\vec{B}$ is defined in terms of vector potential as:

$$\vec{B} = \nabla \times \vec{A}$$

and electric field as:

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

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Four Potential Conservation:

Gives: $\nabla_\alpha A^\alpha = 0$ or $A^\alpha{}_{;\alpha} = 0$

In flat Space:

$$A^\alpha{}_{,\alpha} = 0 \Rightarrow A^0{}_{,0} + A^1{}_{,1} + A^2{}_{,2} + A^3{}_{,3} = 0$$

$$\Rightarrow \frac{\partial(\phi/c)}{c\partial t} + \nabla \cdot \vec{A} = 0$$

$$\Rightarrow \frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \vec{A} = 0$$

Electromagnetic field:

The electromagnetic field contains the combine effects of electric and magnetic fields produced by charged objects.

The area in the surrounding of a charged particle within which it can exert the electrostatic force

force on other particles is the electric field.
Magnetic field is similar to electric field and describes the effects of nearby magnetized material and electric currents at a point.

When charged particles are in motion then magnetic field is also produced such that both fields are orthogonal to each other as well as to the direction of motion.

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Maxwell field equations describe the evolution of electromagnetic field.

Electromagnetic field tensor:

The electromagnetic field tensor also called Faraday's tensor is defined as

$$F^{\alpha\beta} = \nabla^{\alpha} A^{\beta} - \nabla^{\beta} A^{\alpha}$$

where A^{α} is the four vector potential.

Also, $A^\alpha = (\phi/c, \vec{A})$

In flat space:

$$F^{\alpha\beta} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$F^{\alpha\beta}$ is an antisymmetric tensor.

and

$$F_{\alpha\beta} = g_{\alpha\gamma} F^{\gamma\delta} g_{\delta\beta}$$

$$= \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

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Components of $F^{\alpha\beta}$ in Flat space

$F^{\alpha\beta} = \nabla^\alpha A^\beta - \nabla^\beta A^\alpha \rightarrow A^{\beta;\alpha} - A^{\alpha;\beta}$

$F^{\alpha\beta} = g^{\alpha\gamma} \nabla_\gamma A^\beta - g^{\beta\gamma} \nabla_\gamma A^\alpha$

where $\alpha, \beta = 0, 1, 2, 3$

and $F^{00} = 0 = F^{11} = F^{22} = F^{33}$

$F^{01} = g^{0\gamma} \nabla_\gamma A^1 - g^{1\gamma} \nabla_\gamma A^0$

$= g^{00} A^1{}_{,0} - g^{11} A^0{}_{,1}$ $\quad x^0 = ct \ (\partial x^0 = c\,\partial t)$

$F^{01} = (1)\frac{\partial A^1}{c\,\partial t} - (-1)\frac{1}{c}\frac{\partial \phi}{\partial x}$

Recall that,

$\vec{B} = \nabla \times \vec{A}$ and $\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$

$\Rightarrow F^{01} = \frac{1}{c}\left(\frac{\partial \phi}{\partial x} + \frac{\partial A^1}{\partial t}\right) = -\frac{1}{c} E_x$

$\Rightarrow F^{10} = E_x/c$

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

$$= -\left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) - \left(\frac{\partial A^1}{\partial t}\hat{i} + \frac{\partial A^2}{\partial t}\hat{j} + \frac{\partial A^3}{\partial t}\hat{k}\right)$$

$$= -\left(\frac{\partial\phi}{\partial x} + \frac{\partial A^1}{\partial t}\right)\hat{i} \rightarrow E_x = -\left(\frac{\partial\phi}{\partial x} + \frac{\partial A^1}{\partial t}\right)$$

$$-\left(\frac{\partial\phi}{\partial y} + \frac{\partial A^2}{\partial t}\right)\hat{j} \rightarrow E_y = -\left(\frac{\partial\phi}{\partial y} + \frac{\partial A^2}{\partial t}\right)$$

$$-\left(\frac{\partial\phi}{\partial z} + \frac{\partial A^3}{\partial t}\right)\hat{k} \rightarrow E_z = -\left(\frac{\partial\phi}{\partial z} + \frac{\partial A^3}{\partial t}\right)$$

$$\vec{B} = \nabla \times \vec{A}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A^1 & A^2 & A^3 \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial A^3}{\partial y} - \frac{\partial A^2}{\partial z}\right) - \hat{j}\left(\frac{\partial A^3}{\partial x} - \frac{\partial A^1}{\partial z}\right) + \hat{k}\left(\frac{\partial A^2}{\partial x} - \frac{\partial A^1}{\partial y}\right)$$

(B_x, B_y, B_z)

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Similarly,

$F^{02} = -\frac{1}{c}E_y$ and $F^{03} = \frac{1}{c}E_z$

and

$$F^{12} = g^{1r}\nabla_r A^2 - g^{2r}\nabla_r A^1$$

$$= g^{11}A^2_{,1} - g^{22}A^1_{,2}$$

$$= -\frac{\partial A^2}{\partial x} + \frac{\partial A^1}{\partial y} = \frac{\partial A^1}{\partial y} - \frac{\partial A^2}{\partial x}$$

$F^{12} = -B_z \Rightarrow F^{21} = B_z$

$$F^{13} = g^{1r}\nabla_r A^3 - g^{3r}\nabla_r A^1$$

$$= g^{11}A^3_{,1} - g^{33}A^1_{,3}$$

$F^{13} = B_y \Rightarrow F^{31} = -B_y$.

$F^{23} = -B_x$

In Short, we can write

$$F_{ij} = \varepsilon_{ijk} B^k \; ; \; i,j,k = 1,2,3$$

and $$F_{i0} = -\frac{E_i}{c} \; ; \; i = 1,2,3$$

$$\star \; F^{i0} = \frac{E_i}{c}$$

and $$F^{ij} = \varepsilon^{ijk} B_k$$

Using:

$$F_{ij} = \varepsilon_{ijk} B^k$$

{ cyclic 123, permutations 231, 312 }

$$F_{12} = \varepsilon_{12k} B^k = \varepsilon_{123} B^3$$

$$= (1) B^3 = B^3$$

$$F_{12} = g^{3\mu} B_\mu = g^{33} B_3 = (-1) B_3$$

$$= -B_z \Rightarrow F_{21} = B_z$$

$$F_{13} = B_y \text{ and } F_{31} = -B_y$$

and $F_{23} = -B_x$ and $F_{32} = B_x$

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$$F_{i0} = -\frac{E_i}{c}$$

$$\Rightarrow F_{10} = -\frac{E_1}{c} = -\frac{E_x}{c}$$

Similarly

$$F_{20} = -\frac{E_y}{c} \text{ and } F_{30} = -\frac{E_z}{c}$$

$$\Rightarrow F_{01} = \frac{E_x}{c}, \; F_{02} = \frac{E_y}{c}, \; F_{03} = \frac{E_z}{c}$$

Now using:

$$F^{ij} = \varepsilon^{ijk} B_k$$

$$\Rightarrow F^{12} = \varepsilon^{123} B_3$$

$$\Rightarrow F^{12} = g^{11} g^{22} g^{33} \varepsilon_{123} B_3$$

$$= (-1)(-1)(-1)(1) B_z$$

$$= -B_z$$

and $F^{21} = B_z$

Maxwell field equations in Tensor form:

These equations in tensor form are given below:

$$F_{\alpha\beta;\gamma} + F_{\beta\gamma;\alpha} + F_{\gamma\alpha;\beta} = 0$$

$$F^{\alpha\beta}{}_{;\beta} = \mu_0 J^{\alpha}$$

and in flat space covariant derivatives are replaced by partial derivatives.

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$$

$$F^{\alpha\beta}{}_{,\beta} = \mu_0 J^{\alpha}$$

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Now we will derive the four Maxwell equations from tensorial form in flat space.

1) consider,

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$$

for $\alpha = 1$, $\beta = 2$ and $\gamma = 3$ we have

$$\Rightarrow F_{12,3} + F_{23,1} + F_{31,2} = 0$$

$$(-B_z)_{,z} + (-B_x)_{,x} + (-B_y)_{,y} = 0$$

$$\Rightarrow \frac{\partial B_z}{\partial z} + \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = 0$$

$$\Rightarrow \nabla \cdot \vec{B} = 0$$

or $\text{div}\,\vec{B} = 0$

In Short: $F_{jk} = \varepsilon_{jki} B^i$, $F_{ki} = \varepsilon_{kij} B^j$

$$F_{ij} = \varepsilon_{ijk} B^k = B^k$$

when $i \neq j \neq k$ and 'ε_{ijk}' is even permutation

so

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$$

for $\alpha = i$, $\beta = j$, $\gamma = k$

$$\Rightarrow F_{ij,k} + F_{jk,i} + F_{ki,j} = 0$$

$$\Rightarrow B^k{}_{,k} + B^i{}_{,i} + B^j{}_{,j} = 0 \Rightarrow \text{div}\, B = 0$$

2) Now again:

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$$

for $\alpha = i$, $\beta = j$ and $\gamma = 0$

$$\Rightarrow F_{ij,0} + F_{j0,i} + F_{0i,j} = 0$$

now $F_{ij} = \varepsilon_{ijk} B^k$

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$F_{j0} = -E_j/c$ and $F_{0i} = E_i/c$

$$\Rightarrow \varepsilon_{ijk} B^k{}_{,0} + \left(-\frac{E_j}{c}\right)_{,i} + \left(\frac{E_i}{c}\right)_{,j} = 0$$

$$\Rightarrow \frac{1}{c}\frac{\partial B^k}{\partial t} + \frac{1}{c}(E_{i,j} - E_{j,i}) = 0 \quad \left]\begin{array}{l}k^{th} \\ \text{component} \\ \text{of}\end{array}\right.$$

$$\Rightarrow \frac{\partial \vec{B}}{\partial t} + \nabla \times \vec{E} = 0$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

3) Now consider

$$F^{\alpha\beta}{}_{,\beta} = \mu_0 J^\alpha$$

For $\alpha = 0$ and $\beta = i$, we have

$$F^{0i}{}_{,i} = \mu_0 J^0 \quad (\because i = 1,2,3)$$

$$\Rightarrow \left(-\frac{E_i}{c}\right)_{,i} = \mu_0 J^0$$

$$\Rightarrow (-E_i{}_{,i}) = \mu_0 c J^0$$

also here $g_{ii} = -1$ and $g^{ii} = -1$

so $(-E_i)_{,i} = (g^{ii} E_i)_{,i} = E^i{}_{,i}$

$\Rightarrow E^i{}_{,i} = \mu_0 c J^0$

where $J^0 = \rho c$ and $c^2 = \frac{1}{\mu_0 \varepsilon_0}$

$\Rightarrow E^i{}_{,i} = \mu_0 \frac{1}{\mu_0 \varepsilon_0} \rho = \rho/\varepsilon_0$

$\Rightarrow div\, E = \rho/\varepsilon_0$

4) Consider,

$F^{\alpha\beta}{}_{,\beta} = \mu_0 J^\alpha$

for $\alpha = i$

$\Rightarrow F^{i0}{}_{,0} + F^{ij}{}_{,j} = \mu_0 J^i$

As $F^{i0} = E_i/c$ $\quad (E_i = -E^i)$

$\Rightarrow F^{i0} = -E^i/c$

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$\Rightarrow F^{i0}{}_{,0} = \frac{1}{c} \frac{\partial}{\partial t} (-E^i/c)$

$F^{i0}{}_{,0} = -\frac{1}{c^2} \frac{\partial E^i}{\partial t}$ $\quad x^0 = ct$, $\partial x^0 = c\,\partial t$

and $F^{ij} = \varepsilon^{ijk} B_k$

$\Rightarrow F^{ij}{}_{,j} = \varepsilon^{ijk} B_{k,j}$

Implies:

$\frac{-1}{c^2} \frac{\partial E^i}{\partial t} + \varepsilon^{ijk} B_{k,j} = \mu_0 J^i$

$\Rightarrow \varepsilon^{ijk} B_{k,j} = \mu_0 J^i + \frac{1}{c^2} \frac{\partial E^i}{\partial t}$

$\Rightarrow (\nabla \times \vec{B})^i = \mu_0 J^i + \frac{1}{c^2} \frac{\partial E^i}{\partial t}$

$\Rightarrow \nabla \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$

Electromagnetic Energy Momentum Tensor:

It is defined by:

$$T^{\alpha}_{\ \beta} = \frac{1}{4}\delta^{\alpha}_{\beta}\left(F_{\mu\gamma}F^{\mu\gamma}\right) - F_{\mu\beta}F^{\mu\alpha}$$

or

$$T^{\alpha\beta} = \frac{1}{4}g^{\alpha\beta}\left(F_{\mu\nu}F^{\mu\nu}\right) - F^{\alpha\mu}F^{\beta}_{\mu}$$

Trace:

$$T^{\alpha}_{\ \alpha} = T^{0}_{\ 0} + T^{1}_{\ 1} + T^{2}_{\ 2} + T^{3}_{\ 3}$$

$$T^{0}_{\ 0} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - F_{\mu 0}F^{\mu 0}$$

$$T^{1}_{\ 1} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - F_{\mu 1}F^{\mu 1}$$

$$T^{2}_{\ 2} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - F_{\mu 2}F^{\mu 2}$$

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$$T^{3}_{\ 3} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - F_{\mu 3}F^{\mu 3}$$

$$\Rightarrow T^{\alpha}_{\ \alpha} = F_{\mu\nu}F^{\mu\nu} - \left\{F_{\mu 0}F^{\mu 0} + F_{\mu 1}F^{\mu 1} + F_{\mu 2}F^{\mu 2} + F_{\mu 3}F^{\mu 3}\right\}$$

$$= F_{\mu\nu}F^{\mu\nu} - F_{\mu\nu}F^{\mu\nu}$$

$$= 0$$

Components of electromagnetic Energy Momentum tensor in Flat Space:

consider

$$T^{\alpha}_{\beta} = \frac{1}{4}\delta^{\alpha}_{\beta}\left(F_{\mu\nu}F^{\mu\nu}\right) - F_{\mu\beta}F^{\mu\alpha}$$

consider

$$F_{\mu\nu}F^{\mu\nu} = F_{0\nu}F^{0\nu} + F_{1\nu}F^{1\nu} + F_{2\nu}F^{2\nu} + F_{3\nu}F^{3\nu}$$

where $\nu = 0,1,2,3$ in each term

$\Rightarrow F_{\mu\nu}F^{\mu\nu} = 2B^2 - \frac{2}{c^2}E^2$

Now

$T^0_{\ 0} = \frac{1}{4}\left(2B^2 - \frac{2}{c^2}E^2\right) - F_{\mu 0}F^{\mu 0}$

$= \frac{1}{4}\left(2B^2 - \frac{2}{c^2}E^2\right) + \frac{E_x^2}{c^2} + \frac{E_y^2}{c^2} + \frac{E_z^2}{c^2}$

$= \frac{1}{2}B^2 - \frac{1}{2c^2}E^2 + \frac{1}{c^2}E^2 = \frac{1}{2}\left(B^2 + \frac{E^2}{c^2}\right)$

Similarly:

$T^0_{\ 1} = \frac{1}{4}\delta^0_1 F_{\mu\nu}F^{\mu\nu} - F_{\mu 1}F^{\mu 0}$

$= -F_{01}F^{00} - F_{11}F^{10} - F_{21}F^{20} - F_{31}F^{30}$

$= -(B_z)\left(\frac{E_y}{c}\right) - (-B_y)\left(+\frac{E_z}{c}\right)$

$= \frac{1}{c}\left[B_y E_z - B_z E_y\right]$

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Properties of $F^{\mu\nu}$:

1) Antisymmetry $F^{\mu\nu} = -F^{\nu\mu}$

2) It has Six independent components.

3) $F_{\mu\nu}F^{\mu\nu} = 2\left(B^2 - E^2/c^2\right)$

4) Trace:

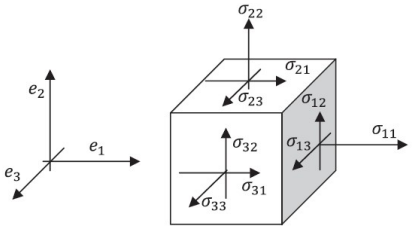
$F = F^{\mu}_{\ \mu} = 0$

The Stress Tensor

Stress is defined as force per unit area. If we take a cube of material and subject it to an arbitrary load we can measure the stress on it in various directions. These measurements will form a second rank tensor; called the stress tensor. Whose each component σ_{ij} is characterized by its sub-indices:

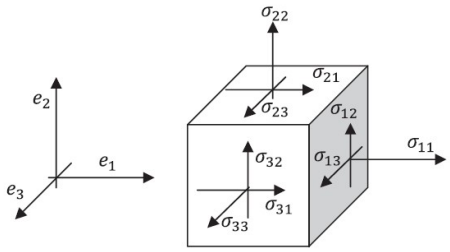
- Index i designates the coordinate plane on which the component acts.
- Index j identifies the coordinate direction in which the component acts.

$$\boldsymbol{\sigma} \equiv \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$



Normal and Shear Stresses:

- σ_{ii} are called the normal stresses acting on plane i.
- Whereas $\sigma_{ij}; i \neq j$ are called the tangential or shear stresses, acting on the plane perpendicular to the i-axis in the direction of j-axis.



The sign criterion for the stress components:

1) $\sigma_{ii} = \begin{cases} \text{positive} \Rightarrow \text{tensile stress} \\ \text{negative} \Rightarrow \text{compressive stress} \end{cases}$

2) $\sigma_{ij}; i \neq j \begin{cases} \text{positive} \Rightarrow \text{positive direction of the } j\text{-axis.} \\ \text{negative} \Rightarrow \text{negative direction of } j\text{-axis.} \end{cases}$

★ tensile stress means material stretches.
★ compressive stress means material compresses.

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Cauchy's Equations of Motion:

Cauchy's first law of motion

According to the principle of conservation of linear momentum, if the body is in static equilibrium it can be demonstrated that the components of the Cauchy stress tensor in every material point in the body satisfy the equilibrium equations.

$$\sigma_{ji,j} + F_i = 0$$

here F_i represents body force component.
★ A body force is a force that acts throughout the volume of the body for example: gravity, electric fields and magnetic fields.

So the Cauchy's equilibrium equations are: (in 3-dimensions)

$$\frac{\partial \sigma_{11}}{\partial x^1} + \frac{\partial \sigma_{21}}{\partial x^2} + \frac{\partial \sigma_{31}}{\partial x^3} + F_1 = 0$$

$$\frac{\partial \sigma_{12}}{\partial x^1} + \frac{\partial \sigma_{22}}{\partial x^2} + \frac{\partial \sigma_{32}}{\partial x^3} + F_2 = 0$$

$$\frac{\partial \sigma_{13}}{\partial x^1} + \frac{\partial \sigma_{23}}{\partial x^2} + \frac{\partial \sigma_{33}}{\partial x^3} + F_3 = 0$$

Cauchy's second law of motion

According to the principle of conservation of angular momentum, equilibrium requires that the summation of moments with respect to an arbitrary point is zero, which leads to the conclusion that the stress tensor is symmetric, thus having only six independent stress components, instead of the original nine in three dimensions.

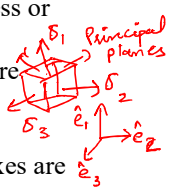
$$\sigma_{ij} = \sigma_{ji}$$

Moment: The moment of a force is a measure of its tendency to cause a body to rotate about a specific point or axis.

Angular momentum is the rotational analog of linear momentum. $\vec{L} = \vec{r} \times \vec{p}$

Principle Stresses: $\sigma = (\sigma_{ij}) = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix}$

- Regardless of the state of stress, it is always possible to choose a special set of axes (principal axes of stress or principal stress directions) so that the shear stress components vanish when the stress components are referred to this system.



- The three planes perpendicular to the principal axes are the principal planes.
- The normal stress components in the principal planes are the principal stresses.

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Principle Stresses:

- The Eigen values of σ_{ij} tensor; represented as σ_1 , σ_2 , σ_3 are referred to as the principle stresses. The Eigen vectors are the principle stress directions.

$$\begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix}$$ → principal stress matrix

- The stress tensor is a symmetric 2nd order tensor so its eigenvalues are real numbers.

For the eigenvalue λ and its corresponding eigenvector v:

$$\sigma_{ij} v = \lambda v$$

$$\Rightarrow (\sigma_{ij} - \lambda I) \cdot v = 0$$

and $|\sigma_{ij} - \lambda I| = 0$ → characteristic equation.

$$\Rightarrow \lambda^3 - I_1 \lambda^2 - I_2 \lambda - I_3 = 0$$

I_1, I_2, I_3 → Invariants

The three eigen values $\lambda_1, \lambda_2, \lambda_3 \rightarrow \sigma_1, \sigma_2, \sigma_3$ are principal stresses.

→ Recall that the invariants associated with a tensor are values which do not change with the change of coordinate system.

Stress Invariants

- Principal stresses are invariants of the stress state: Invariant w.r.t. rotation of the coordinate axes to which the stresses are referred.

The principal stresses are combined to form the stress invariants $I_i; i = 1,2,3$ as:

$$I_1 = Tr(\sigma) = \sigma_{ii} = \sigma_1 + \sigma_2 + \sigma_3$$

$$I_2 = -(\sigma_1\sigma_2 + \sigma_1\sigma_3 + \sigma_2\sigma_3)$$

$$I_3 = \det(\sigma)$$

where $\sigma = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} = (\sigma_{ij})$

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1) Mean Stress:

Given the cauchy stress tensor and its principal stresses, mean stress is defined as

$$\sigma_m = \frac{1}{3} Tr(\sigma_{ij}) = \frac{1}{3}\sigma_{ii} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$$

2) Mean Pressure:

$$\bar{P} = -\sigma_m = -\frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$$

3) A spherical or hydrostatic state of stress:

$$\sigma_1 = \sigma_2 = \sigma_3 \Rightarrow \sigma_{ij} = \begin{bmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & \sigma \end{bmatrix}$$

In a hydrostatic state of stress, the stress tensor is isotropic and, thus, its components are the same in any Cartesian coordinate system. As a consequence, any direction is a principal direction and the stress state (traction vector) is the same in any plane.

Question: A body in equilibrium subject to the following stress state

$$\sigma = (\sigma_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix} \qquad \sigma_{ji,j} + F_i = 0$$

determine body forces.

Solution: here we use the following Eqs:

$$\frac{\partial \sigma_{11}}{\partial x^1} + \frac{\partial \sigma_{21}}{\partial x^2} = -F_1 \quad \text{and} \quad \frac{\partial \sigma_{12}}{\partial x^1} + \frac{\partial \sigma_{22}}{\partial x^2} = -F_2$$

$$\Rightarrow \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(5y) = -F_1$$

$$\Rightarrow F_1 = -y + 5 \Rightarrow F_1 = 5 - y$$

and

$$\frac{\partial}{\partial x}(5y) + \frac{\partial}{\partial y}(4x) = -F_2$$

$$\Rightarrow F_2 = -(0+0) = 0$$

$$\vec{F} = (5-y, 0)$$

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Question: For the stress matrix $(\sigma_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix}$ write normal and shear stresses.

Solution: **Normal stresses:**

$$\sigma_{11} = xy \,, \quad \sigma_{22} = 4x$$

Shear Stresses:

$$\sigma_{12} = \sigma_{21} = 5y$$

Question: Find the constraint for the stress matrix $(\sigma_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix}$ for tensile and compressive stress.

Solution:

For tensile stress: $\sigma_{ii} > 0$

$$\Rightarrow \sigma_{11} > 0 \text{ and } \sigma_{22} > 0$$

$$\Rightarrow xy > 0 \text{ and } 4x > 0$$

$$\Rightarrow x > 0 \text{ and } y > 0$$

For compressive stress:

$\sigma_{ii} < 0$

$\Rightarrow \sigma_{11} < 0$ and $\sigma_{22} < 0$

$\Rightarrow xy < 0$ and $4x < 0$

$\Rightarrow x < 0$ and $y > 0$

$x < 0$ and $y > 0$.

Question: Find the principal stresses for the stress state:

$$(\sigma_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix}$$

Solution: consider:

$$|\sigma_{ij} - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} xy - \lambda & 5y \\ 5y & 4x - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (xy - \lambda)(4x - \lambda) - 25y^2 = 0$$

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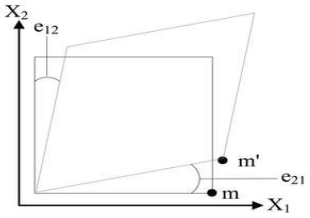
$$\Rightarrow 4x^2y - 4\lambda x - \lambda xy + \lambda^2 - 25y^2 = 0$$

$$\Rightarrow \lambda^2 - (4x + xy)\lambda + 4x^2y - 25y^2 = 0$$

$$\Rightarrow \lambda = \frac{4x + xy \pm \sqrt{(4x+xy)^2 - 4(4x^2y - 25y^2)}}{2}$$

The Strain Tensor

Strain is defined as the relative change in the position of points within a body that has undergone deformation. The classic example in two dimensions is of the square which has been deformed to a parallelepiped.



where $\varepsilon_{ij} = e_{ij} - \tilde{\omega}_{ij}$

ε_{ij} is defined as the strain

$\tilde{\omega}_{ij}$ is defined as rigid body rotation.

and

$$\varepsilon_{ij} = \frac{1}{2}(e_{ij} + e_{ji}) \rightarrow \varepsilon_{ji} = \frac{1}{2}(e_{ji} + e_{ij}) = \varepsilon_{ij}$$

$$\tilde{\omega}_{ij} = \frac{1}{2}(e_{ij} - e_{ji}) \rightarrow \tilde{\omega}_{ji} = \frac{1}{2}(e_{ji} - e_{ij}) = -\tilde{\omega}_{ij}$$

For the two-dimensional case:

$$\begin{pmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{pmatrix} = \begin{pmatrix} e_{11} & \frac{1}{2}(e_{12}+e_{21}) \\ \frac{1}{2}(e_{21}+e_{12}) & e_{22} \end{pmatrix} + \begin{pmatrix} 0 & \frac{1}{2}(e_{12}-e_{21}) \\ \frac{1}{2}(e_{21}-e_{12}) & 0 \end{pmatrix}$$

ε_{ij} $\tilde{\omega}_{ij}$

- ε_{ij} is a symmetric tensor and $\tilde{\omega}_{ij}$ is an antisymmetric tensor; the leading diagonal of $\tilde{\omega}_{ij}$ is always zero.
- The tensor ε_{ij} has Eigen values which are called the principal strains $\varepsilon_1, \varepsilon_2, \varepsilon_3$.

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Question: Given the matrix:

$$e_{ij} = \begin{bmatrix} a & c \\ c & a \end{bmatrix}$$ find ε_{ij} and $\tilde{\omega}_{ij}$.

Solution:

$$\varepsilon_{ij} = \frac{1}{2}(e_{ij} + e_{ji})$$

$$\Rightarrow \varepsilon_{11} = \frac{1}{2}(e_{11} + e_{11}) = a$$

$$\varepsilon_{22} = a$$

$$\varepsilon_{12} = \frac{1}{2}(e_{12} + e_{21}) = \frac{1}{2}(c + c) = c$$

$$\varepsilon_{21} = \frac{1}{2}(c + c) = c$$

$$\Rightarrow \varepsilon_{ij} = \begin{bmatrix} a & c \\ c & a \end{bmatrix} = e_{ij}$$

Similarly: $\tilde{\omega}_{ij} = \frac{1}{2}(e_{ij} - e_{ji})$

$$\Rightarrow \tilde{\omega}_{11} = \tilde{\omega}_{22} = 0$$

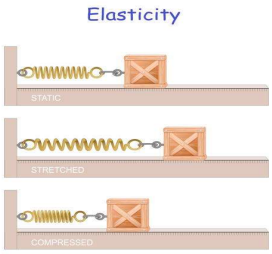
$$\tilde{\omega}_{12} = \frac{1}{2}(e_{12} - e_{21}) = \frac{1}{2}(c - c) = 0 = -\tilde{\omega}_{21}$$

$$\Rightarrow \tilde{\omega}_{ij} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Elasticity

Unlike stress and strain, elasticity is an intrinsic property of a material. The elastic properties of Earth materials affects everything from the variation of density with depth in the planet to the speed at which seismic waves pass through the interior.

If the deformation in a body under stress does not exceed a certain limit, called the elastic limit, the body will return to its initial shape when the stress is removed.



Relation of Elasticity with stress and Strain:

If the amount of stress (σ) is infinitesimally small then the amount of strain (ε), which is also infinitesimal, is linearly proportional to the strain and may be written as:

$$\varepsilon = s\sigma$$

$$\text{or } \sigma = C\varepsilon$$

Where s is the elastic compliance and c is the elastic stiffness. In order to relate two second rank tensors, a fourth rank tensor is necessary.

↓

stress and strain

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Elasticity Tensor:

So, $\varepsilon_{ij} = S_{ijkl}\,\sigma_{kl}$

or $\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl}$

* S_{ijkl} → elasticity tensor → related to change in volume when subjected to an applied force. It is reciprocal of stiffness tensor.

* C_{ijkl} → Stiffness tensor

Components of S_{ijkl}:

In 2D → $2^4 = 16$ → $3^4 = 81$

For the three-dimensional case there are 81 terms in a fourth rank tensor. However, both stress and strain are symmetric tensors; $\sigma_{ij} = \sigma_{ji}$ and $\varepsilon_{ij} = \varepsilon_{ji}$ each only has 6 independent terms.

There are only 6 equations needed to calculate ε_{ij} from σ_{ij} and in each equation there will only be 6 independent terms. Therefore, there can be no more than 36 independent values in S_{ijkl}.

Components of ε_{ij} using: $\varepsilon_{ij} = S_{ijkl}\sigma_{kl}$

To calculate ε_{ij} for the three-dimensional case we would begin like so:

$$\varepsilon_{11} = S_{1111}\sigma_{11} + S_{1112}\sigma_{12} + S_{1113}\sigma_{13} + S_{1121}\sigma_{21} + S_{1122}\sigma_{22} + S_{1123}\sigma_{23} + S_{1131}\sigma_{31} + S_{1132}\sigma_{32} + S_{1133}\sigma_{33}$$

$$\varepsilon_{12} = S_{1211}\sigma_{11} + S_{1212}\sigma_{12} + S_{1213}\sigma_{13} + S_{1221}\sigma_{21} + S_{1222}\sigma_{22} + S_{1223}\sigma_{23} + S_{1231}\sigma_{31} + S_{1232}\sigma_{32} + S_{1233}\sigma_{33}$$

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